1 Non-Abelian anyons from gauging logical gates

Source: arXiv:2602.12228v1, *Non-Abelian Quantum Low-Density Parity Check Codes and Non-Clifford Operations from Gauging Logical Gates via Measurements.*

Background assumed: basic qubit states, tensor products, and quantum measurement. The note introduces stabilizer codes, anyons, gauge fields, and the group theory it uses. Chains and cochains are developed from binary linear algebra in the discussion of Section III.

The paper connects two descriptions of the same quantum system. In quantum error correction, a set of constraints defines a space in which information can be encoded. In many-body physics, those same constraints define a Hamiltonian's ground space, and violations of the constraints are excitations. Gates acting on the encoded information can then be used to construct new phases with different excitations.

The opening follows Section II of the paper. We first understand the toric code and the action of a controlled-Z gate on its excitations. We then explain gauging, derive its circuit and measurement implementation, and show how to correct random measurement outcomes. Finally, we unpack the name $\mathcal{D}(D_4)$ for the resulting phase. Section III will reuse this construction when there is no ordinary spatial lattice.

1.1 Before the construction: states, codes, and anyons

1.1.1 A constraint can define both a code and a ground state

For a qubit, the Pauli operators act as

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle, \qquad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

They obey $X^2 = Z^2 = I$ and $XZ = -ZX$. Pauli operators on different qubits commute. The states $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$ are eigenstates of X, with eigenvalues ± 1.

A **stabilizer** is an operator whose $+1$ eigenspace we require the state to occupy. For example, $Z_1 Z_2|\psi\rangle = |\psi\rangle$ restricts two qubits to the span of $|00\rangle$ and $|11\rangle$. It fixes their parity without fixing the coefficients in $\alpha|00\rangle + \beta|11\rangle$. Those coefficients can carry one encoded qubit. In a Pauli stabilizer code, the defining operators are commuting Pauli products, also called **checks**.

The Hamiltonian $-S$, for a check S with eigenvalues ± 1, assigns energy -1 to its $+1$ eigenspace and $+1$ to its -1 eigenspace. Thus a sum $-\sum_j S_j$ of compatible checks has the code space as its lowest-energy, or **ground**, space. A violated check costs energy. Measuring the checks gives a list of signs called the **syndrome**, which tells us which constraints are violated without reading out the encoded coefficients.

1.1.2 What topological order and non-Abelian anyons mean here

A **phase of matter** describes properties shared by a family of systems, not the exact value of every microscopic coupling. For the phases considered here, there is an **energy gap**: a nonzero energy cost to leave the ground space, persisting as the system grows. Small local changes that keep this gap open do not change the topological order. The characteristic data include the kinds of excitations, their fusion and braiding, and the ground states allowed by the topology of the space.

An **anyon** is a particle-like excitation in two spatial dimensions with a definite topological charge, or *type*. It need not be an added physical particle: in the toric code it is a localized pattern of violated checks. One can create a pair, separate its members, move one around another, and bring them together again. Moving them around one another is **braiding**; bringing them together and measuring their combined

charge is **fusion**. The label 1, called the **vacuum**, means that no topological charge remains. It does not mean that the underlying lattice qubits are absent.

For **Abelian anyons**, two specified types have a unique fusion outcome. Braiding fixed types on a plane changes their state by a phase. That phase can be -1, or another nontrivial complex number; Abelian does not mean that braiding has no effect.

For **non-Abelian anyons**, specifying the positions, individual types, and total charge can still leave a multidimensional space of states. For example, with four anyons and total vacuum charge, the first pair may fuse through different intermediate charges, with the second pair compensating to keep the same total charge. These alternatives label a **fusion space**. Braiding can mix its states, and different sequences of braids can act differently. This information belongs to the collection of anyons; it is not generally a freely accessible qubit attached to each particle.

The word *non-Abelian* refers to noncommuting operations. For two suitable braid operations U_1, U_2, it is possible that $U_1U_2 \neq U_2U_1$. We will first see how gauging makes such a richer description necessary, then identify the resulting theory more precisely.

1.2 Section II.A: from a logical symmetry to non-Abelian anyons

Section II.A motivates the construction before giving its circuit. Its starting point is two copies of an Abelian phase, the toric code. A controlled-Z circuit changes certain excitation types into composites of the two copies. Applying that circuit in only part of the system creates a line across which the labels change. Gauging makes the associated defects part of the new quantum system's excitations.

The paper identifies the resulting phase as $\mathcal{D}(D_4)$. Here D_4 is the eight-element symmetry group of a square and $\mathcal{D}(G)$ denotes the quantum-double topological order associated with a finite group G. We will define that construction after the circuit derivation; for now, the name specifies the phase we aim to build.

1.3 Starting point: two decoupled toric codes

The toric code gives us a model in which both error syndromes and moving excitations can be calculated with Pauli operators. Take a square lattice with periodic boundary conditions: opposite sides are identified, making a torus. Put one qubit on each edge. A vertex v is where four edges meet, and a plaquette p is one elementary square. Introduce

$$A_v = \prod_{e \ni v} X_e, \qquad B_p = \prod_{e \in \partial p} Z_e.$$

The notation $e \ni v$ selects the four edges touching v, and $e \in \partial p$ selects the four sides of the square p. Thus A_v is an X-product shaped like a star and B_p is a Z-product around a square. A star and a plaquette share either zero or two edges. On each shared edge, exchanging X and Z gives a minus sign, so their total sign is always positive. All these checks therefore commute.

The Hamiltonian can be written

$$H_{\mathrm{TC}} = -\sum_v A_v - \sum_p B_p.$$

The ground space is the simultaneous $+1$ eigenspace of every A_v and B_p. It contains several states: these local constraints do not fix all global information. This is what lets the toric code encode qubits.

The checks have bounded size—four qubits each—and each qubit participates in two stars and two plaquettes. This is also an example of the low-density check structure generalized in Section III.

We take two independent copies, called B and C, with Pauli operators

$$X_e^B, Z_e^B, \qquad X_e^C, Z_e^C.$$

Their decoupled Hamiltonian is $H_{\mathrm{TC}}^B + H_{\mathrm{TC}}^C$. The superscripts are labels of the two codes, not powers. In particular, B_p^C means a plaquette check of copy C: the letter B in B_p names the type of check, while the superscript names the copy.

1.3.1 Excitations and string operators

We now identify the excitations by the checks they violate and show how Pauli strings create and move them. This will let us infer the anyon action of a gate by conjugating its string operators.

An e anyon is a vertex where $A_v = -1$. If γ is a path on the direct lattice, then

$$W_e^B(\gamma) = \prod_{e \in \gamma} Z_e^B$$

commutes with all B_p^B and anticommutes only with the two star stabilizers at the endpoints of γ. Acting on a ground state, it therefore creates a pair of e_B anyons at those endpoints. At an interior vertex the path uses two edges, so the two anticommutation signs cancel; at each endpoint there is only one. Appending a Z on the next edge moves the endpoint excitation by one step. Applying the same string twice gives identity and annihilates the pair.

The symbol e in a product over edges is an edge label; the italic anyon name e_B instead denotes an electric charge. Neither charge nor flux here refers to an actual electron or an electromagnetic field.

The **dual lattice** places one vertex at the centre of every square, with a dual edge crossing each original edge. A path $\gamma^\star$ on it connects plaquettes rather than original vertices. The operator

$$W_m^B(\gamma^\star) = \prod_{e \perp \gamma^\star} X_e^B$$

creates a pair of m_B anyons at the two endpoint plaquettes, where $B_p^B = -1$. The subscript $e \perp \gamma^\star$ selects original edges crossed by the dual path. The same cancellation explains why only the endpoints violate checks. The same statements hold for code C.

These excitations have nontrivial braiding even before gauging. Transport an electric charge around a closed direct-lattice loop bounding a region. The Z-string around that loop is the product of the plaquette checks inside: interior edges cancel in pairs. If precisely one magnetic flux lies inside, that product has eigenvalue -1. The charge therefore acquires a minus sign on circling the flux. This is **mutual Abelian statistics**: the types stay fixed, and the braid supplies a scalar phase.

The notation $e_B^2 = 1$ means that fusing two e_B charges leaves vacuum. This matches the cancellation of two identical Pauli strings. The four elementary charges and fluxes satisfy

$$e_B^2 = m_B^2 = e_C^2 = m_C^2 = 1.$$

For example, $m_B e_C$ is a composite excitation carrying an m flux of code B and an e charge of code C. Before gauging, every excitation of the two-copy theory has a unique fusion outcome: this is an Abelian anyon theory.

1.3.2 Logical operators and the bar

A closed string has no endpoint syndrome. If it encloses a disk, it can be written as a product of checks inside that disk, and acts as identity on the ground space. A loop winding around a torus need not bound such a region: it cannot be shrunk to a point and is not a product of checks. It acts nontrivially within the ground-state subspace and is a logical operator. A bar distinguishes such an encoded operation from a single-qubit Pauli. For a selected encoded pair, we write

$$\bar{X}_B,\ \bar{Z}_B,\ \bar{X}_C,\ \bar{Z}_C.$$

For instance, $\bar{X}_B$ is a noncontractible X^B-string (with the precise direct/dual choice depending on convention). It acts as Pauli X on one encoded qubit of code B. The bar does **not** mean complex conjugation. It means "logical operator."

One toric code on a torus encodes **two** qubits, corresponding to its two independent directions of winding. For either encoded qubit, choose a Z loop and a dual X loop crossing once. The operators anticommute, just like one physical qubit's X and Z, but both preserve every check. The single B, C pair of bars displayed below suppresses the other winding label; it does not mean each toric code encodes only one qubit.

1.4 The controlled-Z symmetry

The two codes are placed with a small relative displacement. The circuit U_{CZ} in the paper is a product of physical controlled-Z gates: each horizontal edge qubit of one copy is coupled to a nearby vertical edge qubit of the other copy, and conversely. The geometry is chosen so that the extra Z factors produced by these gates form plaquette stabilizers of the other code.

For bits $b, c \in \{0, 1\}$, controlled-Z acts as $\mathsf{CZ}|bc\rangle = (-1)^{bc}|bc\rangle$: only $|11\rangle$ gets a minus sign. To follow a check or string through a circuit U, we calculate its **conjugate** $USU^\dagger$. If $S|\psi\rangle = |\psi\rangle$, then $(USU^\dagger)U|\psi\rangle = U|\psi\rangle$, so this is the check satisfied by the transformed state. Here everything follows from the one-pair identities

$$\mathsf{CZ}_{12}\,(X_1)\,\mathsf{CZ}_{12} = X_1 Z_2, \qquad \mathsf{CZ}_{12}\,(X_2)\,\mathsf{CZ}_{12} = Z_1 X_2,$$

$$\mathsf{CZ}_{12}\,(Z_1)\,\mathsf{CZ}_{12} = Z_1, \qquad \mathsf{CZ}_{12}\,(Z_2)\,\mathsf{CZ}_{12} = Z_2.$$

There is no dagger on the right in these equations because $\mathsf{CZ}^\dagger = \mathsf{CZ} = \mathsf{CZ}^{-1}$.

1.4.1 Why it preserves the ground space

Consider a B-code star operator. Conjugating each of its X^B factors by the nearby controlled-Z gates produces a pattern of Z^C factors. For the pairing shown in the paper, that pattern is precisely a neighboring C-code plaquette operator. Schematically,

$$U_{\mathsf{CZ}} A_v^B U_{\mathsf{CZ}}^\dagger = A_v^B B_{p(v)}^C.$$

Similarly, a C-code star is sent to itself times a B-code plaquette. Plaquette stabilizers themselves are unchanged because controlled-Z commutes with every Z.

Thus U_{CZ} does *not* leave each term of the microscopic Hamiltonian fixed. Nevertheless, if $|\psi\rangle$ is a ground state, then $A_v^B|\psi\rangle = B_p^C|\psi\rangle = |\psi\rangle$, so

$$(A_v^B B_{p(v)}^C)|\psi\rangle = |\psi\rangle.$$

All transformed stabilizers still have eigenvalue $+1$. Therefore U_{CZ} maps the ground-state subspace to itself. This is what the paper means by an **emergent symmetry of the ground-state subspace**.

1.4.2 Its logical action

The same local conjugation rule applied along noncontractible strings gives

$$\begin{aligned} U_{\mathsf{CZ}}\bar{X}_B U_{\mathsf{CZ}}^\dagger = \bar{X}_B\bar{Z}_C, \quad U_{\mathsf{CZ}}\bar{Z}_B U_{\mathsf{CZ}}^\dagger = \bar{Z}_B, \\ U_{\mathsf{CZ}}\bar{X}_C U_{\mathsf{CZ}}^\dagger = \bar{Z}_B\bar{X}_C, \quad U_{\mathsf{CZ}}\bar{Z}_C U_{\mathsf{CZ}}^\dagger = \bar{Z}_C. \end{aligned}$$

These are exactly the Heisenberg-picture conjugation rules of a controlled-Z gate acting on the two encoded qubits. Since every physical CZ squares to identity,

$$U_{\mathsf{CZ}}^2 = 1.$$

The generated symmetry group is consequently $\mathbb{Z}_2$: its two elements are I and U_{CZ}, and applying the nontrivial one twice returns to identity. The notation names this composition rule. A gate that sends Pauli products to Pauli products under conjugation is called a **Clifford gate**; CZ is one example.

1.5 From the logical gate to an anyon permutation

The logical equation already suggests the local anyon action. More directly, take an m_B string. It is made from X^B operators. Under conjugation by U_{CZ}, every X^B in the string acquires a nearby Z^C. Symbolically,

$$U_{\mathsf{CZ}} W_m^B(\gamma^\star) U_{\mathsf{CZ}}^\dagger = W_m^B(\gamma^\star)\, W_e^C(\gamma).$$

Explicitly, if $\phi(e)$ is the C-edge paired with B-edge e, then conjugation distributes over the product:

$$U_{\mathsf{CZ}}\left(\prod_{e\in\Gamma} X_e^B\right) U_{\mathsf{CZ}}^\dagger = \prod_{e\in\Gamma}(X_e^B Z_{\phi(e)}^C) = \left(\prod_{e\in\Gamma} X_e^B\right)\left(\prod_{e\in\Gamma} Z_{\phi(e)}^C\right).$$

The extra factors therefore form a second string, rather than isolated charges in its interior. The exact path γ is set by the shifted edge pairing, but its endpoints are at the same physical locations as the endpoints of the m_B string. Thus each endpoint now carries both an m_B and an e_C excitation:

$$g(m_B) = m_B e_C.$$

Repeating the argument with B and C exchanged gives

$$g(m_C) = m_C e_B, \qquad g(e_B) = e_B, \qquad g(e_C) = e_C.$$

The map g is an **anyon automorphism**: an invertible relabelling that preserves fusion and braiding. An anyon type is also called a **superselection sector**. This means that an operation confined near one isolated excitation cannot change its total topological charge without supplying another charge. The symmetry circuit acts throughout the system, so its relabelling is compatible with this rule. The labels distinguish excitations physically: for instance, a loop of m_C detects the additional e_C in $m_B e_C$ through a minus sign.

It is useful to write the complete action once. Every two-copy toric-code anyon can be labeled as

$$x = e_B^a m_B^b e_C^c m_C^d, \qquad a, b, c, d \in \{0, 1\}.$$

All exponents are understood modulo two. The symmetry acts as

$$g(x) = e_B^{a+d} m_B^b e_C^{c+b} m_C^d.$$

This formula makes two facts transparent:

1. g respects fusion. Since fusion adds the four exponents modulo two, $g(xy) = g(x)g(y)$.
2. $g^2 = 1$. Applying the map twice adds $2b$ and $2d$, both zero modulo two. In full, $(a, b, c, d) \mapsto (a + d, b, c + b, d) \mapsto (a + 2d, b, c + 2b, d) = (a, b, c, d)$.

An **orbit** is the set of labels reached by repeated symmetry actions. For example, the two-element orbit of m_B is

$$\mathcal{O}_{m_B} = \{m_B,\ m_B e_C\}.$$

1.6 A symmetry patch is a defect line

Until now, U_{CZ} was applied everywhere. A **symmetry patch** means applying the same circuit only in a finite region R. Denote this partial operation by $U_{\mathsf{CZ}}(R)$.

Inside R, an anyon label is related to the outside label by g. Thus an m_B anyon crossing the boundary has the following description:

$$m_B \quad \xrightarrow{\text{cross } \partial R} \quad m_B e_C.$$

The boundary ∂R is therefore not an ordinary boundary between two materials. It is a line across which anyon labels are *twisted* by the permutation g. Such a line is called a **symmetry domain wall** or **branch cut**. An endpoint of an open wall, when the defect construction permits one, is called a twist defect. The boundary of a finite patch itself is closed and has no endpoints.

Extrinsic means imposed from outside. At this stage we chose R by hand and applied an external operation. The defect line is not yet a fluctuating degree of freedom of the Hamiltonian. This distinction is exactly what gauging changes.

1.7 What gauging does

Gauging a global $\mathbb{Z}_2$ symmetry means two related things:

1. Add new gauge degrees of freedom, called the A sublattice in the paper.
2. Replace the global symmetry by a local redundancy, so that a symmetry transformation can be applied independently in different regions.

A redundancy means that different assignments of local variables describe the same physical state. A useful binary example is to give every vertex a local frame bit t_v and every edge $\langle v, w\rangle$ a comparison bit a_{vw}. Changing frames by bits λ_v acts as

$$t_v \mapsto t_v + \lambda_v, \qquad a_{vw} \mapsto a_{vw} + \lambda_v + \lambda_w \pmod 2.$$

The combination $t_v + a_{vw} + t_w$ is unchanged. The gauge field lets neighbouring matter descriptions be compared even if their local frames are chosen independently. In the paper the matter transformation is CZ rather than a single-bit flip, but the role of the added edge variables is the same.

Around a closed loop, the λ's cancel in pairs. The sum of edge bits modulo two—or the product of their signs—is gauge invariant. This loop quantity detects **flux**. A local change of frame can move the chosen string representing a flux, but cannot erase the flux detected by a surrounding loop.

After gauging, the choice of region R is no longer externally specified. The information formerly carried by the imposed branch cut ∂R is carried by a configuration of the A gauge field. In the language of the paper, a closed A-gauge-field loop realizes the boundary of a controlled-Z symmetry patch.

Consequently, crossing the gauge-field loop has the same effect as crossing the old defect line:

$$\text{A-gauge loop}: \qquad m_B \longmapsto m_B e_C, \qquad m_C \longmapsto m_C e_B.$$

The loop may be opened into a string. Its endpoints are now genuine point-like excitations of the gauged theory; the paper calls such a gauge-flux excitation m_A. The string is the dynamical descendant of the old extrinsic defect line. Its drawn path depends on a gauge convention; the endpoint flux is physical and can be detected by braiding around it. "Dynamical" means that gauge configurations and their superpositions are now included in the quantum system, rather than being an externally fixed modification.

Section II.B will show how the A qubits, a cluster state, a symmetry-enrichment circuit, and projection implement this statement in a commuting-projector lattice model. Section II.A only needs the physical consequence: an A flux is an endpoint of a line that performs g.

1.8 Why the resulting braiding is non-Abelian

Take an m_B anyon around an m_A flux once. Topologically, its path crosses the m_A defect string once. Therefore

$$m_B \quad \xrightarrow{\text{braid once around } m_A} \quad m_B e_C.$$

For comparison, in an Abelian anyon theory a full braid of two fixed anyon types changes a state only by a complex number:

$$|\psi\rangle \longmapsto e^{i\theta}|\psi\rangle.$$

In the description inherited from the ungauged codes, the braid exchanges m_B and $m_B e_C$. On that two-label bookkeeping space it has the permutation part

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This is more than a scalar phase. However, after gauging, those two old labels are related by a gauge transformation. The braid does not change one gauge-invariant anyon type into another: it acts on the components used to describe a single new type. In this construction, the two-element orbit gives a dimension-two non-Abelian excitation.

The matrix P illustrates the mechanism but is not alone a proof of noncommuting braids: P commutes with itself. Nor should its two labels be read as an independently measurable local qubit on an isolated anyon. The physical protected information is described by fusion spaces of several excitations. Identifying the full gauged theory as $\mathcal{D}(D_4)$ supplies its fusion rules and the braid operations on those spaces. The quantum-double discussion below will make this identification more concrete.

1.9 The torus diagnostic mentioned in the paper

A torus has two independent noncontractible directions. In the process illustrated in the paper, one creates an anyon pair from vacuum, winds one member around one direction, and brings the pair together. A second pair is used for the other direction. The winding strings cross, so the second process can detect the first even though neither process is confined to a small disk.

For Abelian anyons the crossing contributes a phase; each pair still has its unique fusion outcome. In the non-Abelian construction the winding operations can change a pair's fusion channel. The second pair need not annihilate to vacuum, and a single localized excitation can remain where it fuses. The state also carries global information around the torus: it was not created by an operator confined to a small neighbourhood of that one excitation.

This is why the diagnostic does not mean that a single charge can be created locally from vacuum. It uses the nontrivial topology and the non-Abelian fusion structure together. Later the paper seeks an analogous effect in qLDPC codes without an ordinary spatial torus.

1.10 Where D_4 enters

The anyon relabelling explains why a richer fusion space can appear, but it is not a complete classification of the output phase. The paper uses the gauging construction and its established identification with $\mathcal{D}(D_4)$. The full symmetry and gauging data matter; a list of permutations alone is not a general recipe for identifying every possible gauged phase.

We will return to the finite group D_4 after deriving the lattice constraints. That order lets us connect its abstract flux and charge labels to operators already constructed on qubits.

2 Section II.B in detail: gauging on the lattice

Section II.A explained what gauging is meant to achieve. Section II.B is the constructive part: it gives a local circuit and measurements whose remaining degrees of freedom realize that gauging. The construction has three ingredients:

1. an auxiliary cluster state, built from new qubits a and A;
2. a controlled-controlled-Z circuit that couples this cluster state to the two toric codes; and
3. X-measurements of all the a qubits.

The notation is deliberately suggestive: lower-case a denotes qubits which will be measured away, while upper-case A denotes the gauge qubits which remain. The original two toric codes continue to be called B and C.

2.1 The auxiliary lattice and its cluster state

An **ancilla** is an additional qubit used to assist a construction or measurement. Here some ancillas are discarded and others become part of the output code. A **cluster state** is an entangled state made by preparing $|+\rangle$'s and applying CZ gates along a specified connection graph. We use it because its correlations will turn an operation over a region into an operation along the region's boundary.

Put an a qubit at every vertex of the square lattice and an A qubit at the midpoint of every edge. This vertex–edge lattice is often called a Lieb lattice. An A edge e has two endpoint vertices; write them as v and w.

Initially put every new qubit in $|+\rangle$. The stabilizer of an individual $|+\rangle$ is $X = +1$. Apply a controlled-Z gate between every incident a–A pair, and then apply a Hadamard to every A qubit. The resulting state is a slightly rotated form of the usual two-dimensional cluster state. Its stabilizers are

$$K_v^a = X_v^a \prod_{e \ni v} X_e^A, \qquad K_e^A = Z_v^a Z_e^A Z_w^a. \tag{1}$$

Here and below, $\prod_{e \ni v}$ means the product over the four square lattice edges touching vertex v.

2.1.1 Deriving Eq. (1), rather than memorising it

Two elementary conjugation rules are enough:

$$\mathrm{CZ}_{ij}\, X_i\, \mathrm{CZ}_{ij} = X_i Z_j, \qquad HXH = Z, \quad HZH = X. \tag{2}$$

Before the final Hadamards, the stabilizer which began as X_v^a has passed through one controlled-Z for each neighbouring edge. It is therefore

$$X_v^a \prod_{e \ni v} Z_e^A.$$

The Hadamards on the A qubits exchange their Z's for X's. This gives the first operator in Eq. (1). Similarly, the X_e^A stabilizer acquires a Z^a at each endpoint under the two controlled-Z gates:

$$X_e^A Z_v^a Z_w^a.$$

The final Hadamard changes its X_e^A into Z_e^A, giving the second operator in Eq. (1). This short calculation is useful: all later stabilizer bookkeeping is just this same idea applied to a larger circuit.

2.2 Why this cluster state is the correct resource for gauging

The needed resource is a precise relation between vertex flips and edge operators. We derive it from the cluster checks. The paper describes this state using "0-form" and "1-form" symmetry; here those terms distinguish a symmetry acting on all vertices from symmetries supported on loops. The first is an ordinary, or "0-form'', global symmetry,

$$G_a = \prod_v X_v^a. \tag{3}$$

The second is a loop, or "1-form'', symmetry. For every closed collection of A edges γ, it is

$$G_A(\gamma) = \prod_{e\in\gamma} Z_e^A. \tag{4}$$

For example, G_a commutes with K_e^A because that stabilizer has two Z^a factors, each of which anticommutes with its corresponding X^a. The two minus signs cancel. A closed A loop meets each vertex an even number of times, so the same even-number-of-minus-signs argument shows that $G_A(\gamma)$ commutes with every K_v^a.

The key identity is obtained by multiplying the vertex stabilizers in any connected region R:

$$\prod_{v\in R} K_v^a = \left(\prod_{v\in R} X_v^a\right)\left(\prod_{e\in\partial R} X_e^A\right). \tag{5}$$

Every A-edge strictly inside R occurs in two vertex stabilizers and therefore contributes $(X_e^A)^2 = 1$. Exactly the edges crossing the boundary occur once, which proves Eq. (5).

Each stabilizer in the product has eigenvalue $+1$, so Eq. (5) implies

$$G_a(R)X_A(\partial R)|\text{cluster}\rangle = |\text{cluster}\rangle, \qquad G_a(R)|\text{cluster}\rangle = X_A(\partial R)|\text{cluster}\rangle.$$

Here $G_a(R) = \prod_{v\in R} X_v^a$ and $X_A(\partial R) = \prod_{e\in\partial R} X_e^A$. The boundary in this formula is the **cut**: edges with exactly one endpoint in R. Drawn on the dual lattice, it follows the perimeter of the vertex region. For one vertex, it is the four incident edges; for two neighbouring vertices, their shared edge cancels and six edges remain.

These operators are equal in their action on the cluster state, not as operators on arbitrary states. That distinction will also matter when we simplify the enriched checks after measurement.

The paper calls the resource an SPT, or **symmetry-protected topological**, state. Roughly, its entanglement cannot be removed by a constant-depth circuit built from local gates that each respect the protecting symmetries, although it can be removed if that restriction is dropped. No general SPT classification is needed here: the property used in the derivation is exactly the region-to-boundary identity above. The X^A boundary strings and Z^A strings anticommute on an odd overlap. A closed direct-lattice loop crosses the boundary of a finite vertex region an even number of times, consistent with the global symmetries commuting; individual open segments can have one crossing.

2.3 The general gauging move

Let U be a global symmetry of a matter state $|\psi\rangle$, so

$$U|\psi\rangle = |\psi\rangle. \tag{6}$$

This is stronger than saying that U preserves the code space: a symmetry can move one code state to another. For this preparation we choose an invariant input, for example logical computational-basis zero states in both toric codes for the logical CZ action. This choice will also explain the parity of the measurement outcomes in Section II.C.

The role of the additional circuit Ω is to *attach* that matter symmetry to the cluster-state symmetry. The required algebraic condition is

$$\Omega G_a \Omega^\dagger = G_a U. \tag{7}$$

Conjugating the original ancilla symmetry through the coupling circuit therefore attaches U to it. To relate this global equation to a patch, we use the *controlled-symmetry form* of the circuit constructed below. Let $\mathcal{P}(v)$ be the local set of B, C qubit pairs whose CZ gates are controlled at vertex v, as specified in the next subsection. Define

$$U_v = \prod_{(b,c)\in\mathcal{P}(v)} \mathsf{CZ}_{bc}, \qquad U(R) = \prod_{v\in R} U_v.$$

Each temporary qubit a_v controls the factor U_v. Thus the local calculation will give

$$\Omega G_a(R) \Omega^\dagger = G_a(R) U(R).$$

This property follows from the explicit controls; the global identity (7) and locality alone would not specify this precise patch action.

Let $|\Psi_{\text{enr}}\rangle = \Omega(|\text{cluster}\rangle \otimes |\psi\rangle)$. The circuit acts on a, B, C, leaving the A boundary operator unchanged. Conjugating the cluster identity therefore yields

$$G_a(R) U(R) X_A(\partial R) |\Psi_{\text{enr}}\rangle = |\Psi_{\text{enr}}\rangle.$$

Now measure every temporary qubit in the X basis. First consider **postselection**: keep the branch in which every result is $+1$. This means applying the corresponding projectors and, in that branch, using

$$X_v^a = +1 \quad \text{for every } v. \tag{8}$$

On the surviving state the preceding relation becomes

$$U(R) X_A(\partial R) |\Psi_{\text{gauged}}\rangle = |\Psi_{\text{gauged}}\rangle.$$

Applying the matter symmetry in R now has the same action as the A-operator on its boundary. This realizes the symmetry-defect relation described in Section II.A using physical gauge qubits. The a qubits

have been measured away; the A qubits remain. Section II.C will replace postselection by corrections for every allowed measurement outcome.

For a simpler check on the direction of the controls, take a matter symmetry $U = \prod_v X_v^{\text{matter}}$. Then $\Omega = \prod_v \mathsf{CNOT}_{a_v \to \text{matter}_v}$ works, because conjugation sends X_v^a to $X_v^a X_v^{\text{matter}}$. The ancilla is the control. In our actual construction, the target operation is CZ, so its controlled version is CCZ.

2.4 Attaching the controlled-Z symmetry with CCZ gates

For the present problem $U = U_{\text{CZ}}$, the finite-depth circuit that implements the anyon relabelling described in Section II.A. The appropriate three-qubit gate is

$$\text{CCZ}_{abc}|x_a x_b x_c\rangle = (-1)^{x_a x_b x_c}|x_a x_b x_c\rangle. \tag{9}$$

It is diagonal, and it has the following especially important conjugation identities:

$$\begin{aligned} \text{CCZ}_{abc}\, X_a\, \text{CCZ}_{abc} &= X_a\, \text{CZ}_{bc}, \\ \text{CCZ}_{abc}\, X_b\, \text{CCZ}_{abc} &= X_b\, \text{CZ}_{ac}\,. \end{aligned} \tag{10}$$

The first follows directly by comparing the phase before and after flipping x_a: their ratio is $(-1)^{x_b x_c}$, which is exactly the phase of a CZ_{bc}. The second identity is the same statement with the labels exchanged. Every Z operator is unchanged because CCZ is diagonal.

The paper chooses a local pattern of CCZ gates. Each gate has one a vertex as its control and targets one nearby B edge and one nearby C edge. Denote the set of B–C pairs coupled to vertex v by $\mathcal{P}(v)$, and write

$$\Omega_{aBC} = \prod_v \prod_{(b,c)\in\mathcal{P}(v)} \text{CCZ}_{vbc}\,. \tag{11}$$

The diagrams in the paper specify $\mathcal{P}(v)$. What matters for the derivation is that, across all vertices, each controlled-Z gate in U_{CZ} occurs exactly once. Equation (10) then gives

$$\Omega_{aBC} X_v^a \Omega_{aBC}^\dagger = X_v^a \prod_{(b,c)\in\mathcal{P}(v)} \text{CZ}_{bc}\,. \tag{12}$$

Multiplying Eq. (12) over all vertices yields the promised symmetry enrichment condition,

$$\Omega_{aBC} G_a \Omega_{aBC}^\dagger = G_a U_{\text{CZ}}. \tag{13}$$

This is the central derivation of Section II.B. A CCZ gate is not chosen by analogy: it is chosen because conjugating its X on the a leg literally creates the desired CZ on the B, C legs.

2.5 What survives the X^a measurements

The enriched state still includes the temporary qubits. To identify the output code, we must derive constraints that survive their measurement. The useful criterion follows directly from projection: if $S|\Psi\rangle = |\Psi\rangle$ and S commutes with the measurement projector P, then $SP|\Psi\rangle = P|\Psi\rangle$. This gives two immediate cases.

- A stabilizer with no a-support commutes with the measurement and is retained immediately.
- A stabilizer containing X_v^a also commutes with the X_v^a measurement; after projection its X_v^a can be replaced by $+1$.

An operator containing a lone Z_v^a anticommutes with that measurement, so it cannot be retained by itself. We must multiply such relations together, or rewrite them using other stabilizers, until their a dependence cancels.

2.5.1 Diagonal plaquette relations

The plaquette constraints provide the simplest case and will also be needed to simplify the stars. Recall that a **plaquette** p is one elementary square. Its boundary, written ∂p, is the closed loop made from its four sides. Hence $e \in \partial p$ means that e is one of the four lattice edges lying along the perimeter of p, and $\prod_{e\in\partial p}$ means to multiply one operator for each of those four edges.

For corners v_1, v_2, v_3, v_4, name the four sides $\partial p = \{e_{12}, e_{23}, e_{34}, e_{41}\}$. Then the compact notation expands to

$$\prod_{e\in\partial p} Z_e^B = Z_{e_{12}}^B Z_{e_{23}}^B Z_{e_{34}}^B Z_{e_{41}}^B. \tag{14a}$$

The superscript specifies the layer. Each layer has its own corresponding squares; their positions are shifted relative to one another as in the paper. The same boundary notation applies to each.

We also need a precise meaning of "survives the measurements." Let $|\Psi\rangle$ be the state immediately before measuring the a qubits. Conditioning on a $+1$ result for every X_v^a applies

$$P_+ = \prod_v \frac{1 + X_v^a}{2}. \tag{14b}$$

Thus the unnormalised post-measurement state is $P_+|\Psi\rangle$. If a pre-measurement stabilizer S obeys $S|\Psi\rangle = |\Psi\rangle$ and commutes with P_+, then

$$SP_+|\Psi\rangle = P_+ S|\Psi\rangle = P_+|\Psi\rangle. \tag{14c}$$

So S remains a $+1$ constraint on the projected state. This, rather than simply being "left over," is what we mean by *survives*.

CCZ is diagonal in the computational basis and therefore commutes with every Z operator. The enrichment circuit Ω_{aBC} leaves the old B- and C-plaquette stabilizers unchanged. They act only on B or C qubits, while P_+ acts only on a qubits, so they commute with P_+. Equation (14c) therefore proves that the projected state obeys

$$B_p^B = \prod_{e\in\partial p} Z_e^B, \qquad B_p^C = \prod_{e\in\partial p} Z_e^C. \tag{14}$$

The A-layer relation needs one more step. The pre-measurement cluster stabilizer on an A edge $e = (v, w)$ is

$$K_e^A = Z_v^a Z_e^A Z_w^a. \tag{14d}$$

It too is unchanged by Ω_{aBC}, because CCZ is diagonal. But K_e^A does **not** survive by itself: it contains Z_v^a and Z_w^a, which anticommute with the X^a measurements at their respective vertices. Equivalently, it does not commute with P_+.

Now multiply these edge relations around the closed boundary of a single plaquette. With the labels introduced above,

$$\begin{aligned}\prod_{e\in\partial p} K_e^A &= (Z^a_{v_1} Z^A_{e_{12}} Z^a_{v_2})(Z^a_{v_2} Z^A_{e_{23}} Z^a_{v_3})(Z^a_{v_3} Z^A_{e_{34}} Z^a_{v_4})(Z^a_{v_4} Z^A_{e_{41}} Z^a_{v_1}) \\ &= (Z^a_{v_1})^2 (Z^a_{v_2})^2 (Z^a_{v_3})^2 (Z^a_{v_4})^2\, Z^A_{e_{12}} Z^A_{e_{23}} Z^A_{e_{34}} Z^A_{e_{41}} \\ &= \prod_{e\in\partial p} Z_e^A.\end{aligned} \tag{14e}$$

Every corner lies on exactly two sides of the square. Hence every Z_v^a appears twice and cancels because $(Z_v^a)^2 = 1$. The final operator has no a-support, hence it commutes with P_+. The criterion in Eq. (14c) therefore proves that the projected state satisfies

$$B_p^A = \prod_{e\in\partial p} Z_e^A. \tag{15}$$

This is the magnetic-flux, or plaquette, constraint for the newly created A gauge field. The contrast is essential: an open edge relation K_e^A is incompatible with the X^a measurement, but the closed product of four such relations is compatible because all dependence on the discarded a qubits cancels exactly.

2.5.2 The dressed A-star constraint

The enriched version of the first cluster stabilizer in Eq. (1) is immediate from Eq. (12):

$$\widetilde{K}_v^a = X_v^a \left(\prod_{e\ni v} X_e^A\right) \left(\prod_{(b,c)\in\mathcal{P}(v)} \mathrm{CZ}_{bc}\right). \tag{16}$$

It commutes with the measurement, so applying Eq. (8) gives the surviving constraint

$$\widetilde{A}_v^A = \left(\prod_{e\ni v} X_e^A\right) \left(\prod_{(b,c)\in\mathcal{P}(v)} \mathrm{CZ}_{bc}\right). \tag{17}$$

Without the second factor, this is the usual toric-code star: it flips the four gauge bits adjacent to v. The product of CZ gates dresses that flip by a local transformation of the B, C matter. This is the microscopic form of the slogan that an A-gauge transformation carries the old controlled-Z symmetry along with it.

2.5.3 Why the B- and C-star constraints take more work

This is the only genuinely delicate measurement update in the construction, so we now do it completely for one representative B-star. Every other B-star is a translate or rotation of this calculation. The C-star calculation is the same with B and C interchanged.

Let A_v^B be the ordinary B-code star centered at v. Near it, label the four C-edge qubits on the relevant neighbouring C plaquette by c_1, c_2, c_3, c_4, in cyclic order. The CCZ incidence pattern in the paper has three nearby disposable vertex qubits:

vertex ancilla	controlled-Z factors attached after CCZ conjugation
a_{12}	$\mathrm{CZ}_{a_{12},c_1}\,\mathrm{CZ}_{a_{12},c_2}$
a_3	CZ_{a_3,c_3}
a_4	CZ_{a_4,c_4}.

(18)

The subscripts are labels, not powers. They describe the local pattern: a_{12} is attached to the two C edges c_1, c_2, while a_3 and a_4 are attached to c_3 and c_4, respectively. Let A_3 and A_4 be the two A-edge qubits connecting a_{12} to a_3 and to a_4. This is exactly the part of the incidence diagram used in the paper's rewrite.

For any qubit q, define its computational-basis occupation number by

$$n_q = \frac{1 - Z_q}{2}. \tag{19}$$

This is a projector, with eigenvalue 0 on $|0\rangle$ and 1 on $|1\rangle$, rather than a classical number already known before measurement. For an operator W on another qubit,

$$W^{n_q} := (I - n_q) \otimes I + n_q \otimes W.$$

It applies W only in the $|1\rangle_q$ component and remains a coherent controlled operation on a superposition. Therefore

$$\mathrm{CZ}_{r,s} = Z_r^{n_s}. \tag{20}$$

The enriched B-star is consequently

$$\begin{aligned} S_v^B &= A_v^B\,\mathrm{CZ}_{a_{12},c_1}\,\mathrm{CZ}_{a_{12},c_2}\,\mathrm{CZ}_{a_3,c_3}\,\mathrm{CZ}_{a_4,c_4} \\ &= A_v^B Z_{a_{12}}^{n_{c_1}+n_{c_2}} Z_{a_3}^{n_{c_3}} Z_{a_4}^{n_{c_4}}. \end{aligned} \tag{21}$$

The first line is the post-CCZ stabilizer; the second line is only the identity (20) applied four times. At this stage the operator cannot be kept after an X^a measurement: it contains Z's on the measured qubits a_{12}, a_3, a_4.

The cluster state supplies the way to remove them. Its edge stabilizers on A_3 and A_4 impose the following relations on the enriched state (they are unchanged by the diagonal CCZ circuit):

$$Z_{a_{12}}^a Z_{A_3}^A Z_{a_3}^a = 1, \qquad Z_{a_{12}}^a Z_{A_4}^A Z_{a_4}^a = 1. \tag{22}$$

We write $\doteq$ for equality **on the subspace satisfying these cluster constraints**. It is not equality on every state. Then

$$Z_{a_3}^a \doteq Z_{a_{12}}^a Z_{A_3}^A, \qquad Z_{a_4}^a \doteq Z_{a_{12}}^a Z_{A_4}^A. \tag{23}$$

Substitute Eq. (23) into Eq. (21). The two endpoint a-variables are moved onto the two A edges, and all remaining a-dependence collects at the single vertex a_{12}:

$$S_v^B \doteq A_v^B \left(Z_{a_{12}}^a\right)^{n_{c_1}+n_{c_2}+n_{c_3}+n_{c_4}} \left(Z_{A_3}^A\right)^{n_{c_3}} \left(Z_{A_4}^A\right)^{n_{c_4}}. \tag{24}$$

The first factor in Eq. (24) looks troublesome, but it is actually the existing C-plaquette constraint in disguise. Write $n_{a_{12}} = (1 - Z_{a_{12}}^a)/2$. On a computational-basis state,

$$\begin{aligned} \left(Z_{a_{12}}^a\right)^{\sum_{i=1}^4 n_{c_i}} &= (-1)^{n_{a_{12}} \sum_{i=1}^4 n_{c_i}} \\ &= \prod_{i=1}^4 \left(Z_{c_i}^C\right)^{n_{a_{12}}} \\ &= \left(\prod_{i=1}^4 Z_{c_i}^C\right)^{n_{a_{12}}} = \left(B_p^C\right)^{n_{a_{12}}} = 1. \end{aligned} \tag{25}$$

The penultimate equality uses the fact that $c_1, \ldots, c_4$ form the boundary of that C plaquette p. The final equality is legitimate on the code state because $B_p^C = +1$; Eq. (14c) also proves that this plaquette stabilizer survives the a-measurements. All operators used in the substitution are diagonal on the relevant ancilla and C qubits, so the replacements remain valid inside the controlled powers.

Removing the identity factor in Eq. (25) leaves an operator with no a-support:

$$\begin{aligned} \widetilde{A}_v^B &= A_v^B \left(Z_{A_3}^A\right)^{n_{c_3}} \left(Z_{A_4}^A\right)^{n_{c_4}} \\ &= A_v^B \ \mathrm{CZ}_{A_3,c_3} \mathrm{CZ}_{A_4,c_4}. \end{aligned} \tag{26}$$

Equation (26) is the explicit final dressed B-star represented by the paper's diagram. It commutes with every X^a measurement because it contains no a qubit at all. The old B-star is therefore not discarded; it is converted into a gauge-invariant local constraint whose two control qubits are now physical A-gauge-field edges.

For a different B-star, the names A_3, A_4, c_3, c_4 change according to translation or rotation. A notation that covers every location is

$$\widetilde{A}_v^B = A_v^B \prod_{(A_e,c)\in \mathcal{Q}_B(v)} \mathrm{CZ}_{A_e,c}, \tag{27}$$

where $\mathcal{Q}_B(v)$ is the local pair of A- and C-edge qubits read from the translated incidence pattern. In the representative above, $\mathcal{Q}_B(v) = \{(A_3, c_3), (A_4, c_4)\}$.

Repeating the same four steps for a C-star gives

$$\widetilde{A}_w^C = A_w^C \prod_{(A_e,b)\in \mathcal{Q}_C(w)} \mathrm{CZ}_{A_e,b}. \tag{28}$$

Here the opposite-code edges b belong to the B layer, and the residual a-factor is removed by the neighbouring B-plaquette constraint $B_p^B = +1$. Thus the two derivations are exact layer-swapped copies of each other.

2.6 The final stabilizer structure and what is unusual about it

After the projection, the state is characterized locally by three diagonal plaquette families,

$$B_p^A, \qquad B_p^B, \qquad B_p^C, \tag{29}$$

and three dressed star families,

$$\widetilde{A}_v^A, \qquad \widetilde{A}_v^B, \qquad \widetilde{A}_v^C. \tag{30}$$

The tildes matter. These are no longer Pauli stabilizers in the elementary toric-code sense, because they contain controlled-Z factors. In particular, two bare-looking X-type constraints need not commute as literal operators. Instead, as the paper notes, interchanging two of them can produce a product Q of the diagonal plaquette constraints:

$$S_i S_j = S_j S_i Q. \tag{31}$$

On a physical ground state, every such Q has eigenvalue $+1$, so the two orders act identically there. This is why the construction is consistent even though its local terms are not simply a mutually commuting Pauli stabilizer group. These formulas specify compatible ground-space constraints. They should not be inserted unmodified into a Hamiltonian and called mutually commuting terms on the full Hilbert space. To obtain commuting projectors, one must incorporate the flux constraints appropriately. For example, if P_{flat} projects onto the required plaquette-$+1$ sector and commutes with the stars, then $P_{\text{flat}}(I+S_i)/2$ and $P_{\text{flat}}(I+S_j)/2$ commute whenever their only commutation obstruction is a Q fixed to $+1$ in that sector. A local Hamiltonian uses the corresponding local flux projectors.

The output is the lattice realization of the gauged phase discussed in Section II.A: the A field supplies dynamical symmetry-defect strings, and its flux endpoints enact the controlled-Z-induced relabelling of B, C anyons. The construction is therefore a preparation of the $\mathcal{D}(D_4)$ topological order, not merely three independent toric codes.

2.7 The construction in one line per stage

cluster resource	encodes "symmetry patch = A boundary",
Ω_{aBC}	changes that symmetry into $G_a U_{\text{CZ}}$,
X^a measurement	removes a and leaves A as the gauge field.

The first stage supplies the geometry, the second supplies the desired anyon automorphism, and the third turns the formerly imposed symmetry patch into a dynamical gauge configuration. That is the entire logical skeleton of Section II.B. We now replace its ideal postselection by the actual, random-outcome measurement procedure of Section II.C.

2.8 Section II.C: preparation by measurement and feedforward

2.8.1 What has to be fixed

Up to now we pretended that every measurement of X_v^a gave the outcome $+1$. That is useful for deriving the code, but it is not a preparation protocol: quantum measurements do not allow us to choose their outcomes. Section II.C makes the construction deterministic:

1. Measure all disposable a qubits in the X basis.

2. Treat the pattern of negative outcomes as a syndrome.
3. Apply a Pauli correction on the unmeasured A qubits.

The correction removes *only Abelian electric charges* of the new A gauge field. It does not have to undo, or even identify, any non-Abelian excitation. This is why ordinary decoder-style feedforward is enough.

2.8.2 Keep the measurement record instead of postselecting it

Write the outcome at a-vertex v as a bit $s_v \in \{0, 1\}$:

$$X_v^a = (-1)^{s_v}. \tag{32}$$

Thus $s_v = 0$ denotes the old $+1$ branch, whereas $s_v = 1$ denotes a -1 outcome. The projector onto the complete outcome record $s = \{s_v\}$ is

$$P_s = \prod_v \frac{1 + (-1)^{s_v} X_v^a}{2}. \tag{33}$$

If $|\Psi\rangle$ is the state immediately before these measurements, write the normalised branch state as $|\psi_s\rangle \propto P_s|\Psi\rangle$. The outcome bits are classical information; the a qubits do not remain part of the code.

2.8.3 A negative outcome is exactly an e_A excitation

Before measuring, Eq. (16) gave the enriched vertex stabilizer

$$\widetilde{K}_v^a = X_v^a \widetilde{A}_v^A, \qquad \widetilde{K}_v^a |\Psi\rangle = |\Psi\rangle. \tag{34}$$

Here $\widetilde{A}_v^A$ is the dressed A-star from Eq. (17). It acts only on retained qubits, so it commutes with P_s. Use this and Eq. (34):

$$\begin{aligned} \widetilde{A}_v^A P_s |\Psi\rangle &= X_v^a \widetilde{K}_v^a P_s |\Psi\rangle \\ &= X_v^a P_s |\Psi\rangle \\ &= (-1)^{s_v} P_s |\Psi\rangle. \end{aligned} \tag{35}$$

After normalisation,

$$\widetilde{A}_v^A |\psi_s\rangle = (-1)^{s_v} |\psi_s\rangle. \tag{36}$$

This is the complete syndrome calculation. In a toric code, an electric anyon e at a vertex is defined by a -1 eigenvalue of the star constraint at that vertex. Equation (36) is the identical definition for the new, dressed gauge field: every $s_v = 1$ marks an e_A charge at v. No A-plaquette, B-plaquette, or C-plaquette eigenvalue is flipped. In particular, the random measurements do not create an m_A flux, the excitation responsible for the type-changing braiding in Section II.A.

The new e_A is the electric charge of the gauged $\mathbb{Z}_2$ symmetry. That group has a one-dimensional nontrivial charge action: the nontrivial group element acts by -1. Two such charges combine to the trivial action, so $e_A \times e_A = 1$. This is the reason these byproducts are Abelian. A binary check eigenvalue alone would not prove that an arbitrary excitation is Abelian.

2.8.4 The correction string, derived operator by operator

Choose a collection γ of A-edges. Its boundary $\partial\gamma$ is the set of vertices incident on an odd number of chosen edges:

$$(\partial\gamma)_v = \sum_{e \ni v} \gamma_e \pmod 2, \qquad \gamma_e \in \{0, 1\}. \tag{37}$$

If γ is a path, every interior vertex touches two chosen edges and drops out modulo two; only its endpoints lie in $\partial\gamma$. Apply the Pauli string

$$R_\gamma = \prod_{e:\, \gamma_e = 1} Z_e^A. \tag{38}$$

Each Z_e^A anticommutes with the X_e^A in the dressed star at the endpoints of e, and commutes with all CZ dressing factors because they are diagonal. Hence

$$\widetilde{A}_v^A R_\gamma = (-1)^{(\partial\gamma)_v} R_\gamma \widetilde{A}_v^A. \tag{39}$$

Read Eq. (39) literally: R_γ flips the star eigenvalue precisely at the vertices in $\partial\gamma$, and nowhere else. It commutes with the plaquette constraints because they are products of Z's (or do not act on A). It also commutes with the dressed B- and C-stars: their A-dependent parts are diagonal CZ gates, while their X's act on B or C, not on A.

A **decoder** is a classical rule that takes the syndrome and chooses a correction. In this ideal preparation problem, it need only choose γ so that

$$\partial\gamma = s. \tag{40}$$

In words: pair the vertices with negative measurement outcomes and draw an A-edge path for each pair. Then

$$\begin{aligned} \widetilde{A}_v^A R_\gamma |\psi_s\rangle &= (-1)^{(\partial\gamma)_v} R_\gamma \widetilde{A}_v^A |\psi_s\rangle \\ &= (-1)^{s_v} (-1)^{s_v} R_\gamma |\psi_s\rangle \\ &= R_\gamma |\psi_s\rangle. \end{aligned} \tag{41}$$

Thus $R_\gamma|\psi_s\rangle$ has the desired $+1$ A-star eigenvalue at every vertex. Since the other final constraints were unaffected, it is the desired $\mathcal{D}(D_4)$ ground state in the chosen global sector.

2.8.5 Why the charges can be paired

The even parity of the measurement record follows from the symmetry of the input state in Eq. (6), rather than from the star shape alone. Initially $G_a = +1$. After enrichment, $G_a U$ stabilizes the state. For the controlled-CZ circuit, U also commutes with Ω, so an input with $U = +1$ leaves $G_a = +1$. The measurement outcomes therefore obey

$$\prod_v (-1)^{s_v} = +1, \qquad \sum_v s_v = 0 \pmod 2.$$

There are an even number of negative vertices, so on a connected lattice we can pair them and choose paths joining each pair. This constructs a solution of $\partial\gamma = s$.

This restriction applies to the sector prepared here. Multiplying the dressed A-stars over the closed lattice cancels their edge X's and gives $\prod_v \widetilde{A}_v^A = U$, not an unconditional identity. Other matter sectors can therefore have different total gauge charge. This distinction is consistent with the later single-anyon diagnostic.

For a finite lattice with boundaries, a path may instead end at a boundary that condenses e_A. On a torus, two pairing choices can differ by a closed noncontractible Z_A loop. This is a logical operator, so a decoder must use a convention (or track a logical frame) to prepare a specified ground-state sector.

The paper calls this a finite-depth adaptive circuit. Once a correction is chosen, every factor in Eq. (38) is a single-qubit Z gate, so all of them can be applied simultaneously in one quantum layer, irrespective of path length. Finding a good pairing from a noisy syndrome is the separate classical decoding problem.

2.8.6 The deterministic adaptive circuit

Section II.C is therefore the following procedure:

$$\begin{aligned} \text{prepare cluster state and the two toric-code states} &\longrightarrow \Omega_{aBC} \\ &\longrightarrow \text{measure all } X_v^a \text{ and record } s \\ &\longrightarrow \text{classically choose } \gamma \text{ with } \partial\gamma = s \\ &\longrightarrow R_\gamma = \textstyle\prod_{e\in\gamma} Z_e^A. \end{aligned} \tag{42}$$

Given the input toric-code states, cluster preparation and enrichment have fixed quantum depth, and all X^a measurements are parallel. Preparing the input topological states themselves is a separate resource; one cannot prepare them from a product state by a constant-depth local unitary circuit alone. The final operation depends on the recorded outcomes: that dependence is the *feedforward*. It replaces the unphysical instruction "keep only the all-+1 branch" with a procedure that succeeds for every allowed record in the ideal measurement model. The classical calculation may use results from distant parts of the lattice; constant quantum depth does not imply constant total processing or communication time.

This completes Section II.C. Its role is narrow but crucial: Section II.B constructs the desired phase in an ideal measurement branch; Section II.C shows that branch randomness only creates correctable Abelian e_A excitations, so the preparation is deterministic.

2.9 What exactly is the $\mathcal{D}(D_4)$ quantum double?

This section fills in the name that has appeared throughout the discussion. The short answer is that $\mathcal{D}(D_4)$ is a particular non-Abelian topological order: it is the quantum-double gauge theory whose finite gauge group is the eight-element symmetry group of a square. It has 22 anyon types on the plane and consequently 22 ground states on a torus.

There are two closely related uses of the phrase *quantum double*, which are worth separating.

- Kitaev's **quantum-double lattice model** is a solvable gauge-theory Hamiltonian built from a finite group G.
- $\mathcal{D}(G)$, also called the Drinfeld double of G, denotes the topological order — equivalently, the complete fusion and braiding data — of that model.

Thus the statement "the construction gives $\mathcal{D}(D_4)$" does *not* mean that every displayed local term has to look like the textbook group-element Hamiltonian. It means that, at long distances, the superselection sectors and their fusion and braiding are those of the D_4 quantum double.

2.9.1 Building a gauge model from a finite group

This is the finite-group construction introduced in Kitaev's quantum-double model. We spell out its ingredients to connect the group labels to the star and plaquette constraints already derived.

A **group** is a set of operations with a composition rule: composing any two gives another member, composition is associative, there is an identity 1_G, and every operation has an inverse. A group is **Abelian** if $gh = hg$ for all pairs. "Finite" means it has finitely many elements; $|G|$ denotes that number. For example, $\mathbb{Z}_2$ has the two operations "do nothing" and "flip," with two flips cancelling.

For a finite group G, put a G-valued degree of freedom $|h_e\rangle$, $h_e \in G$, on each oriented lattice edge. Reversing the orientation replaces h_e by h_e^{-1}. Each edge thus has $|G|$ basis states, and a quantum state can superpose different group assignments. The orientation is a convention for recording products. The standard quantum-double Hamiltonian has two kinds of local constraints:

$$H_{D(G)} = -\sum_v A_v - \sum_p B_p. \tag{43}$$

The plaquette term B_p checks **flatness**. If the oriented boundary edges of p carry $h_1, \ldots, h_n$, it keeps precisely the states with

$$h_1 h_2 \cdots h_n = 1_G. \tag{44}$$

To evaluate the loop product, follow the boundary in order, using h_e^{-1} whenever an edge is traversed against its arrow. In a non-Abelian group the order is essential. "Flatness" means that this product is identity, so there is no magnetic flux through that plaquette. The operator B_p is a projector: it returns the configuration if the condition holds and gives zero otherwise.

The vertex term A_v imposes **gauge invariance**. More precisely, A_v is the average over all local gauge transformations:

$$A_v = \frac{1}{|G|} \sum_{g \in G} A_v^g. \tag{45}$$

Under A_v^g, an outgoing edge changes as $h_e \mapsto g h_e$, and an incoming edge changes as $h_e \mapsto h_e g^{-1}$. Left versus right multiplication matters when the group is non-Abelian. Averaging over all g projects onto states unchanged by every local transformation: acting with one more transformation merely permutes the terms in the average. The two requirements, flatness and gauge invariance, generalize the toric code's plaquette and star conditions.

For $G = \mathbb{Z}_2 = \{1, -1\}$, multiplication by the nontrivial element flips an edge bit. The quantum-double operators in (43) are therefore $A_v = (I + A_v^{\mathrm{TC}})/2$ and $B_p = (I + B_p^{\mathrm{TC}})/2$, where the superscript identifies the earlier Pauli checks. The projectors have values 0 or 1, while the Pauli checks have values ± 1; they select the same ground space. Thus

$$\text{toric code} = \mathcal{D}(\mathbb{Z}_2). \tag{46}$$

This is the reason it was natural to describe the two B, C toric codes as an Abelian $\mathbb{Z}_2 \times \mathbb{Z}_2$ starting point.

2.9.2 Flux and charge labels in $\mathcal{D}(G)$

We can now classify excitations by how they depart from flatness and gauge invariance. A flux records the group product around a small surrounding loop. A charge records how a state transforms under local group actions. In the untwisted quantum double, the combined label is

$$\boxed{\text{anyon type} = ([h], \pi).} \tag{47}$$

The first label $[h]$ is the **conjugacy class** of a group element h:

$$[h] = \{ghg^{-1} : g \in G\}. \tag{48}$$

For example, choosing a different local frame at the starting point of a loop changes its product from h to ghg^{-1}: all transformations at intermediate vertices cancel. Such products represent the same physical flux type. A conjugacy class collects precisely these equivalent labels. It is a set of group elements, not a quantum superposition of them.

The second label π is an irreducible representation of the **centralizer**

$$C_h = \{g \in G : gh = hg\}. \tag{49}$$

It is a **subgroup**, a subset closed under the same group operations: the transformations that leave the chosen representative h fixed, since $ghg^{-1} = h$ is equivalent to $gh = hg$. These are the transformations under which a charge can transform while we keep that flux representative fixed.

A **representation** π assigns an invertible complex matrix $\pi(g)$ to every group element, satisfying $\pi(g_1 g_2) = \pi(g_1)\pi(g_2)$ and $\pi(1_G) = I$. Its dimension is the number of components on which those matrices act. It is **irreducible** when there is no smaller nonzero proper subspace preserved by all of them; it describes one indivisible charge type rather than a collection of charge types. Equivalent representations related by one common basis change describe the same charge.

For $\mathbb{Z}_2 = \{1, t\}$, the two irreducible representations are $\pi(t) = +1$ and $\pi(t) = -1$. Both are one-dimensional. They are the neutral and nontrivial electric charge. For a non-Abelian group, an irreducible charge can require matrices rather than scalar signs.

The representation of C_h is the allowed electric charge attached to that flux. A particle with no flux has $h = 1$, hence $C_1 = G$, so its pure electric charges are the ordinary irreducible representations of G.

The **quantum dimension** measures the asymptotic growth of fusion spaces when many copies of an anyon are present. It need not be an integer in a general anyon theory, and is not the dimension of an ordinary independent qubit located on one excitation. Abelian anyons have quantum dimension one. In an untwisted finite-group quantum double, it is the integer

$$d_{([h],\pi)} = |[h]| \dim \pi. \tag{50}$$

This formula identifies the non-Abelian types in this model. A dimension larger than one can arise because a flux has several gauge-related representatives ($|[h]| > 1$), because it carries a multidimensional charge representation ($\dim \pi > 1$), or both.

For an Abelian group all conjugacy classes have one element, every centralizer is the whole group, and every irreducible representation is one-dimensional. Equation (50) then gives $d = 1$ for every anyon. This recovers the fact that all toric-code anyons are Abelian.

2.9.3 The group D_4

We can now evaluate those labels explicitly for the output phase. Think of a square with labelled corners. A quarter-turn r cycles the corners; a reflection s reverses their order. Four quarter-turns or two reflections return every corner to its original position. Reflecting, rotating, and reflecting again reverses the direction of rotation. These facts are written compactly as

$$D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle. \tag{51}$$

The angle-bracket notation means "the group generated by these operations, subject to these relations." Multiplication composes operations, with the rightmost acting first. The relation $srs = r^{-1}$ implies $sr = r^{-1}s \neq rs$, demonstrating noncommutativity. Every element can be written as $r^k s^\epsilon$, for $k = 0, 1, 2, 3$ and $\epsilon = 0, 1$, giving

$$\{1, r, r^2, r^3, s, rs, r^2s, r^3s\}. \tag{52}$$

The notation varies in the literature: some authors call this eight-element group D_8, because it has eight elements. This note follows the notation of the two local papers: D_4 means the symmetry group of the four-sided square, hence $|D_4| = 8$.

Here is one row of the classification before listing the whole table. Conjugating r by a reflection gives $srs^{-1} = r^{-1} = r^3$. Conjugating by a rotation leaves it fixed. Hence its class is $[r] = \{r, r^3\}$, and its centralizer consists only of rotations: $C_r = \{1, r, r^2, r^3\} \cong \mathbb{Z}_4$. This cyclic group has four one-dimensional irreducible representations, with $\pi(r) = 1, i, -1, -i$. Each choice gives an anyon of dimension $2 \times 1 = 2$.

For the identity flux, the centralizer is all of D_4. Its four one-dimensional representations choose $\pi(r) = \pm 1$ and $\pi(s) = \pm 1$. There is also a two-dimensional representation, given by the square's rotation and reflection matrices:

$$\pi(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \pi(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The rotation has two complex eigenvector directions, with eigenvalues $+i$ and $-i$; reflection interchanges them. Thus no one-dimensional subspace is preserved by both matrices, and this representation is irreducible. The squared dimensions $1+1+1+1+4 = 8$ exhaust $|D_4|$, as required by the representation theory of a finite group. These five choices explain the identity-flux row; the other rows are obtained in the same way:

flux class $[h]$	$\lvert[h]\rvert$	C_h	number of irreps of C_h	anyon dimensions in this row
$\{1\}$	1	D_4	5	$1, 1, 1, 1, 2$
$\{r^2\}$	1	D_4	5	$1, 1, 1, 1, 2$
$\{r, r^3\}$	2	$\mathbb{Z}_4$	4	$2, 2, 2, 2$
$\{s, r^2s\}$	2	$\mathbb{Z}_2^2$	4	$2, 2, 2, 2$
$\{rs, r^3s\}$	2	$\mathbb{Z}_2^2$	4	$2, 2, 2, 2$

(53)

Counting the allowed charge choices for each flux class gives $5 + 5 + 4 + 4 + 4 = 22$ pairs $([h], \pi)$, so $\mathcal{D}(D_4)$ has **22 anyon types**. As a check, their dimensions satisfy

$$\sum_a d_a^2 = |D_4|^2 = 8^2 = 64. \tag{54}$$

There are eight dimension-one types and fourteen dimension-two types, so the sum is $8\times1^2+14\times2^2 = 64$. The **total quantum dimension**, defined as $\sqrt{\sum_a d_a^2}$, is therefore $\mathcal{D}_{\text{total}} = 8$. This script letter is distinct from $\mathcal{D}(G)$, the name of the theory. On a torus, the ground-state degeneracy of a quantum double equals the number of anyon types, explaining the number 22 that appears in the paper and in the trapped-ion experiment.

The two-dimensional entries in Eq. (53) are already enough to establish non-Abelian order. For example, the pure charge associated with the unique two-dimensional irrep of D_4 has $d = 2$, even though its flux is trivial. The word "flux" should therefore not be used as a synonym for "non-Abelian anyon": $\mathcal{D}(D_4)$ contains non-Abelian charges, non-Abelian fluxes, and flux-charge composites.

2.9.4 How the Section II.A–II.C derivation reaches this phase

The derivation in this note does not begin by putting eight-level D_4 variables on edges. Instead it starts from qubits and proceeds through

$$\underbrace{\mathcal{D}(\mathbb{Z}_2) \times \mathcal{D}(\mathbb{Z}_2)}_{\text{two Abelian toric codes}} \xrightarrow[\text{anyon-permuting } \mathbb{Z}_2]{\text{gauge}} \mathcal{D}(D_4). \tag{55}$$

The circuit derivation has already established the local mechanism: plaquettes impose flux constraints, and each dressed star ties a gauge transformation to the old CZ action. The comparison with $\mathcal{D}(D_4)$ is a statement about the phase's anyon data, not an assertion that each lattice edge has become an eight-level degree of freedom. In particular, the orbit $\{m_B, m_B e_C\}$ discussed above is consistent with a dimension-two type in the table.

The semidirect-product identity

$$D_4 \cong (\mathbb{Z}_2 \times \mathbb{Z}_2) \rtimes \mathbb{Z}_2 \tag{57}$$

uses the **semidirect product** symbol $\rtimes$. For comparison, $\mathbb{Z}_2 \times \mathbb{Z}_2$ describes two independent flips that commute. In a semidirect product, performing the extra flip can change which of the first operations is being performed. Here the added $\mathbb{Z}_2$ does not remain an independent commuting factor. A useful explicit realization takes two distinct commuting operations x, y and a third operation t, each squaring to identity. Conjugating by t exchanges x and y:

$$x^2 = y^2 = t^2 = 1, \qquad xy = yx, \qquad txt = y, \quad tyt = x. \tag{58}$$

Set $r = xt$ and $s = t$. Then $r^2 = xtxt = xy \neq 1$, $r^4 = (xy)^2 = 1$, and $srs = t(xt)t = tx = r^{-1}$. These are precisely the square-symmetry relations in Eq. (51). This realizes the semidirect product as D_4.

Equation (57) should not be read too literally as a statement about an individual B- or C-edge variable: the original action $m_B \mapsto m_B e_C$, $m_C \mapsto m_C e_B$ mixes magnetic and electric labels in the *anyon theory*. The precise conclusion of the lattice derivation is that, after gauging, the resulting topological order has

the $\mathcal{D}(D_4)$ anyon data. The semidirect product describes the resulting group; identifying this anyon theory requires the specified gauging data, not merely a count of the three input binary labels.

Section II.B provides the microscopic constraints that realize this arrow: the diagonal plaquettes are the flatness-type constraints, while the dressed stars are the gauge-invariance-type constraints. Their CZ decorations are the local implementation of that anyon action. Section II.C then makes the preparation deterministic: a -1 X^a result gives only an Abelian e_A charge, and a Z_A string removes it. In particular, the feedforward corrects preparation byproducts; it does not remove the intended non-Abelian A-flux sectors.

2.9.5 Relationship to arXiv:2305.03766

The local paper arXiv-2305.03766v2, *Non-Abelian Topological Order and Anyons on a Trapped-Ion Processor*, is an experimental realization of the same $\mathcal{D}(D_4)$ topological order. It prepared a 27-qubit kagome-lattice state on Quantinuum's H2 trapped-ion processor and used braiding and torus-sector tests to probe its non-Abelian properties. See also the published-paper entry. The connection is direct at the level of the phase, but the two works use different microscopic presentations:

	this note / arXiv:2602.12228	arXiv:2305.03766
starting point	two toric codes plus an A gauge field	a type-III $\mathbb{Z}_2^3$ SPT construction
geometrical lattice	three square-lattice sublattices	kagome lattice
nontrivial local factor	CCZ enrichment, then CZ-dressed stars	CZ-dressed 12-body stars
measurement byproduct	e_A charges corrected by Z_A strings	Abelian star excitations corrected by Z strings
long-distance order	$\mathcal{D}(D_4)$	$\mathcal{D}(D_4)$

(59)

The word *twisted* in this comparison means that extra configuration-dependent phases modify the gauge transformations. The designation *type III* names the particular twist written below. The apparently different gauge-group language is not a contradiction. The experimental paper writes its kagome Hamiltonian as a **type-III twisted quantum double**

$$\mathcal{D}^{\alpha}(\mathbb{Z}_2^3), \qquad \alpha(a,b,c) = (-1)^{\rho_a \gamma_b \beta_c}, \tag{60}$$

where $a = R^{\rho_a} G^{\gamma_a} B^{\beta_a}$. Without the CZ factors in its star operators, its three colours would be three decoupled toric codes, namely an untwisted $\mathbb{Z}_2^3$ gauge theory. Here each superscript bit records whether a colour generator is present; α is -1 only when $\rho_a = \gamma_b = \beta_c = 1$. This phase is additional data in the gauge theory. The phrase *type III* classifies this particular coupling among three binary factors; the sign function shown in Eq. (60) specifies the twist needed here.

A twist changes the charge rule in a flux sector from an ordinary representation to a **projective representation**:

$$\pi(g)\pi(h) = \omega(g,h)\pi(gh),$$

where $\omega(g,h)$ is a prescribed phase. Even when g and h commute as group elements, their representing matrices can anticommute if $\omega(g,h) = -\omega(h,g)$. In the experiment's red-flux sector, the green and blue operations are represented by Pauli X and Z, which anticommute. No nonzero scalar matrices can do this, so the charge space in that flux sector needs dimension at least two. Thus an Abelian-looking input group can nevertheless produce non-Abelian anyons.

Crucially, the experimental paper gives an explicit anyon table identifying its $\mathcal{D}^{\alpha}(\mathbb{Z}_2^3)$ anyons with the $\mathcal{D}(D_4)$ labels of Eq. (47). This is a **braided equivalence**: after matching the labels, the two descriptions have the same quantum dimensions, fusion rules, and braiding data. A fusion or braiding experiment cannot distinguish the phases merely by choosing one naming convention over the other. It is not a claim that $\mathbb{Z}_2^3 \cong D_4$; those groups are plainly different, since the former is Abelian and the latter is not.

For instance, the experiment uses three dimension-two non-Abelian bosons m_R, m_G, m_B. One of its characteristic fusion rules is

$$m_B \times m_B = 1 + e_R + e_G + e_R e_G. \tag{61}$$

The plus signs list **alternative fusion channels**, not four particles produced at once. A measurement of the fused charge selects one of these outcomes; their probabilities depend on the state before fusion. The vacuum 1 means complete annihilation, while $e_R e_G$ means a composite of two Abelian charges. The four outcomes have total quantum dimension four, matching $d_{m_B}^2 = 2^2$. Braiding m_G around m_B changes their fusion channel; after the pairs are fused, it can leave Abelian e_R charges. This is the same qualitative phenomenon that motivated the "single anyon on a torus" discussion in Section II.A.

Finally, the measurement-and-feedback logic of arXiv:2305.03766 is almost the same as Section II.C. There, -1 outcomes of measured SPT ancillas signal **Abelian** star excitations, which are paired by conditional Z-strings. Here, Eq. (36) showed that -1 outcomes signal Abelian e_A charges, again paired by Z_A-strings. Both protocols exploit the same crucial separation: random measurement outcomes create only correctable Abelian byproducts, whereas the desired non-Abelian topological order remains in the unmeasured system.

2.9.6 What to retain

1. $\mathcal{D}(D_4)$ is the non-Abelian quantum double of the eight-element dihedral group; it has 22 anyon types and 22 ground states on a torus.
2. Anyons of $\mathcal{D}(G)$ are labelled by a flux conjugacy class and a charge representation of its centralizer.
3. Dimension-two anyons are the sharp algebraic signature of the non-Abelian fusion structure anticipated by the defect-gauging argument.
4. The present derivation and arXiv:2305.03766 realize the same phase by different local circuits and lattices.
5. The experimental paper's twisted $\mathcal{D}^{\alpha}(\mathbb{Z}_2^3)$ description is braided-equivalent to $\mathcal{D}(D_4)$, not an assertion that the two underlying groups are isomorphic.

2.10 Section III: from 2D gauging to non-Abelian qLDPC codes

Sections II.A–II.C used two toric codes on a square lattice. Vertices, edges, and plaquettes told us where to put the qubits and how to couple them. Section III asks how to carry out gauging when the input code has no such spatial description. We still know which qubits each check acts on; that incidence data will replace the lattice.

We will proceed from the physical ingredients to the mathematical notation, then work through gauging the transversal SWAP gate. The binary algebra is introduced from scratch, and the same small code will illustrate chains, cochains, and logical operators.

Reading guide. The subsection letters below retain this note's existing organization; they do not all coincide with the paper's letters. In the paper, the toy model is III.A, the general formalism is III.B,

Poincaré-duality gauging is III.C, graph gauging is III.D, and measurement preparation is III.E. Here we explain the homological route at the level of its required structure, then develop the graph route in detail. Equation numbers in this note are also independent of the paper's equation numbers.

2.10.1 What a qLDPC code supplies, and what it does not

A **quantum error-correcting code** stores information in a subspace of the physical qubits' Hilbert space. In a stabilizer code, that subspace is specified by equations $S_j|\psi\rangle = |\psi\rangle$, where the S_j's are commuting Pauli products called **checks** or **stabilizers**. Measuring a check gives $+1$ or -1. A -1 result signals a violated constraint; the list of results is the **syndrome**. It reveals information about errors without measuring the encoded information itself.

A **CSS code** has generators made entirely of X's or entirely of Z's. For example, X_1X_2 and Z_1Z_2 commute: moving one past the other produces two minus signs, which cancel. By contrast, X_1X_2 and Z_2Z_3 anticommute because they overlap on only one qubit. This even-overlap rule is the algebra we will package into a chain complex.

For the purposes of this paper, an Abelian CSS qLDPC code supplies:

- physical qubits;
- a sparse family of X-type checks and a sparse family of Z-type checks;
- commutation between every X-check and every Z-check;
- logical operators and, sometimes, transversal logical Clifford gates.

qLDPC stands for *quantum low-density parity-check*. "Low density" means that each check acts on only $O(1)$ qubits and each qubit occurs in only $O(1)$ checks as the code grows. This is the coding-theory analogue of locality. Here $O(1)$ means bounded by a constant independent of the total number of physical qubits. It does **not** mean that the qubits must lie near one another in a plane or in three-dimensional space.

The 2D toric-code picture had a convenient dictionary:

square-lattice toric code	general CSS qLDPC code
edge	physical qubit
vertex/star	X-check
plaquette	Z-check
which qubits touch a vertex or plaquette	the check–qubit incidence pattern
edges bounding a set of plaquettes	support of a product of Z-checks

(63)

The final row is the subtle one. On a square lattice, "the boundary of a boundary is zero" is visibly why stars and plaquettes commute and why closed strings have no endpoints. In a qLDPC code, the same statements are encoded algebraically. The formalism below introduces chain complexes precisely to record this incidence data without drawing a lattice.

Thus the two notions of locality differ:

$$\text{qLDPC locality} \neq \text{ordinary spatial locality}. \tag{64}$$

That distinction is the source of both the difficulty and the novelty of the paper. The authors cannot simply copy the red A-sublattice of the square-lattice construction. They must decide what the new gauge degrees of freedom should be connected to.

2.10.2 The input is an emergent logical symmetry, not necessarily a Hamiltonian symmetry

The 2D prototype gauged a controlled-Z circuit U_{CZ}. As stressed earlier, this circuit need not leave each microscopic Hamiltonian term unchanged. Rather, it maps stabilizers to products of stabilizers and hence preserves the code subspace. Such a circuit is an **emergent symmetry of the encoded information**.

Exactly the same is true here. Start with two copies, called B and C, of an Abelian CSS qLDPC code. The desired input is a transversal Clifford gate U between them:

$$U = \bigotimes_j U_j, \tag{65}$$

where each U_j has bounded support, often on the corresponding qubits of the two copies. "Transversal" is important operationally: even though the logical action of U can be global, the physical circuit has constant depth. Such a circuit enlarges a check by at most a constant factor. The later measurement and rewriting steps require additional care: their gauge-field strings must also have bounded support.

The paper distinguishes these gates from the code's usual logical Pauli operators. Logical X- and Z-operators are often described algebraically as cycles and cocycles. They are important symmetries, but they are **not** the symmetries being gauged here. The construction instead gauges a finite-depth transversal *Clifford* operation which acts on the encoded degrees of freedom, such as:

$$\mathsf{CZ}, \qquad \mathsf{SWAP}, \qquad \text{or, in a different qudit example, charge conjugation.} \tag{66}$$

For example, SWAP exchanges the two encoded blocks. On physical qubits it is a collection of two-qubit swaps; on logical operators it exchanges the B and C labels. Gauging makes this exchange part of the local constraints of a larger system containing both matter and gauge qubits.

2.10.3 The same three-stage mechanism survives

Although the geometry changes, the logical skeleton from Section II remains unchanged:

$$\begin{aligned} \text{ancilla resource state} &\longrightarrow \text{a system with an ancilla symmetry,} \\ \text{symmetry enrichment } \Omega &\longrightarrow \Omega G_a \Omega^\dagger = G_a U, \\ X^a \text{ measurement} &\longrightarrow \text{constraints on the surviving matter and gauge qubits.} \end{aligned} \tag{68}$$

Here G_a denotes a symmetry of the ancilla resource state, and U is the transversal gate of the B, C code pair that we intend to gauge. The second line is the same symmetry-enrichment equation encountered in Eq. (13), now written without assuming a square lattice.

The new ancillary qubits play the role previously held by the a-vertex and A-edge qubits:

- the measured a-type qubits carry the ancilla symmetry;
- the unmeasured A-type qubits retain the gauge information after measurement;
- Ω ties local changes of that ancilla symmetry record to the logical gate U on the code.

There are two useful ancilla choices. **Homological gauging** copies the input code's incidence structure and uses additional algebraic structure to match a family of transversal gates to the ancilla symmetries.

Graph gauging uses a separately chosen connected graph and matches its one global vertex-flip symmetry to one transversal gate. The toy model below shows why these choices gauge different symmetries. We will compare their mathematical requirements after introducing chains and cochains.

2.10.4 What "non-Abelian qLDPC code" means here

The phrase should be read with care. For the toric code, the non-Abelian claim could be visualized directly through particles moving in a plane and braiding. For a generic qLDPC code, there may be no ambient plane in which to define a conventional braid. The paper does not claim that every such code literally realizes a 2D gas of mobile D_4 anyons.

Instead, it constructs codes with non-Pauli, controlled-Clifford dressed constraints whose ground-state and logical-operator algebra has a non-Abelian character inherited from gauging U. Non-Pauli constraints alone do not establish non-Abelian physics; the later excitation and logical-operator analysis supplies that evidence. In Section V of the paper, the authors provide an operational diagnostic: suitable logical operations can create a state with a single violated gauge-charge constraint in a subsystem-symmetry sector. This is the qLDPC analogue of the "single anyon on a torus" phenomenon from Section II.A.

So the logical order of the paper is:

Section III	construct the gauged qLDPC codes,
Section IV	work out concrete code families,
Section V	show their non-Abelian-like physical signatures,
Section VI	use gauging and ungauging as logical-gate measurement.

(72)

The present introduction promises the construction; it does not yet prove all those later physical claims.

2.11 Section III.A: toy model — the ancilla chooses which symmetry is gauged

The paper's first example is deliberately familiar: take N independent two-dimensional toric-code layers, and take two identical stacks of them. Call the stacks B and C. Thus each layer $\ell = 1, \dots, N$ contains a pair

$$(\text{toric code } B_\ell) \quad \text{and} \quad (\text{toric code } C_\ell). \tag{74}$$

Nothing initially couples distinct layers. The full Hamiltonian is simply

$$H_{\text{in}} = \sum_{\ell=1}^{N} \left(H_{\text{TC}}^{B_\ell} + H_{\text{TC}}^{C_\ell} \right). \tag{75}$$

This is not yet a non-Abelian model; it is just $2N$ independent Abelian toric codes. The advantage of this example is that we already know how to gauge a controlled-Z symmetry for one pair of layers. We can now ask the new question in its simplest form:

Should we gauge each layer's symmetry separately, or only one symmetry acting on the entire stack? (76)

The authors show that both choices are legitimate, but they require different ancillary gauge structures and lead to different physics.

2.11.1 The symmetry content of the input stack

For each layer, let

$$U_\ell = U_{\mathrm{CZ}}^{(\ell)} \tag{77}$$

denote the controlled-Z circuit between B_ℓ and C_ℓ. It is the direct layer-by-layer analogue of the circuit in Section II: within that layer it permutes the appropriate anyon labels and is an emergent symmetry of the encoded subspace.

Because the layers are decoupled, the N operations U_ℓ commute:

$$[U_\ell, U_{\ell'}] = 0 \qquad (\ell \neq \ell'). \tag{78}$$

Each has order two, $U_\ell^2 = 1$. Consequently the stack has a whole family of possible $\mathbb{Z}_2$ symmetries:

$$\left\{ \prod_{\ell=1}^{N} U_\ell^{q_\ell} \;\middle|\; q_\ell \in \{0,1\} \right\} \cong \mathbb{Z}_2^N. \tag{79}$$

Two choices within this family will matter:

$$\begin{aligned} &\text{layer-resolved generators:} && U_1, U_2, \ldots, U_N, \\ &\text{one stack-wide product:} && U_{\mathrm{all}} = \prod_{\ell=1}^{N} U_\ell. \end{aligned} \tag{80}$$

The first list allows us to apply the symmetry to one selected layer. The second applies it to every layer simultaneously. Whether a particular state changes depends on that state; a symmetry can also leave a state fixed. They are not interchangeable pieces of information, even though U_{all} is the product of the first list.

This distinction is the whole toy model in one sentence:

$$\boxed{\begin{array}{c} \text{independent ancilla symmetry generators} \\ \Longleftrightarrow \\ \text{independent input symmetries that can be gauged} \end{array}} \tag{81}$$

2.11.2 Choice 1: one ancillary cluster layer per toric-code layer

First give every physical toric-code layer its own copy of the two-dimensional ancilla resource used in Section II. In each ancillary layer there are a-qubits at vertices and A-qubits on edges; together they form a Lieb-lattice cluster state. The picture is therefore

$$\begin{array}{ccccc} (B_1, C_1) & \longleftrightarrow & \text{ancilla layer } 1 & \longleftrightarrow & U_1, \\ (B_2, C_2) & \longleftrightarrow & \text{ancilla layer } 2 & \longleftrightarrow & U_2, \\ \vdots & & \vdots & & \vdots \\ (B_N, C_N) & \longleftrightarrow & \text{ancilla layer } N & \longleftrightarrow & U_N. \end{array} \tag{82}$$

For a connected two-dimensional cluster layer, the relevant 0-form symmetry is the simultaneous X flip of all its a-vertex qubits. Write it as

$$G^a_\ell = \prod_{v \text{ in ancilla layer } \ell} X^a_{v,\ell}. \tag{83}$$

Because the ancillary layers are disconnected from one another, these are N independent symmetries:

$$\mathcal{G}^{\text{layer}}_{\text{anc}} = \langle G^a_1, \dots, G^a_N \rangle \cong \mathbb{Z}_2^N. \tag{84}$$

This independence is physical. Applying G^a_3, for example, changes only the a-qubits in layer 3. It does not have to be accompanied by an operation in layers 1, 2, 4, and so on.

The enrichment circuit is chosen independently in every layer:

$$\Omega_{\text{layer}} = \prod_{\ell=1}^{N} \Omega_\ell, \qquad \Omega_\ell G^a_\ell \Omega^\dagger_\ell = G^a_\ell U_\ell. \tag{85}$$

Equation (85) is just the Section II relation $\Omega G_a \Omega^\dagger = G_a U_{\text{CZ}}$, repeated N times. After measuring all a-qubits, the A-qubits in each layer become that layer's gauge field. Thus the procedure gauges every U_ℓ independently.

For this decoupled toy input, the output is correspondingly a stack of independently gauged two-dimensional systems: each layer has the same local $\mathcal{D}(D_4)$-type construction discussed in Section II. The important word is *independently*. A defect or gauge transformation can be specified separately in each layer, because the ancilla resource had one symmetry per layer.

The paper calls this the prototype of **homological gauging** because the ancilla inherits the same layer-by-layer structure as the input code. In the later qLDPC generalization, "same structure" becomes "the ancilla and input are described by closely related chain complexes."

2.11.3 Choice 2: one connected three-dimensional ancillary graph

Now keep exactly the same B, C stack, but choose a completely different ancilla resource: a connected three-dimensional cubic cluster state spanning all layers. The a-qubits live at cubic-lattice vertices and the A-qubits live on cubic-lattice edges. The crucial structural difference is that the ancillas are now connected *between* layers.

A connected cluster resource of this kind has one relevant global 0-form symmetry,

$$G^a_{\text{3D}} = \prod_{v \text{ in the full 3D ancilla}} X^a_v. \tag{86}$$

There is no independent factor analogous to G^a_ℓ for an individual horizontal slice. Flipping only one slice is not a symmetry of this three-dimensional cluster state, because the vertical ancilla-edge constraints link that slice to neighbouring slices.

Hence the symmetry matching condition can tie this sole ancilla symmetry only to one input operation:

$$\Omega_{\text{3D}} G^a_{\text{3D}} \Omega^\dagger_{\text{3D}} = G^a_{\text{3D}} U_{\text{all}} = G^a_{\text{3D}} \prod_{\ell=1}^{N} U_\ell. \tag{87}$$

The procedure now gauges the single stack-wide controlled-Z, not the N layer-resolved controlled-Z gates. Measuring the a-qubits leaves a three-dimensional A gauge system, because the unmeasured A-qubits occupy the edges of the connected cubic ancillary lattice.

It is tempting to say that Eq. (87) is merely a simultaneous version of Eq. (85), but that would miss the point. The two outputs differ because their gauge fields have different connectivity:

	layer-resolved choice	connected-3D choice
ancilla 0-form symmetry	$\mathbb{Z}_2^N$	$\mathbb{Z}_2$
gauged input operation	$U_1, \ldots, U_N$ separately	U_{all} only
unmeasured A field	N independent 2D gauge fields	one connected 3D gauge field
resulting interpretation	stack of 2D non-Abelian layers	hybrid 3D/2D non-Abelian system

(88)

The source paper summarizes the second output by saying that its three-dimensional gauge field has non-Abelian mutual statistics with two-dimensional gauge fields of restricted mobility. The toy subsection does not derive those statistics, and neither do we need to derive them yet. The safe conclusion at this point is simply that this is **not** a stack of independent $\mathcal{D}(D_4)$ layers: the connected ancillary field correlates what happens in different layers.

2.11.4 What the toy model tells us

For two layers, independent gauging has four group elements: I, U_1, U_2, U_1U_2. Gauging only the stack-wide product has two: I, U_1U_2. Performing a simultaneous operation does not supply the two independent gauge transformations. The ancillary connectivity determines which of these groups the construction uses.

This observation carries over to codes with no layers. A code may have one transversal gate without admitting separately addressable pieces of it. Graph gauging needs only that one gate. Homological gauging uses additional structure to match a larger family when one is available.

As a check, imagine adding a vertical ancilla edge between two previously disconnected layers. Flipping the temporary vertices of just one layer now anticommutes with that edge's $Z^A Z^a Z^a$ constraint: it flips one endpoint. Flipping both layers gives two sign changes and preserves the constraint. This is the local reason their independent vertex-flip symmetries merge into one.

2.12 Interlude: binary linear algebra over $\mathbb{F}_2$

The next subsection replaces the visible edges, vertices, and plaquettes of the toric code by matrices. The required mathematics is ordinary linear algebra, with one change: all numbers belong to the two-element field

$$\mathbb{F}_2 = \{0, 1\}. \tag{B1}$$

Its addition and multiplication are performed modulo two:

$+$	0	1
0	0	1
1	1	0

$\cdot$	0	1
0	0	0
1	0	1

. (B2)

Thus $1+1=0$. Subtraction is exactly the same operation as addition, because $-1=1$ in $\mathbb{F}_2$. The word *field* means that the usual rules of linear algebra still apply: addition and multiplication distribute, and every nonzero number has an inverse (here, the inverse of 1 is 1).

2.12.1 Vectors are subsets, and addition is XOR

The vector space $\mathbb{F}_2^n$ consists of binary strings $v=(v_1,\dots,v_n)$. Choose labels $e_1,\dots,e_n$ for some objects (for example, qubits). The vector

$$v=\sum_{i=1}^{n} v_i e_i \tag{B3}$$

can be read as the subset of labels for which $v_i=1$. Adding two vectors is bitwise XOR, so it takes the **symmetric difference** of their subsets: an element included twice disappears. For instance,

$$(1,0,1,1)+(0,1,1,0)=(1,1,0,1). \tag{B4}$$

This cancellation is the algebraic form of the Pauli facts $X_i^2=Z_i^2=I$. If a binary vector records the support of a product of Z's, then adding support vectors records multiplication of those Pauli products, with repeated factors cancelling.

2.12.2 Linear maps are binary matrices

A linear map $M:\mathbb{F}_2^n\to\mathbb{F}_2^m$ is an $m\times n$ matrix with entries 0 or 1. It acts by the usual matrix multiplication, but every sum is reduced modulo two. In particular,

$$(Mv)_a=\sum_{i=1}^{n} M_{ai}v_i \pmod 2. \tag{B5}$$

The entry $M_{ai}=1$ says that input label i is incident on, or is selected by, output label a. Hence such a matrix is often called an **incidence matrix**. The value $(Mv)_a=1$ means that output a sees an odd number of selected input labels; value 0 means that it sees an even number.

Three familiar linear-algebra notions will recur:

- The **image** $\operatorname{im} M$ is the set of outputs Mv that can be produced.
- The **kernel** $\ker M=\{v:Mv=0\}$ is the set of input patterns invisible to the map.
- The **rank** is $\dim\operatorname{im} M$. Rank–nullity still holds: $\dim\ker M+\operatorname{rank} M=n$.

As over the real numbers, these quantities can be found by Gaussian elimination. Besides swapping rows, the operation needed is "add one row to another," which is again XOR; there is no distinction between adding and subtracting a row.

2.12.3 The mod-two dot product and transpose

For $u,v\in\mathbb{F}_2^n$, define

$$u\cdot v=\sum_{i=1}^{n} u_i v_i \pmod 2. \tag{B6}$$

It equals 1 precisely when the two corresponding subsets overlap in an odd number of labels. We call u and v **orthogonal** when $u \cdot v = 0$, i.e. when their overlap is even. This is the CSS commutation test: the X-product with support u and the Z-product with support v commute exactly when $u \cdot v = 0$.

The transpose $M^{\mathsf{T}} : \mathbb{F}_2^m \to \mathbb{F}_2^n$ reverses the direction of an incidence matrix and obeys

$$u \cdot (Mv) = (M^{\mathsf{T}} u) \cdot v. \tag{B7}$$

This identity lets us translate freely between "which checks contain this qubit?" and "which qubits are selected by this collection of checks?"

2.12.4 The one equation to watch for

If two maps can be composed,

$$\mathbb{F}_2^r \xrightarrow{N} \mathbb{F}_2^n \xrightarrow{M} \mathbb{F}_2^m, \tag{B8}$$

then $MN = 0$ means: every column of N, after being tested by every row of M, has even overlap. Equivalently, $\operatorname{im} N \subseteq \ker M$. In the next section this becomes $\partial_1 \partial_2 = 0$. It is simultaneously the abstract statement "a boundary has no boundary" and the concrete CSS statement that every X-check and Z-check overlap an even number of times.

Nothing deeper than this binary bookkeeping is being assumed. The new names (*chains*, *cochains*, *boundary maps*) organize these same vectors and matrices so that the toric-code construction can be stated without relying on a lattice picture.

2.13 Section III.B: generalizing gauging to a qLDPC code

The toy model made the central design choice visible by drawing layers: independent ancilla layers gave independent symmetries, whereas one connected ancilla gave one global symmetry. A general qLDPC code need not come with layers, surfaces, or any other useful picture. This subsection replaces the picture with an algebraic description that works for any CSS code.

The paper needs this description for two reasons. It must identify what information the input code encodes, so that a transversal gate has a well-defined logical action. It must also identify relations among checks, because those relations will determine the ancilla symmetries available for gauging. We will build the necessary dictionary in that order: check supports and commutation, logical operators, then relations among checks.

2.13.1 From labelled qubits to a CSS chain complex

A check is specified by the qubits on which it acts. To record all these supports, we need three lists of labels: the physical qubits Q, the X-check generators C_X, and the Z-check generators C_Z. A binary vector on any one of these lists selects a subset of its labels. For a finite set S, write $\mathbb{F}_2[S]$ for the free $\mathbb{F}_2$-vector space on S: it has one **canonical basis vector** $[s]$ for every $s \in S$. Thus the three spaces in this paper are, precisely,

$$D_2 = \mathbb{F}_2[C_Z], \qquad D_1 = \mathbb{F}_2[Q], \qquad D_0 = \mathbb{F}_2[C_X]. \tag{92}$$

For example, an element $b \in D_1$ is a formal binary sum

$$b = \sum_{q \in Q} b_q \, [q], \qquad b_q \in \mathbb{F}_2. \tag{93}$$

It is not a sum of physical qubits: it is a vector whose coefficient b_q records whether the label q is selected. In particular, $\sum_{q \in S}[q]$ means the vector with coefficient one on the subset $S \subseteq Q$ and zero elsewhere. This is exactly the XOR/symmetric- difference addition introduced in the interlude.

Choose two binary incidence matrices H_Z and H_X, or equivalently two linear maps,

$$0 \longrightarrow D_2 \xrightarrow{\partial_2} D_1 \xrightarrow{\partial_1} D_0 \longrightarrow 0, \qquad \begin{aligned} \partial_2[z] &= \sum_{q \in Q} (H_Z)_{qz} \, [q], \\ \partial_1[q] &= \sum_{x \in C_X} (H_X)_{xq} \, [x]. \end{aligned} \tag{94}$$

Here $[z]$ and $[x]$ are canonical basis vectors labelled by $z \in C_Z$ and $x \in C_X$. Therefore $(H_Z)_{qz} = 1$ means that qubit q belongs to Z-check z, while $(H_X)_{xq} = 1$ means that qubit q belongs to X-check x. The arrows in Eq. (94) are simply the two incidence matrices written as linear maps.

To turn a binary list into an operator, put a Pauli on every selected qubit. We use a plain symbol b for a Z-support vector and a bold symbol $\mathbf{a}$ for an X-support vector:

$$\mathsf{Z}(b) := \prod_{q \in Q} Z_q^{b_q}, \qquad \mathsf{X}(\mathbf{a}) := \prod_{q \in Q} X_q^{a_q}. \tag{95}$$

Exponent zero means identity, and exponent one means the indicated Pauli. The bold notation anticipates the cochains defined below; at this point, $\mathbf{a}$ is simply another list of bits. The notation $\mathsf{Z}(b)$ and Z_b will mean the same operator, as will $\mathsf{X}(\mathbf{a})$ and $X_{\mathbf{a}}$.

Our convention places Z-check supports in the **columns** of H_Z and X-check supports in the **rows** of H_X. Some coding texts put both kinds of check in rows; their matrix called H_Z is the transpose of ours.

2.13.2 Why the chain condition is exactly CSS commutation

An arbitrary pair of incidence matrices need not define a quantum code: their proposed checks might anticommute. The equation that makes them a chain complex will enforce precisely the commutation condition needed for the code space to exist.

Let $d_0 = H_X^{\mathsf{T}}$. Applied to a vector selecting one X-check label x, it returns that check's support. Denote that label-selector by $\mathbf{x}$. The actual checks are

$$S_z^Z = \mathsf{Z}(\partial_2[z]), \qquad S_x^X = \mathsf{X}(d_0 \mathbf{x}), \tag{96}$$

and a CSS stabilizer Hamiltonian is

$$H_{\mathrm{CSS}} = -\sum_{x \in C_X} S_x^X - \sum_{z \in C_Z} S_z^Z. \tag{97}$$

The paper writes the natural pairing of a cochain with a chain as evaluation: $\mathbf{a}(b) = \sum_q a_q b_q \pmod 2$. The Pauli commutation rule is exactly

$$\mathsf{X}(\mathbf{a})\mathsf{Z}(b) = (-1)^{\mathbf{a}(b)}\mathsf{Z}(b)\mathsf{X}(\mathbf{a}). \tag{98}$$

Thus S_x^X and S_z^Z commute precisely when $(d_0\mathbf{x})(\partial_2[z]) = 0$. By the transpose identity from Eq. (B7), this pairing is the (x, z) entry of the product $H_X H_Z$, equivalently the coefficient of $[x]$ in $\partial_1\partial_2[z]$. All CSS checks commute if and only if

$$\partial_1\partial_2 = 0 \qquad \text{(equivalently, } H_X H_Z = 0 \text{ over } \mathbb{F}_2\text{)}. \tag{99}$$

Read Eq. (99) from right to left. For one Z-check z, $\partial_2[z]$ is its binary support vector. Applying ∂_1 then counts, modulo two, how many qubits of that support belong to each X-check. The zero vector says that all those numbers are even.

The toric code is the standard geometric realization:

	basis label	support vector and operator
C_X	v (vertex)	$d_0\mathbf{v} = \sum_{e\ni v}\mathbf{e}, \quad S_v^X = \prod_{e\ni v} X_e$
C_Z	p (plaquette)	$\partial_2[p] = \sum_{e\in\partial p}[e], \quad S_p^Z = \prod_{e\in\partial p} Z_e$

(100)

The equality $\partial_1\partial_2 = 0$ says that a vertex and a square plaquette share zero or two edges. For a general CSS qLDPC code there may be no geometric vertex or plaquette, but the same matrix equation remains the complete commutation condition.

2.13.3 A five-qubit example to use throughout

The following small code lets us check each new definition directly on Pauli operators. We will first verify commutation, then return to these same checks to identify stabilizers and logical actions.

Let $Q = \{e_1, \ldots, e_5\}$, $C_Z = \{\beta\}$, and $C_X = \{\alpha_1, \alpha_2, \alpha_3\}$. Bracketed symbols below are the canonical basis vectors introduced above, whereas unbracketed symbols are labels. Choose the Z-check support vector to be

$$\partial_2[\beta] = [e_1] + [e_2] + [e_4] + [e_5]. \tag{101}$$

The corresponding physical stabilizer is not the vector on the left; it is the Pauli operator obtained from it using Eq. (95):

$$S_\beta^Z = \mathsf{Z}(\partial_2[\beta]) = Z_{e_1} Z_{e_2} Z_{e_4} Z_{e_5}. \tag{102}$$

Assume the other incidence map has the values

$$\begin{aligned} &\partial_1[e_1] = [\alpha_1] + [\alpha_2], \quad \partial_1[e_2] = [\alpha_2] + [\alpha_3], \\ &\partial_1[e_4] = [\alpha_1] + [\alpha_3], \quad \partial_1[e_5] = 0, \qquad \partial_1[e_3] = 0. \end{aligned} \tag{103}$$

Linearity over $\mathbb{F}_2$ now gives

$$\begin{aligned} \partial_1\partial_2[\beta] &= ([\alpha_1] + [\alpha_2]) + ([\alpha_2] + [\alpha_3]) + ([\alpha_1] + [\alpha_3]) + 0 \\ &= 2[\alpha_1] + 2[\alpha_2] + 2[\alpha_3] = 0 \qquad \text{in } D_0 \text{ over } \mathbb{F}_2. \end{aligned} \tag{104}$$

This says that every X-check overlaps the support of S^Z_β an even number of times. For example, α_1 sees e_1, e_4, and α_2 sees e_1, e_2. Let $\alpha_i \in D^0$ denote the indicator cochain for the label α_i, as defined generally below. Consequently the physical X-check operators $S^X_{\alpha_i} = \mathsf{X}(d_0\alpha_i)$ commute with S^Z_β. Explicitly, the three checks are $X_{e_1}X_{e_4}$, $X_{e_1}X_{e_2}$, and $X_{e_2}X_{e_4}$. Their product is identity, so only two are independent. This small example illustrates the algebra; it is not a useful error-correcting code, since qubit e_3 is unchecked.

2.13.4 Cochains describe the complementary parity tests

The map ∂_1 already tells us which X-checks detect a chosen Z-operator. We also need to ask which Z-checks detect a chosen X-operator. The paper uses the dual, or cochain, description for this second calculation. It reuses the same incidence data with the arrows reversed.

A chain selects objects; a cochain asks a parity question about that selection. For example, if a chain selects qubits e_1, e_2, e_4, the coordinate cochain $\mathbf{e}_2$ returns 1, while $\mathbf{e}_3$ returns 0. The sum $\mathbf{e}_1 + \mathbf{e}_2$ returns $1 + 1 = 0$: it asks whether an odd number of those two qubits were selected.

Because the value on a sum is determined by the values on its individual labels, this is a linear function. All such functions form the cochain space. The paper denotes these spaces by upper indices:

$$D^0, \qquad D^1, \qquad D^2, \tag{105}$$

and defines them by

$$D^n = \mathrm{Hom}_{\mathbb{F}_2}(D_n, \mathbb{F}_2). \tag{106}$$

Read Hom as "the set of homomorphisms," which in this vector-space setting means **linear maps**. Thus $\mathbf{a} \in D^n$ is a function $\mathbf{a} : D_n \to \mathbb{F}_2$ satisfying

$$\mathbf{a}(u + v) = \mathbf{a}(u) + \mathbf{a}(v), \qquad \mathbf{a}(cu) = c\,\mathbf{a}(u) \quad (u, v \in D_n,\ c \in \mathbb{F}_2). \tag{107}$$

Such functions form a vector space, with addition defined pointwise. For example, $(\mathbf{a} + \mathbf{b})(u) = \mathbf{a}(u) + \mathbf{b}(u)$. The paper uses boldface for cochains: for every qubit label q, the 1-cochain $\mathbf{q} \in D^1$ is the indicator function $\mathbf{q}([q']) = \delta_{q,q'}$. Hence $\mathbf{a} = \sum_q a_q \mathbf{q}$ is a row-vector way to select qubits, whereas $b = \sum_q b_q [q]$ is the corresponding column-vector chain.

In precisely the paper's notation, the associated cochain complex is

$$0 \longleftarrow D^2 \xleftarrow{d_1} D^1 \xleftarrow{d_0} D^0 \longleftarrow 0, \qquad d_n = \partial^{\mathsf{T}}_{n+1}. \tag{108}$$

The superscript on D^n reminds us that this is a cochain space, not the chain space D_n. The left-pointing arrows make the reversal of direction visible. In the bases introduced above, the co-boundary maps act as

$$d_0\mathbf{x} = \sum_{q \in Q} (H_X)_{xq}\mathbf{q}, \qquad d_1\mathbf{q} = \sum_{z \in C_Z} (H_Z)_{qz}\mathbf{z}, \tag{109}$$

where $\mathbf{x} \in D^0$ and $\mathbf{z} \in D^2$ are the analogous indicator cochains for check labels. Thus d_0 turns the cochain for one X-check label into the cochain selecting precisely the qubits in that check. While ∂_1 asks

"which X-checks touch this qubit?", d_0 asks "which qubits are selected by this collection of X-checks?" Likewise, $d_1 : D^1 \to D^2$ takes a 1-cochain selecting qubits and returns a 2-cochain on Z-check labels. For a Z-check z,

$$(d_1\mathbf{a})([z]) = \mathbf{a}(\partial_2[z]). \tag{109a}$$

It equals 1 precisely when the selected qubits have **odd overlap** with the support of z. Thus d_1 asks "which Z-checks have odd overlap with this selected collection of qubits?" The shorter phrase "which Z-checks use these qubits?" is only an intuition: a Z-check containing two selected qubits gives 0 because all counting is modulo two. In the toric code, applying d_0 to one vertex cochain selects its incident edges—the support of the star operator.

Accordingly, the paper represents:

chain language	Pauli operator
$b = \sum_{e\in Q} b_e[e] \in D_1$	$\mathsf{Z}(b) = \prod_{e\in Q}(Z_e)^{b_e}$
$\mathbf{a} = \sum_{e\in Q} a_e\mathbf{e} \in D^1$	$\mathsf{X}(\mathbf{a}) = \prod_{e\in Q}(X_e)^{a_e}$

(110)

and Eq. (99) becomes $d_1 d_0 = 0$ after transposition. In the paper's function notation this is immediate: for $\mathbf{x} \in D^0$ and $p \in D_2$, $(d_1 d_0 \mathbf{x})(p) = \mathbf{x}(\partial_1\partial_2 p) = 0$.

2.13.5 Logical operators: why we need both H_1 and H^1

The maps now tell us which checks a Pauli operator would violate. The next step is to identify which operators can change the **encoded information** while leaving every check satisfied. These are the logical operators on which the paper's transversal gates act.

For any code state $|\psi\rangle$, every check obeys $S_j|\psi\rangle = |\psi\rangle$. If an operator L commutes with every check, then

$$S_j L|\psi\rangle = LS_j|\psi\rangle = L|\psi\rangle.$$

Thus $L|\psi\rangle$ is still a code state. If L anticommutes with a check instead, the same calculation gives a minus sign: that check is violated, and the state has left the code space.

Commuting with the checks is therefore necessary for a Pauli logical operator, but it does not yet tell us whether its logical action is nontrivial. Every product of checks also commutes with them, yet acts as identity on **every** code state. We must distinguish these stabilizer products from operators that have a nontrivial action on the encoded information. This distinction leads to the quotients H_1 and H^1.

2.13.5.1 The Z side: chains, cycles, and boundaries Start with a physical operator $Z_b = \prod_q Z_q^{b_q}$, whose support is a chain $b \in D_1$. It automatically commutes with all Z-checks, because products of Z's commute with one another. Only the X-checks need testing. For the check labelled x, the sign is

$$S_x^X Z_b = (-1)^{(d_0\mathbf{x})(b)} Z_b S_x^X = (-1)^{\mathbf{x}(\partial_1 b)} Z_b S_x^X.$$

The bit $\mathbf{x}(\partial_1 b)$ is the x-component of $\partial_1 b$. Therefore $\partial_1 b$ **is the syndrome of the Z-operator against the X-checks**. To commute with every check, we need

$$\partial_1 b = 0, \qquad b \in \ker \partial_1.$$

Such a support is called a **1-cycle**. It need not be a visible loop: the word means that the boundary map gives zero.

Which of these operators are stabilizer products? A vector $p \in D_2$ selects Z-check generators. Multiplying those generators adds their supports, giving

$$\prod_{z \in C_Z} (S_z^Z)^{p_z} = Z_{\partial_2 p}.$$

Their supports are precisely the **1-boundaries** $\operatorname{im} \partial_2$. Every boundary is a cycle because $\partial_1 \partial_2 = 0$: a stabilizer product cannot violate a check. So $\ker \partial_1$ supplies the allowed Z-actions, and $\operatorname{im} \partial_2$ identifies those acting as identity.

2.13.5.2 The X side: cochains, cocycles, and coboundaries Now start with $X_{\mathbf{a}} = \prod_q X_q^{a_q}$, whose support is a cochain $\mathbf{a} \in D^1$. It automatically commutes with all X-checks. This time only the Z-checks need testing. For a Z-check labelled z,

$$X_{\mathbf{a}} S_z^Z = (-1)^{\mathbf{a}(\partial_2 [z])} S_z^Z X_{\mathbf{a}} = (-1)^{(d_1 \mathbf{a})([z])} S_z^Z X_{\mathbf{a}}.$$

Thus $d_1\mathbf{a}$ **is the syndrome of the X-operator against the Z-checks**. The zero-syndrome supports obey

$$d_1 \mathbf{a} = 0, \qquad \mathbf{a} \in \ker d_1, \tag{110g}$$

and are called **1-cocycles**.

A cochain $\lambda \in D^0$ selects X-check generators: its bit λ_x says whether to include check x. Multiplying those generators gives

$$\prod_{x \in C_X} (S_x^X)^{\lambda_x} = X_{d_0 \lambda}.$$

Their supports form $\operatorname{im} d_0$, the **1-coboundaries**. For example, selecting only α_2 in our five-qubit code gives $d_0 \alpha_2 = \mathbf{e}_1 + \mathbf{e}_2$, the support of $X_{e_1} X_{e_2}$. Every coboundary is a cocycle because $d_1 d_0 = 0$.

2.13.5.3 The quotient groups physical operators by their logical action Suppose two allowed Z-operators have supports related by $b' = b + \partial_2 p$. On any code state,

$$Z_{b'} |\psi\rangle = Z_b Z_{\partial_2 p} |\psi\rangle = Z_b |\psi\rangle.$$

They are different physical operators, but have exactly the same logical action. The same argument holds for X-operators whose supports differ by $d_0\mathbf{x}$. A **quotient** groups all representatives of one logical action together. This gives

$$\boxed{\begin{aligned} H_1(D_\bullet) &= \frac{\ker \partial_1}{\operatorname{im} \partial_2} \quad \text{labels logical } Z \text{ actions,} \\ H^1(D_\bullet) &= \frac{\ker d_1}{\operatorname{im} d_0} \quad \text{labels logical } X \text{ actions.} \end{aligned}} \tag{115}$$

2.13.5.4 Cosets and quotient spaces: the precise definition To make the slash in Eq. (115) precise, we first define a quotient for arbitrary binary vector spaces. We will then substitute the cycle and boundary spaces. This separates the mathematical definition from its physical interpretation without introducing a second construction.

Let V be a vector space over $\mathbb{F}_2$, and let $W \subseteq V$ be a **linear subspace**. This means that W contains zero and is closed under addition and scalar multiplication. In particular, adding two members of W gives another member of W.

Equivalence. Declare two vectors equivalent when their difference belongs to W:

$$v \sim v' \quad \Longleftrightarrow \quad v' - v \in W.$$

Over $\mathbb{F}_2$, subtraction and addition coincide, so this can also be written $v' + v \in W$. Equivalently, $v' = v + w$ for some $w \in W$. For the code, W will consist of stabilizer supports: adding an element of W changes the physical representative but not its logical action.

This rule is an **equivalence relation**. Each vector is equivalent to itself because $v - v = 0 \in W$. If $v' - v \in W$, then $v - v' \in W$, so equivalence is symmetric. Finally, if $v' - v \in W$ and $v'' - v' \in W$, their sum is $v'' - v \in W$, so equivalence is transitive. These properties allow us to group vectors into nonoverlapping equivalence classes.

Cosets. The equivalence class of v is the set

$$[v] := v + W := \{v + w \mid w \in W\}.$$

It is called the **coset of W containing** v. To construct it, add v to every vector in W and collect the results. The vector v is a *representative* of this set, not the set itself. It belongs to its coset because $v = v + 0$ and $0 \in W$.

Two representatives give the same coset exactly when they are equivalent:

$$v + W = v' + W \quad \Longleftrightarrow \quad v' - v \in W.$$

For example, if $v' = v + w_0$ with $w_0 \in W$, then $v' + w = v + (w_0 + w)$. As w runs through W, so does $w_0 + w$, and the two cosets contain exactly the same vectors. Conversely, equality of the cosets puts v' in $v + W$, which means $v' = v + w_0$ for some $w_0 \in W$.

Any two cosets are therefore either equal or disjoint. If they shared a vector, that vector would be equivalent to both representatives; by transitivity the representatives would be equivalent to each other. Every vector belongs to a coset, so the distinct cosets partition V.

The quotient. Define

$$V/W := \{v + W \mid v \in V\}.$$

This is the set of all **distinct cosets**. An element of V/W is a whole class $[v]$, whereas an element of V is an individual vector v. In particular, V/W is not obtained by deleting the vectors of W from V, and it is not the union of the cosets—that union is just V again. It treats each coset as one new object.

A small example makes the distinction explicit. Let

$$V = \mathbb{F}_2^2 = \{(0,0),(1,0),(0,1),(1,1)\}, \qquad W = \{(0,0),(1,1)\}.$$

Adding W to each representative gives only two distinct sets:

$$(0,0) + W = (1,1) + W = \{(0,0),(1,1)\},$$
$$(1,0) + W = (0,1) + W = \{(1,0),(0,1)\}.$$

Thus

$$V/W = \{\{(0,0),(1,1)\},\ \{(1,0),(0,1)\}\}.$$

There are four vectors in V, but only two elements in V/W. The outer braces collect the two classes; the inner braces list the vectors representing each class. Here adding $(1,1)$ toggles both bits, so a class retains only whether their parity is even or odd.

Vector-space operations. The quotient is itself a vector space when we define

$$[v] + [u] := [v + u], \qquad c[v] := [cv], \qquad c \in \mathbb{F}_2.$$

To add two classes, choose a representative of each, add the representatives, and take the class of the result. This must give the same answer for any choice of representatives. Indeed, replacing v by $v + w$ and u by $u + w'$ changes their sum by $w + w' \in W$, so it stays in the same class. Similarly, replacing v by $v + w$ changes cv by $cw \in W$, so scalar multiplication is well-defined. The vector-space rules then follow from those in V.

The zero vector of this new space is

$$[0] = 0 + W = W.$$

Thus its zero is an entire coset, containing every vector of W. For example, $[0] + [v] = [v]$, and over $\mathbb{F}_2$, $[v] + [v] = [0]$. In the two-bit example, adding the odd-parity class to itself gives the even-parity class.

For finite-dimensional spaces,

$$\dim(V/W) = \dim V - \dim W.$$

One way to see this is to start with a basis of W and extend it to a basis of V. The basis vectors already in W all become the zero class. The classes of the additional basis vectors form a basis of the quotient. Over $\mathbb{F}_2$, the quotient therefore has $2^{\dim V - \dim W}$ elements. It is also an abelian group under its addition, which explains the paper's term *homology group*.

2.13.5.5 Applying the definition to logical operators For the two logical-operator spaces, the substitutions are

Quotient	V: allowed supports	W: stabilizer supports
H_1	$\ker \partial_1$: Z-operators commuting with all checks	$\operatorname{im} \partial_2$: products of Z-checks
H^1	$\ker d_1$: X-operators commuting with all checks	$\operatorname{im} d_0$: products of X-checks

The prerequisite $W \subseteq V$ follows from $\partial_1 \partial_2 = 0$ and $d_1 d_0 = 0$, respectively. The cosets are explicitly

$$[b] = \{b + \partial_2 p \mid p \in D_2\}, \qquad [\mathbf{a}] = \{\mathbf{a} + d_0 \lambda \mid \lambda \in D^0\}. \tag{110m}$$

Each zero class contains all the supports of the corresponding stabilizer products. All of those operators act as identity on the encoded space. Each nonzero class collects the physical representatives of one nontrivial logical action. Adding classes corresponds to multiplying logical operators: $Z_b Z_c = Z_{b+c}$, and similarly on the X side.

In $[b]$ here, brackets denote a coset; in the earlier notation $[q]$ for an individual qubit label, they denote a basis vector. A basis vector can be a representative of a coset, but those are different mathematical objects.

The association of the **lower** index with Z and the **upper** index with X comes from our choice to record Z-supports as chains and X-supports as cochains in Eq. (95). Lower-index homology follows the chain maps ∂; upper-index cohomology follows their transposes d. The association follows from that convention and the commutation calculations above, not from the position of the index by itself. A text that interchanges the initial X/Z convention will interchange the roles of the two quotients. The 1 in both cases refers to supports on the physical qubits, which occupy degree one of the complex; it is not a count of logical qubits.

Question	Z-type operator	X-type operator
What records its support?	Chain $b \in D_1$	Cochain $\mathbf{a} \in D^1$
Which checks can detect it?	X-checks	Z-checks
When does it preserve the code space?	$\partial_1 b = 0$: cycle	$d_1 \mathbf{a} = 0$: cocycle
When is it a stabilizer product?	$b \in \operatorname{im} \partial_2$: boundary	$\mathbf{a} \in \operatorname{im} d_0$: coboundary
What labels distinct logical actions?	H_1	H^1

2.13.5.6 Both quotients in the five-qubit code The checks in our running example are

$$S^Z = Z_{e_1} Z_{e_2} Z_{e_4} Z_{e_5}, \qquad S^X_{\alpha_1} = X_{e_1} X_{e_4}, \quad S^X_{\alpha_2} = X_{e_1} X_{e_2}, \quad S^X_{\alpha_3} = X_{e_2} X_{e_4}.$$

Consider Z_{e_5}. No X-check acts on e_5, so $\partial_1 [e_5] = 0$. It is not a Z-stabilizer product: the only such products are I and S^Z. Therefore $[e_5]$, viewed as a support chain, represents a nonzero element of H_1. Its two representatives are

$$\{[e_5],\ [e_1]+[e_2]+[e_4]\}.$$

Indeed, $Z_{e_5}S^Z = Z_{e_1}Z_{e_2}Z_{e_4}$; these two physical operators agree on every code state.

For the X side, consider $X_{e_1}X_{e_5}$. It overlaps the sole Z-check twice, so $d_1(\mathbf{e}_1+\mathbf{e}_5)=0$. It cannot be an X-stabilizer product because none of the X-checks acts on e_5. It therefore represents a nonzero element of H^1. Adding each possible coboundary gives its four representatives:

$$\{\mathbf{e}_1+\mathbf{e}_5,\quad \mathbf{e}_4+\mathbf{e}_5,\quad \mathbf{e}_2+\mathbf{e}_5,\quad \mathbf{e}_1+\mathbf{e}_2+\mathbf{e}_4+\mathbf{e}_5\}.$$

For instance, multiplying by $S^X_{\alpha_2}$ changes $X_{e_1}X_{e_5}$ into $X_{e_2}X_{e_5}$ without changing its logical action.

These representatives let us identify a logical Pauli pair:

$$\overline{Z} = Z_{e_5}, \qquad \overline{X} = X_{e_1}X_{e_5}, \qquad \overline{X}\,\overline{Z} = -\overline{Z}\,\overline{X}.$$

They anticommute because their supports overlap once, on e_5. Both preserve the code space, yet act on the encoded information like Pauli X and Z on one qubit. In particular, $\overline{X}$ exchanges the $+1$ and -1 eigenspaces of $\overline{Z}$. This is what connects the two abstract quotients to the familiar logical bit flip and phase flip. The unchecked qubit e_3 supplies another logical pair, X_{e_3}, Z_{e_3}; the small example encodes two qubits.

For contrast, Z_{e_1} violates the α_1 and α_2 checks, while X_{e_1} violates S^Z. Neither support belongs to the relevant kernel, so neither defines an element of the logical quotient.

In this example, $\ker\partial_1$ has dimension three and $\operatorname{im}\partial_2$ has dimension one, so $\dim H_1 = 2$. Likewise, $\ker d_1$ has dimension four and $\operatorname{im} d_0$ has dimension two, so $\dim H^1 = 2$. Both quotients therefore have four classes: identity and three nontrivial logical actions. Choosing two independent classes supplies two logical generators; their sum is the third nonzero class, not a third encoded qubit. More generally, a CSS code encoding k qubits has $\dim H_1 = \dim H^1 = k$.

2.13.6 Relations among checks: why gauging also needs H^0

The degree-one quotients describe what can act on encoded information. The paper also needs to know which **products of checks equal identity**. This is different information: it concerns dependencies in the list of check generators. For the ancillary system, these dependencies will become the symmetries to which the transversal gates are attached.

A cochain $\mu \in D^0$ selects X-check labels. Their product has support $d_0\mu$. It equals identity precisely when

$$d_0\mu = 0, \qquad H^0 = \ker d_0. \tag{110o}$$

Such a selection is a **0-cocycle**. There is no preceding space in this three-term complex, so there are no degree-zero coboundaries to quotient out. In the five-qubit example, $\mu = \alpha_1+\alpha_2+\alpha_3$ is a nonzero 0-cocycle: multiplying all three X-checks puts two X's on each of e_1, e_2, e_4, leaving identity.

The degree matters here. A 1-cocycle selects **qubits** for an X-operator that preserves the code space. A 0-cocycle selects **check labels** whose product is identity. It is the latter calculation that we next apply to the ancilla complex.

2.13.7 From the input code to the ancilla

We will use this dictionary twice:

$$\begin{array}{rcl} D_\bullet & : & \text{the input } B, C \text{ codes,} \\ A_\bullet & : & \text{the ancillary } a, A \text{ system.} \end{array} \tag{111}$$

The subscript $\bullet$ means the whole sequence of spaces and maps. There is no assumption that these two sequences coincide. In particular, the graph construction will use an ordinary graph for $A_\bullet$, even when $D_\bullet$ has no useful geometric interpretation.

2.14 Section III.C: symmetries and gauging in qLDPC codes

The input code's logical operators are now identified: H_1 labels logical Z-actions, and H^1 labels logical X-actions. The paper aims to gauge a transversal gate that acts on those operators, such as exchanging their B and C labels under SWAP.

To do so, it introduces a separate ancilla system. We will build its cluster state from incidence data, identify the ancilla-only symmetries by the same kernel calculations as above, and then attach a matter gate to each symmetry. This is where the chain-complex description becomes a gauging construction.

2.14.1 The ancilla is another chain complex

Replace the square-lattice ancilla by a three-term chain complex

$$0 \longrightarrow A_2 \xrightarrow{\partial_2^a} A_1 \xrightarrow{\partial_1^a} A_0 \longrightarrow 0. \tag{116}$$

Let V_A label a basis of A_0 and E_A a basis of A_1. The construction places one temporary ancilla qubit a_v at each $v \in V_A$, and one surviving gauge-field qubit A_e at each $e \in E_A$. The labels "vertex" and "edge" are only mnemonic: for a general qLDPC complex, they mean a 0-chain and a 1-chain basis label, not a location in ordinary space.

For a vertex label v, let $\mathbf{v} \in A^0$ be its indicator cochain. For an edge label e, let $[e] \in A_1$ be its chain basis vector. The ancilla cluster state is the simultaneous $+1$ eigenspace of the following two kinds of stabilizer:

$$K_v^a = X_{a_v} X^A_{d_0^a \mathbf{v}}, \qquad K_e^A = Z_{A_e} Z^a_{\partial_1^a [e]}. \tag{117}$$

This compact expression has a concrete meaning. If $d_0^a \mathbf{v} = \sum_e c_e \mathbf{e}$, then $X^A_{d_0^a \mathbf{v}} = \prod_e (X_{A_e})^{c_e}$: it acts on all gauge-field qubits incident on v. If $\partial_1^a [e] = \sum_v r_v [v]$, then $Z^a_{\partial_1^a [e]} = \prod_v (Z_{a_v})^{r_v}$: it acts on all temporary ancillas incident on e. These are cluster-state stabilizers in the same Hadamard-rotated convention used in Section II: applying a Hadamard to every A qubit turns them into the usual X-on-one-qubit, Z-on-its-neighbours form. Their incidence graph is bipartite, with the a and A qubits as its two vertex sets.

2.14.1.1 Reading Eq. (117) one edge at a time It is worth slowing down here, because the superscripts and the maps in Eq. (117) encode three different ideas at once. The superscript a means "temporary vertex-ancilla qubit," whereas the superscript A means "surviving gauge-field qubit." Thus X_{a_v} flips the qubit placed at the vertex label v, while X_{A_e} flips the qubit placed at the edge label e.

The subscript $d_0^a\mathbf{v}$ does *not* name one qubit. It names a binary 1-cochain, and X^A turns that binary list into a product of X's on the selected A qubits.

For an ordinary graph, suppose that the edges incident on v are $e_1, \ldots, e_r$. Then the coboundary of the indicator cochain of v is

$$d_0^a\mathbf{v} = \mathbf{e}_1 + \cdots + \mathbf{e}_r \quad \text{in } A^1,$$

where addition is in $\mathbb{F}_2$. Consequently Eq. (117)'s first operator is nothing more mysterious than

$$K_v^a = X_{a_v} X_{A_{e_1}} \cdots X_{A_{e_r}} = X_{a_v} \prod_{e \ni v} X_{A_e}.$$

It is one stabilizer for vertex v: it flips that vertex ancilla and every gauge-field qubit on an edge touching it.

Now take one edge e with endpoints u and w. Its boundary is

$$\partial_1^a[e] = [u] + [w] \quad \text{in } A_0.$$

The second operator of Eq. (117) therefore reads

$$K_e^A = Z_{A_e} Z_{a_u} Z_{a_w}.$$

It is one stabilizer for edge e: it compares the Z-parity of the gauge-field qubit on the edge with the Z-parity of the two temporary ancillas at its endpoints. In particular, for the one-edge graph

```
u  --- e ---  w
```

the complete set of cluster stabilizers is

$$K_u^a = X_{a_u} X_{A_e}, \qquad K_w^a = X_{a_w} X_{A_e}, \qquad K_e^A = Z_{A_e} Z_{a_u} Z_{a_w}.$$

This tiny example is also a useful check that Eq. (117) is consistent. For example, K_u^a and K_e^A overlap on two qubits: on a_u, where they contain X and Z, and on A_e, where they again contain X and Z. Each overlap supplies a minus sign on swapping the operators, and the two signs cancel. Hence the stabilizers commute.

The same two-overlap argument works for any incidence matrix: an incident pair has two sign changes, and a nonincident pair has none.

2.14.2 The two ancilla symmetries, and why cocycles matter

The cluster stabilizers couple temporary a qubits to surviving A qubits. Certain products cancel one kind of qubit entirely, leaving a symmetry supported only on the other kind. Finding those products tells us which ancilla operations can carry the matter symmetry and which gauge constraints can survive measurement. The cycle and cocycle conditions express exactly these cancellations.

First, if $\eta \in \ker \partial_1^a \subseteq A_1$ is an ancilla 1-cycle, then

$$G_A(\eta) = Z^A_\eta \tag{118}$$

is a symmetry. Indeed, its possible sign when commuting past K^a_v is the mod-two overlap $(d^a_0\mathbf{v})(\eta) = \mathbf{v}(\partial^a_1\eta) = 0$. These are the ancilla's 1-cycle symmetries and become the familiar magnetic-type structure of the gauge field.

Second, if $\mu \in \ker d^a_0 \subseteq A^0$ is a 0-cocycle, then

$$G_a(\mu) = X^a_\mu = \prod_{v\in V_A} (X_{a_v})^{\mu([v])} \tag{119}$$

is a symmetry acting only on the temporary a qubits. To see it, commute this operator through K^A_e. The sign is $\mu(\partial^a_1[e]) = (d^a_0\mu)([e]) = 0$. It clearly commutes with all K^a_v, which are X-type on the a qubits. These are called **0-cocycle subsystem symmetries** because a nonzero μ can select only part of the set of a qubits.

Why is the kernel condition more than a formal label? After measuring every a_v in the X basis and correcting to the $+1$ branch, the first family of cluster stabilizers becomes the X-check family $X^A_{d^a_0\mathbf{v}}$ on the remaining A qubits. Multiplying these checks according to a cocycle gives

$$\prod_{v\in V_A} \left(X^A_{d^a_0\mathbf{v}}\right)^{\mu([v])} = X^A_{d^a_0\mu} = I \qquad (\mu \in \ker d^a_0). \tag{120}$$

Thus μ labels a relation, or redundancy, among these pure gauge X-checks. This identity describes the ancilla alone; after enrichment, the same product also contains the attached matter gate. On a single connected square lattice, the familiar example is the product of all star checks, which is the identity. In the layer-resolved toy model, each independent layer supplied one such relation. This is exactly why the number and shape of 0-cocycles determine how many symmetry actions the ancilla can keep track of.

2.14.3 Symmetry enrichment: attaching an input-code gate to each cocycle

So far the ancilla and the input codes are independent. Gauging requires a circuit that links them: each selected ancilla symmetry must acquire the transversal gate we want to gauge. The following equation specifies that link and will guide both choices of ancilla in the next subsection.

Let $\Omega_{a,BC}$ denote the finite-depth circuit that entangles the ancilla with the two input codes. The central condition of the subsection is

$$\Omega_{a,BC}\, X^a_\mu\, \Omega^\dagger_{a,BC} = X^a_\mu\, U_\mu \qquad \text{for every } \mu \in \ker d^a_0. \tag{121}$$

The left side asks what happens to an exact ancilla symmetry under the entangling circuit. The right side says that it remains a symmetry, but is now **decorated** by a transversal Clifford operation U_μ on the B, C codes. This is the abstract qLDPC version of the partial controlled-Z patch in Section II: a transformation in the ancilla/gauge sector is tied to a type-changing symmetry action in the matter sector.

Equation (121) must hold for every cocycle, not merely for one chosen generator. It is allowed that several cocycles produce the same gate, or that some produce $U_\mu = I$. At least one must produce a nontrivial encoded action, otherwise the gauging is physically trivial. Conjugation also forces the

attached gates to compose consistently: because $X^a_{\mu+\nu} = X^a_\mu X^a_\nu$, the encoded actions must satisfy the corresponding product rule on the code space (up to stabilizers, which act trivially there).

Once Eq. (121) is satisfied, the temporary a qubits are projected into $|+\rangle$ states. As in Section II.C, actual X-measurements can give -1 outcomes; a classically chosen Pauli correction returns the desired stabilizer sector. The paper later proves this deterministic feedforward step in the general setting. What remains after removing the a qubits is a gauge-field code on A, coupled to the input-code symmetry that has been gauged.

2.15 Section III.D: two approaches to gauging qLDPC codes

The enrichment condition (121) tells us what the coupling circuit must do, but does not choose the ancilla. We can now revisit the toy model's choice between independent layers and a connected graph in algebraic terms. The homological route reuses the input code's structure; the graph route builds an ancilla for one chosen gate. Comparing their requirements explains why the paper develops both.

	homological gauging	graph gauging
ancilla complex	$A_\bullet \simeq D_\bullet$	$A_\bullet$ is a connected graph complex
0-cocycle symmetries	a structured family	one global generator
what is gauged	a compatible collection of gates	one transversal gate
extra input structure	local Poincare duality	none beyond the chosen gate
canonical example later	addressable transversal CZ's	transversal SWAP

(123)

The word "structured" in the second row is deliberately cautious: the relevant number is not simply the number of physical ancilla qubits, but the dimension and support structure of $\ker d^a_0$. These cocycles label the independent relations among the pure ancilla X-checks, and therefore the independent symmetry sectors that can be enriched.

2.15.1 Homological gauging: preserve the input's higher-dimensional structure

The homological route applies when the input complex $D_\bullet$ has a suitable algebraic version of **Poincare duality**. The paper requires a *local isomorphism* between

$$H^2(D_\bullet) \quad \text{and} \quad H_0(D_\bullet). \tag{124}$$

These two new spaces are quotients at the ends of the complex:

$$H^2 = D^2/\operatorname{im} d_1, \qquad H_0 = D_0/\operatorname{im} \partial_1.$$

In H^2, two patterns on Z-check labels are equivalent if their difference is the syndrome of an X-operator. In H_0, two patterns on X-check labels are equivalent if their difference is the syndrome of a Z-operator. On a connected toric-code lattice, creating a string changes the syndrome at its two endpoints; the total parity is unchanged. These quotients record such parity information that physical Pauli errors cannot change.

For the gauging construction, the identification must come from suitable local maps and a compatible **cup product**, a rule for combining cochains that the paper develops in its Section III.C. An abstract vector-space isomorphism alone does not provide those ingredients. An isomorphism says that these two spaces carry the same independent information; *local* says that the identification can be implemented

using bounded-support incidence data, rather than by a dense matrix reaching across the whole code. It is much stronger than the bare numerical statement $\dim H^2 = \dim H_0$. The later construction uses this local dictionary to turn the check-redundancy data of one type into the local support data needed to define transversal controlled gates.

For intuition, a two-dimensional toric-code lattice has a dual lattice: vertices of one lattice correspond locally to plaquettes of the other. Poincare duality abstracts precisely the part of this picture that survives without drawing a surface. A generic qLDPC code need not possess such a local primal–dual dictionary, even though its chain complex still obeys $\partial_1\partial_2 = 0$.

One can already read the gate formula without mastering the cup product. A vector $\mathcal{M} \in H_2 = \ker \partial_2$ selects Z-checks whose product is identity. On a torus, selecting every plaquette gives an example: each edge occurs twice. Given the paper's extra cup-product rule,

$$c_{ee'} = \int_{\mathcal{M}} \mathbf{e} \smile \mathbf{e}' \in \mathbb{F}_2, \qquad U_{\mathsf{CZ}}[\mathcal{M}] = \prod_{e,e'} (\mathsf{CZ}_{B_e, C_{e'}})^{c_{ee'}}.$$

The cup product $\mathbf{e} \smile \mathbf{e}'$ assigns a bit to each Z-check label. The integral adds those bits over the checks selected by $\mathcal{M}$, modulo two. The resulting bit says whether to put a CZ gate between B_e and $C_{e'}$. For instance, if precisely two selected checks contribute 1, their contributions cancel and this pair receives no CZ. The hard part is constructing a local rule whose gates preserve the code space; the existence of a chain complex by itself does not supply such a rule. This is why the paper imposes extra algebraic conditions for this route.

When the dictionary does exist, the authors choose an ancilla that mirrors the input complex,

$$A_\bullet \simeq D_\bullet. \tag{125}$$

This does not mean that the ancilla qubits are the input qubits. It means that their check–qubit incidence pattern is an isomorphic copy of the one used by B and C. Consequently its 0-cocycles have the same structured family of relations that the duality can connect to collections of transversal gates on B, C. Eq. (121) can then enrich different μ's by different, compatible gates. This is the sense in which the resulting symmetries are **addressable**: one is not limited to gauging only the product of all available gates at once.

The paper makes this concrete using transversal controlled-Z gates. The role of the additional structure is:

local duality and a compatible cup product $\implies$ structured 0-cocycles of $A_\bullet \simeq D_\bullet$ $\implies$ a compatible family of enric (126)

2.15.2 Graph gauging: retain only one global symmetry

Suppose instead that the input code offers one useful transversal gate, but does not come with a local Poincare-duality structure or a decomposition into independently addressable pieces. The authors then choose $A_\bullet$ to be the chain complex of a **connected graph**:

$$0 \longrightarrow A_1 \xrightarrow{\partial_1^a} A_0 \longrightarrow 0. \tag{127}$$

There is no A_2 in an ordinary graph. Put a temporary a qubit on each graph vertex and a surviving A gauge qubit on each graph edge, as before. Over $\mathbb{F}_2$, a connected graph has one independent 0-cocycle, represented by the uniform vertex cochain

$$\mu_{\text{all}} = \sum_{v \in V_A} \mathbf{v}, \qquad d_0^a \mu_{\text{all}} = 0. \tag{128}$$

Each edge contributes once at each endpoint, so the two contributions cancel. Below we will prove that connectedness rules out any other independent 0-cocycle.

The enrichment condition now contains only one nontrivial instruction:

$$\Omega_{a,BC} X^a_{\mu_{\text{all}}} \Omega^\dagger_{a,BC} = X^a_{\mu_{\text{all}}} U. \tag{129}$$

This construction can gauge that one gate U, but it cannot separately gauge a family $U_1, U_2, ...$: the necessary independent 0-cocycle symmetries are absent. This is exactly the distinction already visible in the stacked-toric-code toy model, now expressed without layers.

There is an important universal example. Any two identical quantum codes have a physical transversal swap:

$$U_{\mathsf{SWAP}} = \prod_{q \in Q} \mathsf{SWAP}(B_q, C_q). \tag{130}$$

It exchanges every physical qubit of copy B with its counterpart in copy C, so it maps each B stabilizer to the corresponding C stabilizer and vice versa. It is therefore an emergent symmetry of the two- copy code space. Graph gauging is available whenever this one global gate can be matched by a suitable symmetry-enrichment circuit; it requires no Poincare duality of $D_\bullet$.

"Choose a graph" does not mean choose one arbitrarily. Its incidence matrix must still let a finite-depth $\Omega_{a,BC}$ satisfy Eq. (129), and it must retain bounded-weight checks if the final code is to remain qLDPC. Hence the appropriate graph depends on the input complex and on the gate being gauged.

2.16 Section III.E: non-Abelian codes from gauging with graph states

We now develop the graph route. Keep the two input CSS codes and choose an order-two transversal gate U, with SWAP as the worked example. We need a connected graph, an enrichment circuit satisfying (121), and bounds on the supports of the resulting constraints. The graph can be chosen even when the input code also admits homological gauging; that extra structure is not needed for this route.

2.16.1 The graph ancilla, written without a picture

Choose a finite connected graph $G = (V, E)$. Its chain complex is just

$$0 \longrightarrow A_1 = \mathbb{F}_2[E] \xrightarrow{\partial_1^a} A_0 = \mathbb{F}_2[V] \longrightarrow 0. \tag{134}$$

For an edge $e = \langle v, w \rangle$, the boundary map is

$$\partial_1^a [e] = [v] + [w]. \tag{135}$$

The ancilla has two kinds of qubit:

- a temporary qubit a_v at every graph vertex $v \in V$;
- a gauge-field qubit A_e at every graph edge $e \in E$.

The notation is the same as before, but now it has an ordinary graph interpretation. Let $\mathbf{v} \in A^0$ be the indicator cochain at vertex v. Since $d_0^a = (\partial_1^a)^\mathsf{T}$, $d_0^a \mathbf{v}$ selects exactly the graph edges incident on v. The graph cluster state is stabilized by

$$K_v^a = X_{a_v} \prod_{e \ni v} X_{A_e}, \qquad K_{\langle v,w \rangle}^A = Z_{A_{\langle v,w \rangle}} Z_{a_v} Z_{a_w}. \tag{136}$$

This is Eq. (117) specialized to an ordinary graph. The first stabilizer ties a vertex qubit to the incident gauge-field qubits. The second ties an edge gauge qubit to the two endpoint vertex qubits. For an endpoint–edge pair, the two stabilizers have two anticommuting overlaps, one on the vertex qubit and one on the edge qubit, so they commute overall.

2.16.2 Why connectedness produces exactly one symmetry to gauge

The temporary vertex qubits have a global X-symmetry

$$X_{\mu_{\text{all}}}^a = \prod_{v \in V} X_{a_v}, \qquad \mu_{\text{all}} = \sum_{v \in V} \mathbf{v}. \tag{137}$$

It is a 0-cocycle because each edge sees both of its endpoints:

$$(d_0^a \mu_{\text{all}})([\langle v, w \rangle]) = 1 + 1 = 0 \qquad \text{in } \mathbb{F}_2. \tag{138}$$

Conversely, let $\mu \in \ker d_0^a$. Equation (138), with 1's replaced by the values of μ, says that the values of μ at the endpoints of every edge are equal. If the graph is connected, that equality propagates along paths, so all vertices have the same value. Therefore

$$H^0(A_\bullet) = \ker d_0^a = \text{span}_{\mathbb{F}_2}\{\mu_{\text{all}}\} \cong \mathbb{Z}_2. \tag{139}$$

A graph with r connected components has r independent uniform cocycles, one per component. The authors deliberately take $r = 1$, because the graph construction is intended to gauge one global gate rather than an independently addressable family.

2.16.3 Symmetry enrichment becomes a group-commutator test

The graph's sole nontrivial cocycle must be linked to the desired input-code gate. In the notation of Section III.C, the condition is

$$\Omega_{a,BC} X_{\mu_{\text{all}}}^a \Omega_{a,BC}^\dagger = X_{\mu_{\text{all}}}^a U. \tag{140}$$

Because $(X_{\mu_{\text{all}}}^a)^2 = I$, multiply Eq. (140) on the left by $X_{\mu_{\text{all}}}^a$ to obtain the equivalent form used in the paper:

$$X_{\mu_{\text{all}}}^a \, \Omega_{a,BC} \, X_{\mu_{\text{all}}}^a \, \Omega_{a,BC}^\dagger = U. \tag{141}$$

The left-hand side is a group commutator: it measures the failure of $\Omega_{a,BC}$ to commute with the global ancilla symmetry. Here that commutator equals the desired matter operation: flipping every temporary ancilla vertex performs the transversal gate U on the two matter codes. After the a qubits are measured and removed, this relation is what makes U a gauged, rather than externally imposed, symmetry.

2.16.4 The three graph requirements

The paper's requirements can now be stated sharply.

Requirement	Reason
Connected graph	$\ker d_0^a \cong \mathbb{Z}_2$, so the ancilla supplies exactly the one global symmetry intended for gauging.
Enrichment circuit Ω	It must satisfy Eq. (140), or equivalently Eq. (141), so the global cocycle symmetry implements the chosen transversal gate U.
LDPC preservation	Every final stabilizer must have bounded weight, and every qubit must occur in only a bounded number of final stabilizers. This constrains both graph degrees and the locality/sparsity of Ω.

The final row is essential. A connected graph always supplies the desired one-dimensional symmetry, but a dense graph can give a high-weight vertex stabilizer K_v^a, and a nonlocal enrichment circuit can spread one input check across too many ancilla qubits. Either failure would destroy the LDPC property, even though the abstract symmetry condition was correct.

The next subsection gives the promised worked example: gauge U_{SWAP}. It will introduce a sparse assignment from input code data to graph vertices, construct $\Omega_{a,BC}$ explicitly, and derive the resulting non-Abelian stabilizers.

2.17 Section III.E.1: example - gauging the transversal SWAP gate

We now work through the paper's first graph-state example. Its target symmetry is the transversal interchange of the two input copies:

$$U_{\mathsf{SWAP}} = \prod_{e \in Q} \mathsf{SWAP}_{B_e, C_e}. \tag{143}$$

Here Q is the set of input qubits. Each $e \in Q$ labels one qubit in copy B and its counterpart in copy C. The two-qubit operator SWAP_{B_e,C_e} exchanges their states:

$$\mathsf{SWAP}_{B_e,C_e}(|b\rangle_{B_e}|c\rangle_{C_e}) = |c\rangle_{B_e}|b\rangle_{C_e}. \tag{144}$$

The product in Eq. (143) is unambiguous because its factors act on disjoint pairs. The word *transversal* means precisely this: the gate is a product of identical, pairwise local gates, one per input-qubit label. If the two copies use the same CSS code, this gate exchanges the entire two code blocks without spreading a single physical error into many qubits within either block.

The aim is not merely to implement Eq. (143). It is to make it the global $\mathbb{Z}_2$ symmetry which is subsequently gauged by the graph-state ancilla. This is analogous to Section II: there the gauged symmetry was controlled-Z, while here it is exchange of the two copies.

2.17.1 Assigning the controls: what ϕ means

Keep three sets of labels apart:

Labels	Qubits	Role
$e \in Q$	B_e, C_e	Input pair to be swapped
$v \in V$	a_v	Temporary control, later measured
$f \in E$	A_f	Surviving gauge qubit on a graph edge

Choose a vertex v_e for each input pair and write

$$\phi(e) = v_e : \qquad a_{v_e} \text{ controls the swap of } B_e \text{ and } C_e. \tag{151}$$

For example, $\phi(e_1) = u$, $\phi(e_2) = w$, and $\phi(e_3) = u$ means that a_u controls two swaps and a_w controls one. The assignment must have bounded load: no vertex may control a number of input pairs that grows with the code size.

To translate to the paper, $[e] \in D_1$ is the basis chain for a qubit, and $\mathbf{e} \in D^1$ is its coordinate cochain. Both refer to the same qubit label. The paper writes the assignment as

$$\phi : D^1 \oplus D_1 \longrightarrow D_0. \tag{150}$$

This packages its use on both X- and Z-check supports. At the basis level the physical assignment is the same in both cases. The output labels in D_0 are understood as vertices of the chosen ancilla graph. To make that convention explicit, we take a labelled copy of these X-check labels among the graph vertices; their indicator cochains as ancilla vertices belong to A^0, not D^0.

The paper assumes the controller is within bounded distance of the input qubit in the check-incidence graph. Its stronger example condition,

$$\phi(e) \in \operatorname{supp}(\partial_1[e]), \tag{152}$$

means to choose an X-check that contains e. For a geometric edge, this is a choice of one endpoint. The stronger condition is an optional locality choice, not part of the symmetry-enrichment identity itself.

2.17.2 The enrichment circuit is a collection of controlled swaps

For a temporary ancilla vertex v, define its computational-basis occupation operator by

$$n_v^a := \frac{I - Z_v^a}{2}. \tag{153}$$

It has eigenvalue 0 on $|0\rangle_a$ and 1 on $|1\rangle_a$. Thus $W^{n_v^a}$, for an involution $W^2 = I$, means "apply W if the a_v qubit is $|1\rangle$, and otherwise do nothing." As an operator this is

$$W^{n_v^a} = |0\rangle\langle 0|_{a_v} \otimes I + |1\rangle\langle 1|_{a_v} \otimes W.$$

It therefore also acts coherently when the control is in a superposition; it does not measure the control. This exponent notation is used throughout the paper.

The paper's enrichment circuit is

$$\Omega_{a,BC} = \prod_{v,e} \left(\mathsf{SWAP}_{B_e,C_e}\right)^{n_v^a \int_{\phi(\mathbf{e})} v}. \tag{154}$$

The integral is not a continuum integral. It is the $\mathbb{F}_2$ pairing between a 0-chain and a 0-cochain:

$$\int_{\phi(\mathbf{e})} v = v(\phi(\mathbf{e})) = \begin{cases} 1, & v = \phi(e), \\ 0, & v \neq \phi(e). \end{cases} \tag{155}$$

Consequently, only one term in the product over v is active for a given qubit e. With the basis-level assignment $v_e = \phi(e)$, Eq. (154) is simply

$$\boxed{\Omega_{a,BC} = \prod_{e \in Q} \left(\mathsf{SWAP}_{B_e,C_e}\right)^{n_{v_e}^a}.} \tag{156}$$

It is a product of Fredkin gates: temporary qubit a_{v_e} controls a swap of the two matter qubits B_e, C_e. Controlled swaps for different e's commute, even when they share their control, because their target pairs are disjoint. This makes the circuit particularly transparent.

2.17.3 Why this circuit enriches the global cocycle symmetry

The unique nontrivial graph cocycle is the operation $X^a_{\mu_{\mathrm{all}}} = \prod_{v \in V} X^a_v$. Conjugating the occupation projector by its control flip gives

$$X^a_v n^a_v X^a_v = I - n^a_v. \tag{157}$$

Use this in one controlled-swap factor. Since $\mathsf{SWAP}^2 = I$,

$$X^a_{v_e} \left(\mathsf{SWAP}_{B_e,C_e}\right)^{n^a_{v_e}} X^a_{v_e} \left(\mathsf{SWAP}_{B_e,C_e}\right)^{n^a_{v_e}} = \mathsf{SWAP}_{B_e,C_e}. \tag{158}$$

Multiplying this result over all e gives exactly the required commutator condition from Eq. (141):

$$X^a_{\mu_{\mathrm{all}}}\, \Omega_{a,BC}\, X^a_{\mu_{\mathrm{all}}}\, \Omega^\dagger_{a,BC} = \prod_e \mathsf{SWAP}_{B_e,C_e} = U_{\mathsf{SWAP}}. \tag{159}$$

This is the central calculation of the example. Globally flipping every temporary ancilla reverses every control bit. Each controlled swap is then applied exactly once in the commutator, so the global ancilla symmetry acts on the matter as layer exchange. If a qubit were assigned to an even number of controlling vertices, its swaps would cancel pairwise; this is the paper's condition that $\int_{\phi(\mathbf{e})} \mu = 1$ for each e. Choosing one vertex $\phi(e)$ per qubit is the simplest way to satisfy it.

2.17.4 Conjugating the input stabilizers before measurement

The preceding calculation establishes that the circuit attaches the desired SWAP symmetry. To determine the resulting code, we must also follow each check through that circuit and then through measurement. Conjugation gives the checks of the enriched state before any ancilla is removed. For one input qubit, controlled swap changes a Pauli on the B copy as

$$\begin{aligned} \Omega X_e^B \Omega^\dagger &= X_e^B \left(X_e^B X_e^C\right)^{n^a_{\phi(e)}}, \\ \Omega Z_e^B \Omega^\dagger &= Z_e^B \left(Z_e^B Z_e^C\right)^{n^a_{\phi(e)}}. \end{aligned} \tag{160}$$

Check the two possible control values. If $n^a_{\phi(e)} = 0$, the right side is the original B Pauli. If it is one, the extra product turns X_e^B into X_e^C, or Z_e^B into Z_e^C, exactly as a swap should. The corresponding formulas for a C Pauli have the same correction factor.

Therefore an X-check labelled by the 0-cochain v becomes

$$\widetilde{S}^X_{B,v} = X^B_{d_0 v} \prod_{\mathbf{e}\in d_0 v} \left(X^B_{\mathbf{e}} X^C_{\mathbf{e}}\right)^{n^a_{\phi(\mathbf{e})}}, \tag{161}$$

This equation is simply the original B-check with the one-qubit rule in Eq. (160) applied to every qubit in that check. To read it, start with the 0-cochain $v \in D^0$. Its coboundary

$$d_0 v \in D^1$$

is a binary list of input qubits: its value on the basis chain $[e]$ is either zero or one. The shorthand

$$X^B_{d_0 v} = \prod_{e\in Q} \left(X^B_e\right)^{(d_0 v)([e])}$$

therefore means "put X_e^B on precisely those qubits for which $(d_0 v)([e]) = 1$." In the displayed formula, $\mathbf{e} \in d_0 v$ is only a compact shorthand for the same condition. A cochain is not literally a set: the product is over the *support* of the binary cochain $d_0 v$.

For a small graph-like example, suppose v selects a vertex whose check contains three input qubits e_1, e_2, e_3. Then

$$d_0 v = \mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3, \qquad X^B_{d_0 v} = X^B_{e_1} X^B_{e_2} X^B_{e_3}.$$

Equation (161) says that this becomes

$$\begin{aligned} \widetilde{S}^X_{B,v} = {} & X^B_{e_1} X^B_{e_2} X^B_{e_3} \\ & \times \left(X^B_{e_1} X^C_{e_1}\right)^{n^a_{\phi(e_1)}} \left(X^B_{e_2} X^C_{e_2}\right)^{n^a_{\phi(e_2)}} \left(X^B_{e_3} X^C_{e_3}\right)^{n^a_{\phi(e_3)}}. \end{aligned}$$

For any one qubit e_i, the corresponding temporary-control value decides what happens. If $n^a_{\phi(e_i)} = 0$, its extra factor is the identity and the check still contains $X^B_{e_i}$. If $n^a_{\phi(e_i)} = 1$, then

$$X^B_{e_i} \left(X^B_{e_i} X^C_{e_i}\right) = X^C_{e_i}.$$

That local part of the stabilizer has been swapped from copy B to copy C. The compact formula in Eq. (161) says this independently for every qubit in the original check. A general v can label a product of check generators rather than one independent generator; if $d_0 v = 0$, it labels the identity. This is why the paper uses all 0-cochains rather than choosing an independent check basis at this stage.

The corresponding C-check has exactly the same correction factors, with B and C exchanged. Similarly, a Z-check labelled by a 2-chain $[p]$ becomes

$$\widetilde{S}^Z_{B,[p]} = Z^B_{\partial_2[p]} \prod_{e \in \partial_2[p]} \left(Z^B_e Z^C_e\right)^{n^a_{\phi(e)}}. \tag{162}$$

The temporary-vertex graph stabilizer has an equally important change. A control flip X^a_v toggles every controlled swap assigned to v, so

$$\widetilde{K}^a_v = X^a_v X^A_{d^a_0 v} \mathsf{SWAP}_{\phi^T(v)}, \qquad \mathsf{SWAP}_{\phi^T(v)} := \prod_{e:\, \phi(e)=v} \mathsf{SWAP}_{B_e, C_e}. \tag{163}$$

Here d^a_0 belongs to the ancilla graph. It selects all gauge edges incident on v. The paper's transpose notation $\phi^{\mathsf{T}}(v)$ means the list of input pairs assigned to that controller.

For example, if

$$\phi(e_1) = v, \qquad \phi(e_3) = v,$$

then

$$\mathsf{SWAP}_{\phi^{\mathsf{T}}(v)} = \mathsf{SWAP}_{B_{e_1}, C_{e_1}} \mathsf{SWAP}_{B_{e_3}, C_{e_3}}.$$

If, in addition, v has the two incident graph edges f_1, f_2, the full stabilizer is

$$\widetilde{K}^a_v = X^a_v X^A_{f_1} X^A_{f_2} \mathsf{SWAP}_{B_{e_1}, C_{e_1}} \mathsf{SWAP}_{B_{e_3}, C_{e_3}}.$$

The new factor follows from the same control-flip identity as (158), now applied only to the swaps controlled at v. Thus a local gauge transformation is tied to exchanging the matter pairs assigned to that vertex.

Finally, the graph-edge stabilizers $Z^A_f Z^a_{\partial f}$ are unchanged because $\Omega_{a,BC}$ is diagonal in every control Z^a_v.

2.17.5 Measuring the temporary qubits: how strings of Z^A appear

The enriched checks still act on temporary qubits. We need constraints involving only the surviving A, B, C system. An X^a factor can be replaced by its measured eigenvalue; a Z^a-dependent factor cannot. We handle these two cases separately. The second case will explain the path-dependent gauge-field factors in the paper's final formulas.

Vertex stabilizers. First consider only the graph-cluster vertex stabilizer $\widetilde{K}^a_v$ in Eq. (163). Its X-measurement has a particularly direct effect: it removes X^a_v and leaves a constraint on the surviving A, B, C qubits.

The protocol now measures every a_v in the X basis and, after the standard Pauli-frame correction, takes the outcomes to be $+1$. Thus the factor X_v^a can be removed from Eq. (163), leaving the gauged vertex constraint

$$G_v^A = X^A_{d_0^a v}\,\mathsf{SWAP}_{\phi^T(v)}. \tag{164}$$

Thus G_v^A is precisely what the graph cluster state's vertex stabilizer becomes after symmetry enrichment and the X-measurement (with postselection or Pauli-frame correction) removes the temporary qubit a_v.

It is called a constraint because physical code states must satisfy

$$G_v^A|\Psi_{\text{gauged}}\rangle = |\Psi_{\text{gauged}}\rangle.$$

And it is called gauged because it is now a local relation involving only surviving degrees of freedom:

- an X transformation on gauge-field edges adjacent to v;
- the associated swaps of B, C matter qubits.

Without enrichment, the surviving relation would have been only

$$X^A_{d_0^a v} = 1,$$

which is an ordinary pure-$\mathbb{Z}_2$ gauge-type vertex constraint. Enrichment dresses it by layer exchange.

A useful global check is to multiply all these local constraints. Every A-edge appears at both of its endpoints, so its two X^A's cancel. If every input qubit is assigned to exactly one vertex, the SWAP factors multiply to the global transversal swap:

$$\prod_{v\in V} G_v^A = U_{\mathsf{SWAP}}.$$

The all-$+1$ branch therefore lies in the $+1$ eigenspace of the whole-block SWAP. For deterministic preparation with corrections on A alone, we take a symmetric input, for example the same pure encoded state in both copies. For general input data, the global symmetry sector and the measurement record must be retained: edge Z^A corrections flip gauge charges in pairs and cannot change the eigenvalue of their total product.

This is the essential gauging idea: the former global $B \leftrightarrow C$ symmetry has been distributed across local vertex constraints of the enlarged A, B, C system.

B- and C-code stabilizers. We now return to a different family of stabilizers: the dressed input-code checks in Eqs. (161) and (162). Their measurement problem is not solved merely by setting an X_v^a factor to $+1$, because they contain Z^a-dependent occupation projectors. They must first be rewritten using the graph-edge stabilizers.

The exponents $n_v^a = (I - Z_v^a)/2$ are operators, not known classical bits. An X-measurement does not determine their values: X_v^a and n_v^a do not commute. In particular, we cannot substitute either 0 or 1 for them after projecting onto $|+\rangle$.

The paper therefore first changes to an equivalent set of stabilizer generators in which the unwanted temporary-ancilla dependence occurs in pairs such as $Z_u^a Z_w^a$. The graph cluster-state stabilizers then replace each such pair by a string of Z^A's on the surviving edge qubits. Only after this rewriting is

it safe to measure and discard the a qubits. Here "equivalent'' means *equal on the prepared stabilizer subspace*, after multiplication by already satisfied input and cluster stabilizers; it does not mean that the displayed operators are unconstrained identities on the whole Hilbert space. This distinction is important.

The key cluster-state relation on any graph path $\gamma_{u,w}$ from vertices u to w is

$$\prod_{f\in\gamma_{u,w}} (Z_f^A Z_{\partial f}^a) = Z_{\gamma_{u,w}}^A Z_u^a Z_w^a \doteq I. \tag{165}$$

Read Eq. (165) directly as a product of stabilizers. Every factor $Z_f^A Z_{\partial f}^a$ in the product is a graph-edge stabilizer, so it acts as $+1$ on the premeasurement cluster state. Their product must therefore also act as $+1$ on that state.

Now take a simple path from u to w. Every temporary vertex in the interior of the path belongs to exactly two successive graph edges. Its Z^a operator consequently occurs twice in the product and cancels:

$$(Z^a_{\text{interior vertex}})^2 = I.$$

The two endpoint vertices u, w occur only once, so their Z^a's do *not* cancel. Each edge of the path also occurs once, leaving one Z^A on each surviving edge qubit. Thus it is the combined operator

$$Z_{\gamma_{u,w}}^A Z_u^a Z_w^a$$

that is equivalent to the identity on the premeasurement stabilizer subspace. Hence,

$$Z_u^a Z_w^a \doteq Z_{\gamma_{u,w}}^A. \tag{166}$$

The symbol $\doteq$ means equality on that stabilized subspace, not an operator identity on every state. Equation (166) is the precise reason that a pair of unwanted temporary-ancilla Z's can be replaced by a string of Z's on the surviving A edge qubits. The notation $\langle u, w\rangle$ in the paper denotes a chosen such path; it need not be a single graph edge.

Rewriting one matter check. Temporarily use the simpler symbol s for one input X-check. If

$$E_s := \operatorname{supp}(s),$$

then its checks in the two copies are

$$S_{B,s}^X = \prod_{e\in E_s} X_e^B, \qquad S_{C,s}^X = \prod_{e\in E_s} X_e^C.$$

In the paper's notation, s is the basis label v, its coordinate cochain is v, and $E_s = \operatorname{supp}(d_0 v)$. The paper also uses the same label s for a vertex of the ancilla graph. This identification is implicit in the paper; formally one could instead introduce a map sending each check label to a chosen graph vertex.

For each input qubit e, the map ϕ chooses one ancilla vertex $\phi(e)$ whose temporary qubit controls the SWAP of B_e and C_e. This controller need not equal s. Indeed, one physical qubit can belong to several

X-checks, whereas the simplifying assumption in the paper assigns it only one controller $\phi(e)$. If e belongs to two checks s and t, one cannot in general have both $\phi(e) = s$ and $\phi(e) = t$.

Let

$$P_e := X_e^B X_e^C.$$

Equation (161) says that after symmetry enrichment the dressed B-check is

$$\widetilde{S}_{B,s}^X = S_{B,s}^X \prod_{e \in E_s} P_e^{n^a_{\phi(e)}}. \tag{167a}$$

Only the controller occupations $n^a_{\phi(e)}$ occur here; no operator on a_s has appeared yet. We now introduce one in a way that does not change the enriched state.

First define

$$Q_s := S_{B,s}^X S_{C,s}^X = \prod_{e \in E_s} P_e. \tag{167b}$$

The two original checks give $Q_s = +1$ before enrichment. More importantly, this remains true afterwards. A SWAP on B_e, C_e leaves $P_e = X_e^B X_e^C$ invariant, so every controlled-SWAP in the enrichment circuit commutes with Q_s. If $|\Psi_{\text{enr}}\rangle = \Omega|\Psi_0\rangle$, then

$$Q_s|\Psi_{\text{enr}}\rangle = \Omega Q_s|\Psi_0\rangle = |\Psi_{\text{enr}}\rangle. \tag{167c}$$

Since n_s^a acts on an ancilla and commutes with Q_s, the controlled operator

$$Q_s^{n_s^a} = (I - n_s^a) + n_s^a Q_s$$

also acts as the identity on the enriched state. We may therefore multiply Eq. (167a) by it. Using Eq. (167b), this gives the equivalent representative

$$\widetilde{S}_{B,s}^X \doteq S_{B,s}^X \prod_{e \in E_s} P_e^{n^a_{\phi(e)} + n_s^a}. \tag{167d}$$

This multiplication is the precise step that introduces the previously absent operator on a_s.

For each e, the following is an operator identity:

$$P_e^{n^a_{\phi(e)} + n_s^a} = \left(Z^a_{\phi(e)} Z_s^a\right)^{m_e^B + m_e^C}, \qquad m_e^B = \frac{I - X_e^B}{2}, \tag{167e}$$

with the exponent understood modulo two. To verify it, work in a joint X_e^B, X_e^C eigenstate. If their eigenvalues agree, then $P_e = +1$ and both sides are the identity. If they disagree, then $P_e = -1$, and the left side is $(-1)^{n^a_{\phi(e)} + n_s^a} = Z^a_{\phi(e)} Z_s^a$, exactly the right side.

We now have two temporary-ancilla Z's. The second one is essential: a finite graph path has two endpoints, so the cluster-state path relation cannot replace a single $Z^a_{\phi(e)}$ by a finite A-string. For each $e \in E_s$, choose a path in the ancilla graph

$$\gamma_{e,s} : \ \phi(e) \longrightarrow s.$$

The paper denotes it by $\langle \phi(e), s \rangle$. Equation (166) gives

$$Z^a_{\phi(e)} Z^a_s \doteq Z^A_{\gamma_{e,s}} := \prod_{f \in \gamma_{e,s}} Z^A_f. \tag{167f}$$

Substituting this path relation removes every temporary a qubit. The path runs from the controller $\phi(e)$ to the chosen vertex for the whole check s. If these coincide, the path is empty and its Z^A product is identity.

For an X-check labelled v, the result is

$$\begin{aligned} S^{X,\text{gauged}}_{B,v} &= X^B_{d_0 v} \prod_{\mathbf{e} \in d_0 v} \left(Z^A_{\langle \phi(\mathbf{e}), v \rangle} \right)^{m^B_{\mathbf{e}} + m^C_{\mathbf{e}}}, \\ S^{X,\text{gauged}}_{C,v} &= X^C_{d_0 v} \prod_{\mathbf{e} \in d_0 v} \left(Z^A_{\langle \phi(\mathbf{e}), v \rangle} \right)^{m^B_{\mathbf{e}} + m^C_{\mathbf{e}}}, \end{aligned} \tag{167}$$

Here $m^B_{\mathbf{e}} = (I - X^B_{\mathbf{e}})/2$, and similarly for C. The sum in each exponent is modulo two.

The Z-type checks have the parallel form

$$\begin{aligned} S^{Z,\text{gauged}}_{B,[p]} &= Z^B_{\partial_2 [p]} \prod_{e \in \partial_2 [p]} \left(Z^A_{\langle \phi(e), v(p) \rangle} \right)^{n^B_e + n^C_e}, \\ S^{Z,\text{gauged}}_{C,[p]} &= Z^C_{\partial_2 [p]} \prod_{e \in \partial_2 [p]} \left(Z^A_{\langle \phi(e), v(p) \rangle} \right)^{n^B_e + n^C_e}, \end{aligned} \tag{168}$$

where $n^B_e = (I - Z^B_e)/2$, and likewise for C. Let us unpack the choice of $v(p)$ without using geometrical language. A basis element $[p] \in D_2$ labels one Z-check, and

$$\partial_2 [p] = \sum_{e \text{ acted on by } p} [e]$$

lists the physical qubits on which that check acts. For each such qubit, $\partial_1 [e] \in D_0$ lists the X-check labels that also act on e. The paper's "closure'' of p is therefore the set

$$\text{cl}(p) := \bigcup_{e \in \text{supp}(\partial_2 [p])} \text{supp}(\partial_1 [e]). \tag{168a}$$

In words,

$$\boxed{\text{cl}(p) = \{X\text{-check labels touching at least one qubit of the } Z\text{-check } p\}.} \tag{168b}$$

To choose $v(p)$, pick one physical qubit e on which the Z-check p acts, then pick an X-check v that also acts on that qubit. Call the selected X-check label $v(p)$. Because the ancilla-graph vertices are indexed

by X-check labels, $v(p)$ also names an ancilla vertex. In the check–qubit incidence graph (often called a Tanner graph), this is simply the two-step choice

$$p \text{ (a } Z\text{-check)} \longrightarrow e \text{ (a data qubit)} \longrightarrow v(p) \text{ (an } X\text{-check/ancilla vertex).} \tag{168c}$$

One must not replace the union in Eq. (168a) by $\mathrm{supp}(\partial_1\partial_2[p])$. The chain-complex identity $\partial_1\partial_2 = 0$ makes that support empty: the incident X-check labels cancel modulo two after all qubits of the Z-check are summed. The closure instead collects the supports of the individual $\partial_1[e]$'s *before* those cancellations. Geometrically, for a square toric-code plaquette, $\partial_2[p]$ gives its four boundary edges, $\mathrm{cl}(p)$ gives its four corner vertices, and $v(p)$ is one chosen corner. No literal face or corner is required in a general qLDPC code.

The notation in Eqs. (167)–(168) deserves one last reading. For example, if $m_e^B + m_e^C = 0$, the associated path contributes nothing. If it is one, multiply Z^A over every graph edge on the chosen path from $\phi(e)$ to v for an X-check, or to $v(p)$ for a Z-check. These are finite products of *controlled-Pauli* factors; as whole operators they need not be bare Pauli products. In a fixed matter eigenvalue sector each factor reduces to either I or an ordinary Z^A string. Together with Eq. (164), they are the stabilizers of the gauged model written in the paper's occupation-number convention.

These formulas specify constraints satisfied by the prepared state. Their derivation used other constraints set to $+1$, so it does not by itself prove that the displayed non-Pauli operators commute on the entire Hilbert space. As in Section II, constructing commuting Hamiltonian projectors requires keeping the other constraints, including gauge-flux conditions, in the argument.

2.17.6 Closed paths and the surviving gauge constraints

There is one more family to keep track of. Multiply the unchanged edge stabilizers along a **closed** graph path η. Every temporary vertex occurs an even number of times, so all its Z^a's cancel:

$$\prod_{f\in\eta} K_f^A = Z_\eta^A, \qquad \partial_1^a \eta = 0.$$

These operators survive the measurement and have eigenvalue $+1$ in the prepared branch. They express zero gauge flux around the selected loops. If two paths between the same endpoints differ by such a cycle, their Z^A strings agree in that sector. This explains why choosing paths is compatible with the preceding replacement rules.

An ordinary graph has no specified faces: $A_2 = 0$. To speak of local magnetic checks such as the paper's $Z_{\partial p}^A$, one must specify which short cycles are used as plaquette boundaries. Longer independent cycles can instead label global flux sectors. The particular cluster-state preparation fixes their eigenvalues too, but that does not make each one a bounded-weight check of the code Hamiltonian.

2.17.7 What must be checked to obtain a qLDPC family

The formulas expose four concrete requirements:

- A bounded number of graph edges meet each vertex, keeping the gauge transformation G_v^A small.
- A bounded number of input pairs share each controller, keeping its SWAP product small and the enrichment circuit at constant depth.
- The paths dressing each matter check have bounded length, and each gauge edge is used by only a bounded number of such checks.

- The chosen magnetic-check generators have bounded weight and bounded participation per qubit.

For example, a weight-four input check dressed by four paths of length at most three acts on at most eight matter qubits and twelve gauge qubits. Its weight is therefore at most twenty, independent of code size. If those paths instead grow with the code, the original weight-four check no longer has a constant bound. Short paths alone also do not prevent many checks from sharing the same gauge edge; their overlap must be controlled.

The paper proposes adding direct connections between each controller $\phi(e)$ and the reference vertex of every check containing e. This makes the required paths short. The resulting graph must still be checked for bounded degrees, bounded use of each edge, connectedness, and suitable loops. The authors leave the choice of graph dependent on the input code and the desired properties.

This separates the algebraic construction from code quality. The controlled swaps satisfy the symmetry-enrichment identity, and the path calculation derives constraints on the surviving qubits. Establishing bounded check weights is an additional task; establishing useful distance and encoding rate is another. Here **distance** measures the smallest support of a nontrivial logical operation, and **encoding rate** is the number of encoded qubits divided by the number of physical qubits. The LDPC condition alone does not guarantee either quantity is large.

2.18 Section IV.A: non-Abelian fractons from gauging transversal CZ gates

The SWAP example established the graph-gauging mechanism in its most universal form. This example is more directly connected to quantum computation: it gauges a transversal CZ, the logical Clifford operator whose measurement will later prepare a magic state. It also lets us follow the construction on a concrete three-dimensional lattice.

There are two things to keep distinct from the outset. The **ALP code** is an Abelian CSS code with fracton/subsystem structure. The construction then takes two copies of that code and gauges a symmetry between the copies. The gauged code is the non-Abelian object. In this first pass we will follow the graph construction through its local circuit and its resulting constraints. The later subsections on addressable gates and cup products are deliberately postponed: they answer how to gauge a family of CZ gates, not what the basic CZ-gauging mechanism is.

2.18.1 The unusual qubit placement in the ALP code

The anisotropic lineon-planeon (ALP) model lives on a periodic cubic lattice. Its CSS chain complex is

$$0 \longrightarrow D_2 \xrightarrow{\partial_2} D_1 \xrightarrow{\partial_1} D_0 \longrightarrow 0. \tag{169}$$

The word *anisotropic* is literal: the three spatial directions are not treated alike. The bases have the following geometric interpretation:

space	basis object	role
D_2	cubes c	Z-check labels
D_1	plaquettes in xy planes and edges parallel to z	physical qubits
D_0	vertices v	X-check labels.

(170)

The middle line is the surprise. A horizontal *xy plaquette* carries a qubit, whereas a vertical *z edge* carries a qubit. There are no qubits on horizontal edges. This is why the ALP stabilizers do not look like the usual cubic-lattice toric-code stabilizers.

The drawings below are reproduced directly from the paper's TikZ source, with the geometry unchanged. The left drawing gives a cube Z-check; the right drawing gives a vertex X-check. A letter marks a qubit on which that Pauli acts. In particular, a letter inside a horizontal face means a **face qubit**, not a qubit at a hidden vertex.

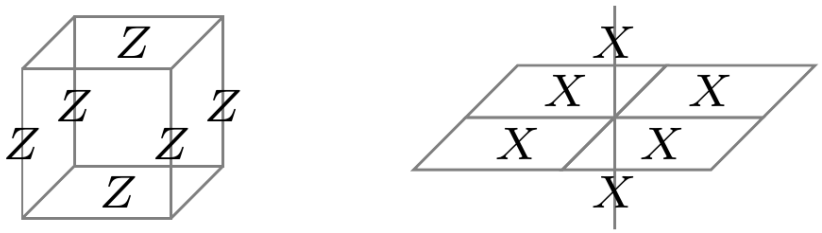


Figure 1: ALP cube Z check (left) and vertex X check (right), reproduced from the paper.

Source: Section IV.A, equation labelled `eq:qubit-ALP-stabilizers` *in the paper's source. Each check acts on six qubits.*

To make these perspective drawings unambiguous, label a vertex by integer coordinates (i, j, k). Write $p_{i,j,k}$ for the horizontal square with corners $(i, j, k), (i+1, j, k), (i, j+1, k), (i+1, j+1, k)$, and $\ell_{i,j,k}$ for the vertical edge from (i, j, k) to $(i, j, k+1)$. Coordinates are understood periodically. The cube $c_{i,j,k}$ then has the following check:

$$B_{i,j,k} = Z_{p_{i,j,k}} Z_{p_{i,j,k+1}} \prod_{\alpha,\beta\in\{0,1\}} Z_{\ell_{i+\alpha,j+\beta,k}}.$$

The two face factors sit on the bottom and top; the other four factors sit on the upright edges. At a vertex, the check is instead

$$A_{i,j,k} = X_{\ell_{i,j,k-1}} X_{\ell_{i,j,k}} \prod_{\alpha,\beta\in\{0,1\}} X_{p_{i-\alpha,j-\beta,k}}.$$

Here the two edge factors are below and above the vertex, and the four horizontal squares meet at that vertex. These formulas specify every qubit in the drawings without requiring the reader to reconstruct depth from a flat image.

The chain maps encode exactly these incidences. The map ∂_2 sends a cube to the sum of its two horizontal faces and four vertical edges. The map ∂_1 sends a face qubit to its four corner vertices, and an edge qubit to its two endpoints. This is a **code chain complex**: its degree records a qubit's role in the checks, not necessarily the spatial dimension of the object carrying it. Both kinds of qubit belong to D_1, even though a square is geometrically two-dimensional.

For example, applying $\partial_1\partial_2$ to a cube counts each of its eight corners twice: once from a horizontal face, once from a vertical edge. Over $\mathbb{F}_2$, these contributions cancel. Equivalently, an incident vertex check and that cube check share two qubits, so commuting them produces $(-1)^2 = +1$. This is the concrete content of the chain-complex condition in this model.

In the paper's compact notation, the two stabilizer families are therefore

$$A_v = X_{d_0 v}, \qquad B_c = Z_{\partial_2[c]}. \tag{171}$$

The chain-complex condition $\partial_1 \partial_2 = 0$, or dually $d_1 d_0 = 0$, guarantees that each A_v and B_c overlap on an even number of qubits and hence commute. This establishes that the checks define a CSS code. Gauging will additionally require a symmetry preserving that code, which we verify below.

An excitation is a check with eigenvalue -1, rather than the ground-state value $+1$. Restricted mobility means that transporting an isolated excitation in some directions requires creating additional excitations. A *lineon* can move along a line; a *planeon* can move within a plane. For a concrete first example, $Z_{\ell_{i,j,k}}$ anticommutes with the vertex checks at the two ends of that edge. A product of Z's along a vertical segment leaves defects only at its endpoints, because the interior signs flip twice. In contrast, $Z_{p_{i,j,k}}$ creates four vertex defects at that square's corners. This already shows why vertical and horizontal operations have different effects. The broader term *fracton order* includes systems with such restricted-mobility excitations; it does not mean that every excitation is completely immobile.

Here each check has weight six, and each qubit occurs in a bounded number of checks, independently of lattice size. That is the low-density condition in qLDPC. "Subsystem" in the discussion of spatial symmetries should not be confused with a subsystem error-correcting code with noncommuting gauge checks: the starting ALP code is a commuting CSS stabilizer code.

For an $L_x \times L_y \times L_z$ periodic lattice, the paper reports $2(L_x + L_y - 1)$ logical qubits and distance scaling as $\min(L_x, L_y, L_z)$. This is a useful reminder that it is a subsystem-like 3D code rather than a conventional topological code with a constant number of logical qubits.

2.18.2 The transversal CZ symmetry between two copies

The next step is to identify a gate that preserves the two input code spaces. This matters because gauging an arbitrary collection of physical gates would not automatically produce a logical-symmetry measurement.

Make two identical ALP blocks, named B and C. In the paper's original drawing, blue dots belong to B, orange dots to C, and black segments specify the CZ pairs. Repeat the displayed pattern in every cubic cell.

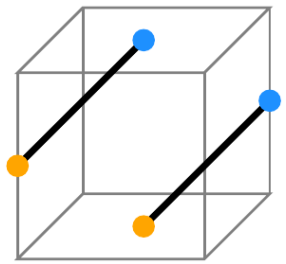


Figure 2: Original ALP transversal CZ pairing from Section IV.A of the paper.

Source: the paper's equation labelled `eq:cubic-CZ-sym`*. Colours distinguish code blocks, not face qubits from edge qubits.*

Writing $\mathcal{P}$ for this matching gives

$$U_{\mathsf{CZ}} = \prod_{(e,e')\in\mathcal{P}} \mathsf{CZ}_{B_e,C_{e'}}. \tag{172}$$

In this transversal matching, each physical qubit participates in exactly one two-qubit gate between blocks. Merely having a bounded number of gates per qubit would describe a bounded-depth circuit, a weaker condition. The paired qubits need not be drawn directly on top of one another: the pairing is a local matching determined by the cubic unit cell. This is a point at which the paper's 3D perspective matters more than an attempted planar picture.

The elementary conjugation rules are

$$\mathsf{CZ}_{B,C} X_B \mathsf{CZ}_{B,C} = X_B Z_C, \qquad \mathsf{CZ}_{B,C} X_C \mathsf{CZ}_{B,C} = Z_B X_C, \tag{173}$$

while both Z_B and Z_C are unchanged. The chosen pairing makes the extra Z factors fit the ALP check structure. The code-preserving property concerns the complete gate: a *single physical* X_e, Z_e, or $\mathsf{CZ}_{B_e,C_{e'}}$ is generally **not** a symmetry of the code. For example, a lone Z_e anticommutes with the X-checks incident on that qubit and creates syndrome defects. Equation (173) is only a local operator identity; it does not say that its individual factors preserve check eigenvalues.

The coordinate convention above makes it possible to check the symmetry explicitly. In cell (i,j,k), the two pairs in the drawing are

$$(C_{p_{i,j,k}}, B_{\ell_{i+1,j+1,k}}), \qquad (C_{\ell_{i,j,k}}, B_{p_{i,j,k+1}}).$$

Consider the entire B-vertex check at (i,j,k). Its two vertical edge qubits acquire C-block Z's on the horizontal faces $p_{i-1,j-1,k-1}$ and $p_{i-1,j-1,k}$. Its four face qubits acquire Z's on the four vertical edges of the same cube. The six new factors are therefore **exactly one complete cube check**:

$$\begin{aligned} U_{\mathsf{CZ}} A^B_{i,j,k} U^\dagger_{\mathsf{CZ}} &= A^B_{i,j,k} B^C_{i-1,j-1,k-1}, \\ U_{\mathsf{CZ}} A^C_{i,j,k} U^\dagger_{\mathsf{CZ}} &= B^B_{i,j,k} A^C_{i,j,k}. \end{aligned} \tag{174}$$

For the second identity, the two C-edge qubits attach Z's to the bottom and top B-faces of $c_{i,j,k}$; the four C-face qubits attach them to its four upright B-edges. This verifies both directions of the matching. The mechanism is the assembly of complete checks, rather than a claim that each physical X or each elementary gate preserves the code separately. For products of checks, repeated Pauli factors cancel modulo two in the usual way.

On a code state, both factors on either right-hand side have eigenvalue $+1$. Thus the transformed check still fixes the state: the transversal gate maps the stabilizer group to itself. It can change the encoded logical state while keeping the physical state inside the codespace.

The Z-checks themselves do not acquire anything, because CZ is diagonal:

$$U_{\mathsf{CZ}} B^B_c U^\dagger_{\mathsf{CZ}} = B^B_c, \qquad U_{\mathsf{CZ}} B^C_c U^\dagger_{\mathsf{CZ}} = B^C_c. \tag{175}$$

In binary language, the test for a proposed matching is that its added Z support be in $\operatorname{im}\partial_2$ for every X-check support in $\operatorname{im} d_0$.

2.18.3 The cubic ancilla graph and the controlled-CZ enrichment

Having verified the matter symmetry, we now introduce the ancillary system that will turn its global action into local gauge constraints. Put temporary qubits a_v on every cubic-lattice vertex and gauge-field qubits A_f on every cubic-lattice edge $f = \langle v, w \rangle$. After making the graph cluster state (the paper applies a Hadamard to the A sublattice), its convenient stabilizer generators are

$$K_v^a = X_v^a \prod_{f \ni v} X_f^A, \qquad K_f^A = Z_f^A Z_v^a Z_w^a. \tag{176}$$

Because the cubic graph is connected, the temporary vertices possess exactly one global X symmetry,

$$X_{\text{all}}^a = \prod_v X_v^a. \tag{177}$$

The enrichment must tie this single symmetry to Eq. (172). It uses CCZ, the three-qubit controlled-controlled-Z gate. Its action is diagonal in the computational basis:

$$\mathsf{CCZ}_{a,B,C} |r, b, c\rangle = (-1)^{rbc} |r, b, c\rangle. \tag{178}$$

Thus when the temporary control is $|0\rangle_a$, it does nothing; when it is $|1\rangle_a$, it applies CZ to the selected B, C pair. The paper's original enrichment drawing shows exactly which two pairs are controlled by the purple vertex:

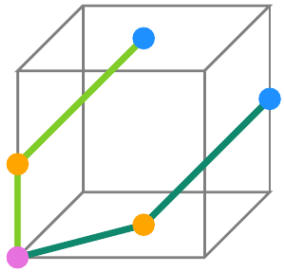


Figure 3: Original ALP CCZ enrichment: two green paths connect the purple control to the two matter pairs.

Source: the paper's equation labelled `eq:cubic-symm-enrich`*. Each shade of green denotes one three-qubit gate, not two successive two-qubit gates.*

At $a_{i,j,k}$, these are the two pairs listed above: the lower C-face with the far B-edge, and the near C-edge with the upper B-face. There are exactly two CCZ's per control in this pattern. This assignment is repeated at every vertex.

The paper denotes this local assignment by $\phi^{\mathsf{T}}(v)$. In this example $\phi^{\mathsf{T}}(v)$ is a set of **pairs** (e, e'), not merely a set of single qubits as in the SWAP example. In a basis-independent form, the enrichment circuit is

$$\Omega = \prod_v \prod_{(e,e') \in \phi^{\mathsf{T}}(v)} \mathsf{CCZ}_{a_v, B_e, C_{e'}}. \tag{179}$$

The map must be local and surjective in the paper's sense: every required physical CZ in U_{CZ} is assigned to an odd number of controls, preferably exactly one, and each control reaches only $O(1)$ nearby pairs. This is the CZ version of the sparse assignment $e \mapsto \phi(e)$ in the SWAP construction.

To see the symmetry enrichment explicitly, use the one-gate identity

$$X_a \, \mathsf{CCZ}_{a,B,C} \, X_a \, \mathsf{CCZ}_{a,B,C} = \mathsf{CZ}_{B,C}. \tag{180}$$

It follows just by checking $a = 0, 1$: flipping the control interchanges the "apply CZ" and "do nothing" branches. Multiply Eq. (180) over all temporary vertices and their assigned pairs. The odd-assignment condition then gives

$$X^a_{\text{all}} \Omega X^a_{\text{all}} \Omega^\dagger = U_{\mathsf{CZ}}. \tag{181}$$

This is the same group-commutator test used for graph-gauged SWAP, with controlled swaps replaced by controlled CZ gates. It is the one equation that shows the cubic ancilla is gauging the desired logical symmetry rather than merely adding extra qubits.

2.18.4 Gauge constraints, matter checks and commuting projectors

Measure every temporary a_v qubit in the X basis. For the moment, consider the all-$+1$ branch; the next subsection of this note explains how a decoder and feedforward correct the actual random signs. From Eq. (176), the temporary-vertex stabilizer becomes the surviving gauge constraint

$$\widetilde{K}^A_v = \left(\prod_{f \ni v} X^A_f \right) \left(\prod_{(e,e') \in \phi^{\mathsf{T}}(v)} \mathsf{CZ}_{B_e, C_{e'}} \right). \tag{182}$$

The first factor is an ordinary X-type gauge transformation on the six edges incident to a cubic vertex. The second factor simultaneously performs the local piece of the original B-C symmetry. This is the concrete form of the slogan "a gauge transformation carries the old symmetry."

The matter checks become dressed as well. The useful one-qubit identity is

$$\mathsf{CCZ}_{a,B,C} X_B \mathsf{CCZ}_{a,B,C} = X_B \mathsf{CZ}_{a,C}, \tag{183}$$

and the analogous equation holds with B and C exchanged. Therefore each X-type ALP check picks up controlled-Z factors involving temporary ancillas. Before measurement, those factors can be traded, using products of the edge relations K^A_f, for operators on paths of surviving A-edge qubits - the path-replacement mechanism developed in the graph-gauged SWAP example. The original ALP Z-checks themselves remain unchanged, since CCZ is diagonal and they act only on matter qubits. Products of K^A_f around a closed ancilla loop also survive: every temporary Z^a_v occurs twice, leaving only the loop product of Z^A_f. On the cubic lattice the elementary examples are the four-edge plaquette flux checks. The complicated dressing belongs to the matter X-checks and the combined gauge constraints; it should not obscure these simple unchanged Z-type operators.

This explains why the paper presents the final stabilizers mainly as 3D figures rather than as a short one-line formula. Three distinct incidence structures must be superimposed: the six-qubit ALP checks, the cubic ancilla graph, and the local B-C pair assignment. The final operators are local, but "local" now means local in that combined three-layer incidence structure, not local in a single planar projection.

There is a second subtlety. The raw dressed generators can commute only **up to** the gauged U_{CZ} symmetry. To define a Hamiltonian with mutually commuting projectors, the paper symmetrizes them with respect to that $\mathbb{Z}_2$ action. Commutation on the constrained codespace is weaker than commutation as operators on the entire Hilbert space; a commuting-projector Hamiltonian requires the latter. A bare matter-check label can be changed by the gauge transformation, so the gauge-invariant term must include both members of its symmetry orbit.

For a concrete illustration of an orbit, the transversal symmetry above sends A_v^B to $A_v^B B_c^C$ and back again, while fixing B_c^C. Here an *orbit* simply means the set of operators obtained by repeatedly applying the symmetry. The simultaneous positive-eigenvalue projector

$$Q = \frac{I + A_v^B}{2} \frac{I + B_c^C}{2}$$

is invariant under this exchange: replacing A_v^B by $A_v^B B_c^C$ leaves Q unchanged, because $B_c^C(I + B_c^C) = I + B_c^C$. The condition $B_c^C = +1$ makes the two representatives of the vertex check equivalent. This elementary example explains why imposing the relevant constraints can turn a symmetry orbit into one well-defined projector. For the gauged operators one must also include the gauge constraints; the paper's appendix on graph-gauged ALP fractons carries out the local dressing calculation. Simply averaging two arbitrary operators is not, by itself, a proof that the result is a projector or that different terms commute.

2.18.5 From the ALP construction to logical measurement

The SWAP example taught the general graph procedure. This ALP example adds the specific ingredient needed later for computation:

$$\text{temporary control} \quad + \quad \mathsf{CCZ} \quad \Longrightarrow \quad \text{gauged transversal logical } \mathsf{CZ}. \tag{184}$$

The later ungauging protocol reverses the ancilla construction and measures the gauge qubits. Equation (181) then reappears as a projector onto an eigenvalue of the logical CZ acting on the original B, C blocks. On appropriate logical input states, that Clifford measurement prepares a non-stabilizer state. Thus the CCZ gates used here are not being claimed as directly applied logical non-Clifford gates; rather, they are the local measurement gadget from which a non-Clifford *resource state* emerges after gauging and ungauging.

The construction therefore connects four ingredients:

1. The ALP code has qubits on xy plaquettes and z edges, giving six-qubit cube and vertex checks.
2. Two copies possess a local transversal CZ matching.
3. A cubic graph cluster state plus local CCZ enrichment satisfies the global commutator identity (181).
4. Measuring the temporary ancillas turns this into a non-Abelian gauge constraint (182); ungauging later converts the same construction into a logical CZ measurement.

2.19 Returning to Section III.E: measurement preparation of non-Abelian qLDPC codes

Before using the ALP construction for computation, we return to the measurement-preparation argument of Section III.E. It supplies the missing operational step in the example: handling random

measurement signs. The gauging discussion has used an idealized instruction: "project every temporary ancilla a_v onto $|+\rangle$." A physical X-basis measurement cannot promise that outcome. It returns a random number

$$x_v \in \{+1, -1\}, \qquad X_v^a \longmapsto x_v. \tag{185}$$

This section explains why that randomness does not turn gauging into a postselected protocol. For preparation from the symmetry-invariant input sector, and assuming that the Abelian code on the ancilla complex has a decoder, the random local signs can be removed deterministically by a Pauli correction on the surviving A qubits. The word *deterministic* means that every possible measurement record is accepted and corrected; it does not mean that individual measurement outcomes cease to be random.

2.19.1 First ignore the matter codes: measuring a graph cluster state

It is useful to strip the problem to its simplest form. Before the measurement, the temporary-vertex graph-cluster stabilizer is

$$K_v^a = X_v^a X_{d_0^a v}^A. \tag{186}$$

For an ordinary graph, $X_{d_0^a v}^A$ is the product of X on all A-edge qubits incident to vertex v. Since $K_v^a = +1$ on the cluster state, measuring $X_v^a = x_v$ leaves the surviving A qubits with the signed check

$$x_v X_{d_0^a v}^A = +1. \tag{187}$$

The usual Z-type ancilla checks, labelled schematically by $Z_{\partial_2[p]}^A$, are unaffected. Thus the only difference between the desired Abelian ancilla code and the actually prepared one is the list of signs $\{x_v\}$ in Eq. (187).

Turn these signs into a binary syndrome vector,

$$s = \sum_v s_v[v] \in A_0, \qquad s_v = \begin{cases} 0, & x_v = +1, \\ 1, & x_v = -1. \end{cases} \tag{188}$$

Why are there an even number of -1 outcomes in the clean target sector? Multiply every premeasurement vertex stabilizer in Eq. (186). Each A-edge occurs at the two ends of that edge, so its two factors cancel:

$$\prod_v K_v^a = \left(\prod_v X_v^a\right)\left(\prod_{f \in A_1} (X_f^A)^2\right) = \prod_v X_v^a = X_{\text{all}}^a. \tag{189}$$

All the K_v^a's have eigenvalue $+1$, hence the cluster state has $X_{\text{all}}^a = +1$. The individual X_v^a measurements commute, so their outcomes must obey

$$\prod_v x_v = +1. \tag{190}$$

A product of signs is $+1$ precisely when an even number of its factors are negative. For three temporary qubits, for example, $(+,+,+)$ and $(+,-,-)$ are allowed, whereas $(-,+,+)$ is not. After symmetry enrichment the corresponding global relation is $X^a_{\mathrm{all}}U = +1$. In the clean target-preparation sector, the input matter state has $U = +1$, so the same even-parity conclusion follows. An odd parity would instead record the opposite global symmetry sector, not an ordinary local syndrome that an A-edge correction can erase.

This is exactly the familiar CSS situation: a -1 outcome says that the corresponding X-check has the wrong sign. On a 2D surface-code graph, one would draw the -1 vertices and connect them pairwise by a path of edges. In a general qLDPC complex there may be no literal picture, but the same binary linear-algebra question remains: find a 1-chain

$$r \in A_1 \quad \text{such that} \quad \partial^a_1 r = s. \tag{191}$$

For ideal measurements, finding any such chain suffices to correct these signs. In error correction with physical noise, a decoder uses a noise model to seek a likely error representative; choosing the logical class then also matters. It is not necessary for this discussion to know how that algorithm works internally.

The corresponding quantum correction is the Pauli product

$$Z^A_r := \prod_{f \in A_1} (Z^A_f)^{r(f)}. \tag{192}$$

Why does it fix the signs? A factor Z^A_f anticommutes with an X-check precisely when edge f occurs in that check. Therefore

$$Z^A_r X^A_{d^a_0 v} Z^A_r = (-1)^{v(\partial^a_1 r)} X^A_{d^a_0 v} = (-1)^{s_v} X^A_{d^a_0 v}. \tag{193}$$

Because $x_v = (-1)^{s_v}$, conjugating Eq. (187) by Z^A_r changes it to the desired $+X^A_{d^a_0 v} = +1$ check. The correction commutes with all Z-type ancilla checks, since it consists only of Z's. This is the full deterministic-preparation argument for the Abelian ancilla code.

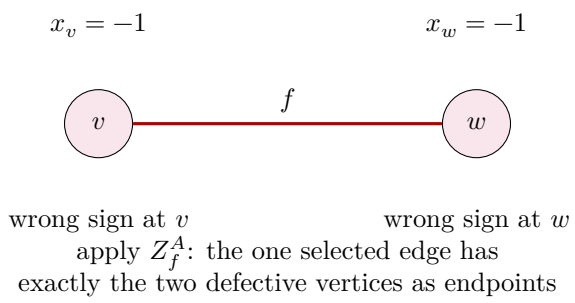
Worked nearest-neighbour example: two -1 outcomes.

Take two neighbouring vertices v and w of the cubic ancilla graph, joined by one surviving A-edge $f = \langle v, w \rangle$. Suppose that the two temporary measurements give $x_v = x_w = -1$, while every other temporary measurement gives $+1$. This is the smallest nontrivial syndrome compatible with a closed, periodic graph:

$$s = [v] + [w] = \partial^a_1 [f]. \tag{194}$$

The decoder can therefore choose the shortest possible correction chain, $r = [f]$. Its quantum correction contains just **one** single-qubit Pauli:

$$Z^A_r = Z^A_f. \tag{195}$$



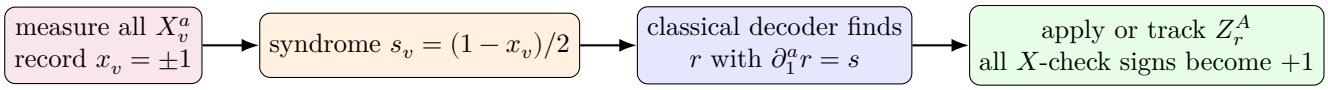
The two signed checks before correction are

$$-X^A_{d_0^a v} = +1, \qquad -X^A_{d_0^a w} = +1. \tag{196}$$

Edge f occurs once in each of the two vertex checks. Hence

$$Z_f^A X^A_{d_0^a v} Z_f^A = -X^A_{d_0^a v}, \qquad Z_f^A X^A_{d_0^a w} Z_f^A = -X^A_{d_0^a w}. \tag{197}$$

The correction flips both minus signs to plus signs at once. At every other vertex u, the edge f does not occur in $d_0^a u$, so the u-check is unchanged. In the gauged code, exactly the same calculation applies to $-X^A_{d_0^a v}\mathcal{U}_v$ and $-X^A_{d_0^a w}\mathcal{U}_w$, because Z_f^A commutes with the matter dressing $\mathcal{U}$. This is the most concrete example of the general rule $\partial_1^a r = s$.



2.19.2 Why exactly the same decoder works after gauging

Now restore the matter blocks B, C and the enrichment circuit. After the X_a measurements, the gauged vertex constraint has the general form

$$\mathcal{G}_v[x_v] = x_v X^A_{d_0^a v}\,\mathcal{U}_v. \tag{198}$$

The factor $\mathcal{U}_v$ is the local piece of the old transversal symmetry. In the SWAP example it is a product of SWAP_{B_e,C_e}'s; in the ALP CZ example it is a product of $\mathsf{CZ}_{B_e,C_{e'}}$'s. Crucially, the random outcome x_v occurs in exactly the same place as it did in Eq. (187): it merely multiplies the X^A factor by a sign.

The correction Z_r^A has two equally important properties:

1. By Eq. (193), it flips precisely the unwanted signs in $\mathcal{G}_v[x_v]$.
2. It commutes with $\mathcal{U}_v$, because Z_r^A acts on the surviving gauge qubits whereas $\mathcal{U}_v$ acts on B, C. It also commutes with the other Z-type and matter constraints.

Hence the same correction selected by the *Abelian* ancilla decoder turns

$$\mathcal{G}_v[x_v] \quad \longmapsto \quad \mathcal{G}_v[+1] \tag{199}$$

without changing the non-Abelian matter dressing. This is the central observation of the paper's subsection. Measurement randomness creates only an Abelian syndrome, even though the final code has non-Abelian structure.

2.19.3 What "feedforward" means here

The decoder may be a nonlocal classical computation: it reads the full list of outcomes before deciding on r. But once r is known, every factor of Z_r^A is a one-qubit Pauli. They can all be performed in one quantum layer. Often one does not physically apply them at all. One instead stores the binary vector r as a **Pauli frame** and flips the interpretation of later measurements accordingly.

This is why the paper calls the preparation a finite-depth *adaptive* circuit: the quantum circuit has bounded depth, while a classical decoder and a classically chosen Pauli frame connect the measurement step to later operations. This is a deterministic state-preparation statement assuming an appropriate decoder exists. It is not yet a complete proof of fault tolerance against noisy gates and noisy measurements; the paper identifies that stronger question as future work.

2.19.4 What to retain for the computation section

The ideal $|+\rangle_a$ projection used in a formal gauging derivation is therefore operationally replaced by

$$\text{measure } X_v^a \quad \longrightarrow \quad \text{decode the sign syndrome} \quad \longrightarrow \quad \text{update by } Z_r^A. \tag{200}$$

This prepares the desired $+1$ constraints in the invariant input sector. The next section asks a different question: what happens for an **arbitrary logical input**, possibly a superposition of symmetry sectors? Then the global products of outcomes contain information about that input and must be retained.

Indeed, an edge correction changes a sign syndrome by $\partial_1^a r$. For any cocycle μ, its effect on the corresponding outcome parity is zero because $\mu(\partial_1^a r) = (d_0^a \mu)(r) = 0$. Thus local feedforward cannot erase a logical symmetry outcome. This is why deterministic preparation and logical measurement are compatible, but are not the same protocol on the same class of inputs.

2.20 Section VI.A: logical Clifford measurement from gauging and ungauging

We now reach the computational claim of the paper. Gauging was introduced as a way to make a transversal symmetry local and dynamical. **Ungauging** reverses that enlargement: undo the ancilla cluster entangler and measure away the surviving A gauge qubits. The surprising result is that the combined operation is a measurement of the logical symmetry that was gauged.

The paper writes the derivation for a transversal CZ in the homological construction, where there can be several cocycle symmetries. The conclusion is broader. In the connected-graph construction there is only one independent cocycle, so the same derivation reduces to a measurement of its chosen involutory symmetry, such as SWAP or the CZ considered here.

2.20.1 The claim in one line

Let U_μ be the transversal logical Clifford symmetry with $U_\mu^\dagger = U_\mu$ and $U_\mu^2 = I$, associated with an ancilla 0-cocycle $\mu \in \ker d_0^a$. Gauging and then ungauging applies the projector

$$P_{x_\mu} = \frac{I + x_\mu U_\mu}{2} \tag{201}$$

to the two input code blocks. Here $x_\mu = \pm 1$ is obtained from the temporary-ancilla measurement outcomes. Equation (201) is precisely the projective measurement of the Hermitian involution U_μ: it selects its $+1$ or -1 eigenspace.

For the connected graph used in the SWAP and ALP examples, $\ker d_0^a$ is one-dimensional: it contains zero and the uniform nonzero cocycle μ_{all}. The formula is then simply

$$\frac{I + x\,U}{2}. \tag{202}$$

The rest of this section derives this statement. The logical information never has to be decoded into individual physical qubits; all operations are performed on the qLDPC blocks and the ancillary qubits.

Where the zero-coboundary condition enters the proof. A vertex pattern μ has zero coboundary when $d_0^a \mu = 0$: it produces no edge flips. This condition has two distinct roles. First, two expansion labels ϕ, ϕ_0 give the same measured edge string exactly when their difference $\mu = \phi + \phi_0$ has zero coboundary. Second, these patterns are ancillary symmetries, and the enrichment circuit is required to attach logical matter gates to them. The first statement follows from linear algebra; the second is a design condition on Ω, Eq. (121), not a consequence of linear algebra alone. The offset correction connects the two statements by leaving a sum indexed by these symmetry patterns.

2.20.2 Ingredients and notation

Let $|\Psi\rangle_{BC}$ be an arbitrary state in the two input code spaces. Let C_A be the Clifford circuit which prepares the ancilla cluster state from

$$|0\rangle_A^{\otimes} |+\rangle_a^{\otimes}. \tag{203}$$

In the Hadamard convention used here, C_A is a product of CNOT gates from temporary controls a_v to the adjacent A qubits.

The enrichment circuit and its required properties. The unitary $\Omega = \Omega_{a,BC}$ couples the temporary vertex qubits to the matter blocks B, C, and acts trivially on the surviving edge register A. In the controlled-CZ construction considered here, it is a product of CCZ gates, with one temporary qubit as control and a pair of matter qubits as targets. More explicitly, it has the controlled form

$$\Omega = \sum_{r \in \{0,1\}^{|V_A|}} |r\rangle_a \langle r|_a \otimes W_r^{BC} \otimes I_A,$$

where W_r^{BC} is the matter CZ circuit selected by the control string r. The cluster CNOTs also leave these control bits unchanged and act only on A. Consequently,

$$[\Omega, C_A] = 0.$$

This compatibility is what allows us to move the cluster circuit past enrichment in Step 1; acting on different targets alone would not be sufficient for arbitrary circuits with shared controls.

The essential **symmetry-enrichment condition** concerns the kernel of the ancillary coboundary map, $d_0^a : A^0 \to A^1$, whose notation is explained below. A label $\mu \in A^0$ assigns a bit to each temporary vertex, and

$$X_\mu^a := \prod_v (X_{a_v})^{\mu(v)}.$$

For **every** zero-coboundary label $\mu \in \ker d_0^a$, we require

$$\boxed{\Omega^\dagger X_\mu^a \Omega = X_\mu^a U_\mu, \qquad d_0^a \mu = 0.}$$

Here U_μ acts only on B, C, preserves their input codespace, and implements the intended transversal logical Clifford symmetry on that space. At least one such symmetry must have a nontrivial logical action for the logical measurement to be nontrivial. Zero coboundary identifies an ancillary symmetry pattern; it does **not**, by itself, prove that the attached matter operation is logical. That is a design requirement on Ω and the input codes.

The paper states this requirement with the conjugation direction $\Omega X_\mu^a \Omega^\dagger = X_\mu^a U_\mu$ in its Eq. (23), corresponding to Eq. (121) of this note. For the CCZ circuit here, $\Omega^\dagger = \Omega$, so both directions coincide. The paper explicitly uses the inverse-conjugation identity above immediately after Eq. (124) in the logical measurement derivation. Its Eqs. (36)–(39) construct the enrichment circuit and verify the attached code symmetry.

Conjugation preserves products, while the ancilla patterns satisfy $X_{\mu+\nu}^a = X_\mu^a X_\nu^a$. Therefore the attached operators compose consistently:

$$U_0 = I, \qquad U_{\mu+\nu} = U_\mu U_\nu, \qquad U_\mu^2 = I, \qquad [U_\mu, U_\nu] = 0.$$

These identities hold for the physical CZ representatives used here, and hence for their restrictions to the codespace. Unitarity and $U_\mu^2 = I$ also imply $U_\mu^\dagger = U_\mu$. These are the properties that let Step 3 rewrite a sum over kernel labels as a product of commuting measurement projectors $(I + x_{\mu_i} U_{\mu_i})/2$.

For a connected graph, the kernel contains only the zero pattern and the all-ones pattern. The nontrivial requirement is then simply

$$\Omega^\dagger X_{\text{all}}^a \Omega = X_{\text{all}}^a U_{\text{all}}.$$

For an arbitrary label ϕ outside the kernel, the CCZ circuit still attaches a physical CZ product V_ϕ:

$$\Omega^\dagger X_\phi^a \Omega = X_\phi^a V_\phi.$$

However, V_ϕ need not preserve the codespace. This explains the role of the offset correction in Step 2: the measured record initially leaves labels in $\phi_0 + \ker d_0^a$; multiplication by $X_{\phi_0}^a$ leaves only kernel labels, precisely where the logical symmetry guarantee applies. The kernel is that of d_0^a, not of the chosen label ϕ_0.

The complete measurement procedure. With these ingredients, the operations are:

1. Apply C_A, then the enrichment circuit Ω.
2. Measure each a_v in the X basis and record x_v.
3. Apply $C_A^\dagger$, then measure each A_e in the Z basis, recording its bit $z(e)$.
4. Solve $d_0^a \phi_0 = z$, and apply $X_{\phi_0}^a$.
5. Apply $\Omega^\dagger$. The ancillary registers now factor from the matter state and can be discarded.

The measured a qubits must remain available for the later gates. An ideal nondestructive X measurement leaves them in the recorded X-eigenstates. With destructive readout, reprepare those same $|x_v\rangle_X$

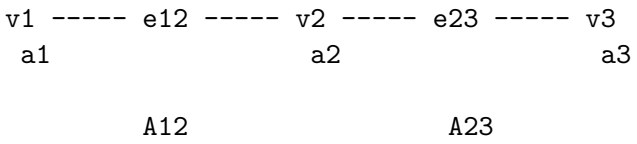
states before step 3. Discarding them without this replacement would not implement the displayed protocol.

The correction in step 4 removes the offset ϕ_0, not the logical measurement outcome. It is required independently of whether x_{ϕ_0} is positive or negative: that scalar is only a global phase. A software frame can replace the physical correction only if its propagation through the subsequent enrichment inverse is accounted for.

We next recall the ancillary chain complex, then derive why these operations produce a projector on the original encoded state.

2.20.3 Chains and cochains on the ancillary register

The calculation uses two different maps on the ancillary register. A boundary locates the endpoints of a correction string; a coboundary tells us which gauge qubits are flipped by a choice of temporary controls. Keeping these roles separate is easiest on a three-vertex graph:



Each vertex v carries a temporary qubit a_v, and each edge e carries a surviving gauge qubit A_e. This is the same vertex-and-edge picture used in ordinary lattice gauging. The qLDPC construction permits the graph to be an abstract incidence structure rather than a planar drawing, but the following bookkeeping is identical.

The symbols A_0 and A_1 mean, respectively, the binary vector spaces of collections of vertices and collections of edges:

mathematical object	plain-English meaning	physical qubits
A_0	collections of vertices	a_v live at vertices
A_1	collections of edges	A_e live on edges.

Because the coefficients are in $\mathbb{F}_2$, a collection means "mark some objects," and marking an object twice cancels: $1 + 1 = 0 \pmod 2$. For example,

$$r = e_{12} + e_{23} \in A_1$$

means "mark both edges," while

$$s = v_1 + v_3 \in A_0$$

means "mark vertices v_1 and v_3." These are **chains**.

The boundary map turns a marked set of edges into the marked endpoints left over:

$$\partial_1^a(e_{12}) = v_1 + v_2, \qquad \partial_1^a(e_{23}) = v_2 + v_3,$$

and therefore

$$\partial_1^a(e_{12} + e_{23}) = (v_1 + v_2) + (v_2 + v_3) = v_1 + v_3.$$

The interior vertex v_2 cancels because the two selected edges meet there. This is the familiar fact that a path has endpoints but its interior has no boundary. In measurement decoding, the equation

$$\partial_1^a r = s$$

means: choose an edge string r whose endpoints are exactly the vertices with unwanted measurement signs s.

Now introduce a 0-cochain $\phi \in A^0$. For intuition, draw it as another assignment of bits to vertices:

```
vertex:       v1   v2   v3
phi(v):        1    0    1
```

The coboundary $d_0^a\phi$ assigns bits to edges. The endpoint rule follows from its definition, rather than being an additional assumption. For an ordinary graph edge $e = \langle u, w\rangle$,

$$\begin{aligned}(d_0^a\phi)(e) &= \phi(\partial_1^a[e]) \\ &= \phi([u] + [w]) \\ &= \phi(u) + \phi(w) \pmod 2.\end{aligned}$$

The first equality defines the coboundary as the dual of the boundary; the second uses the edge's two endpoints; the third uses the linearity of a cochain. This is the evaluation form of $d_0^a = (\partial_1^a)^{\mathsf{T}}$. Thus, on an edge, the coboundary asks whether the endpoint bits differ:

$$(d_0^a\phi)(e_{12}) = \phi(v_1) + \phi(v_2) = 1 + 0 = 1,$$

$$(d_0^a\phi)(e_{23}) = \phi(v_2) + \phi(v_3) = 0 + 1 = 1.$$

Thus $d_0^a\phi$ marks the edges crossing from a vertex with bit 0 to one with bit 1:

```
vertex bits phi:       1 ---- 0 ---- 1
edge bits d phi:           1       1
```

The distinction between the two maps is therefore

$$\partial_1^a : \underbrace{\text{marked edges}}_{\text{chain}} \longrightarrow \underbrace{\text{their endpoints}}_{\text{chain}},$$

$$d_0^a : \underbrace{\text{vertex-bit assignment}}_{\text{cochain}} \longrightarrow \underbrace{\text{edge-bit assignment}}_{\text{cochain}}.$$

In the measurement-and-ungauging calculation, ϕ is used only to choose a term in an expansion. Its coboundary specifies the surviving edge qubits on which to apply X:

$$X^A_{d_0^a\phi} = \prod_{e:\,(d_0^a\phi)(e)=1} X_{A_e}.$$

For the displayed example, this gives

$$X^A_{d_0^a\phi} = X_{A_{12}} X_{A_{23}}, \qquad X^A_{d_0^a\phi}|0\rangle_A = |1\rangle_{A_{12}}|1\rangle_{A_{23}}.$$

Thus a subsequent Z_A measurement records two minus signs. Its edge-outcome pattern is written as the 1-cochain $z = d_0^a\phi$.

The precise definitions underlying these bit assignments are

$$A^0 = \mathrm{Hom}(A_0, \mathbb{F}_2), \qquad A^1 = \mathrm{Hom}(A_1, \mathbb{F}_2).$$

Formally, cochains are linear functions. But because there is a preferred basis of vertices and edges, such a function is completely specified by one bit per vertex or edge. That is why a cochain can be drawn just like a marked collection. For a first reading, retain the following rule:

```
chain:    "Which vertices or edges have I selected?"
cochain:  "What bit do I assign to each vertex or edge?"
```

They look alike in coordinates, but they participate in different maps:

$$\partial : \text{chains} \longrightarrow \text{chains}, \qquad d : \text{cochains} \longrightarrow \text{cochains}.$$

For this three-vertex example, the two maps are the matrices

$$\partial_1^a = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad d_0^a = (\partial_1^a)^{\mathsf{T}} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Multiplying the first by the edge-chain column $(1,1)^{\mathsf{T}}$ gives the endpoint column $(1,0,1)^{\mathsf{T}}$. Multiplying the second by the vertex-cochain column $(1,0,1)^{\mathsf{T}}$ gives the edge bits $(1,1)^{\mathsf{T}}$. All matrix arithmetic is modulo two. The transpose relation follows from the definition

$$(d_0^a\phi)(r) = \phi(\partial_1^a r).$$

The left side evaluates an edge cochain on an edge chain; the right side evaluates the vertex cochain on that chain's boundary. For a single edge, this equation says to add the two endpoint bits. For a path, all interior bits cancel.

In homological gauging, A_0, A_1 still label temporary and surviving qubits, but the incidence matrix need not be that of an ordinary graph. An A qubit can then couple to a different set of temporary controls. The matrix equations, dual-space definitions and calculation below still apply; the statement "add the two endpoint bits" is specific to a graph. Consequently $\ker d_0^a$ can describe subsystem symmetries rather than just a uniform bit assignment.

2.20.4 Step 1: expand the fixed X_a-measurement record

For a record $x_v = \pm 1$, the one-qubit measurement projector is

$$\Pi_v(x_v) = \frac{I + x_v X_v^a}{2}. \tag{204}$$

The cluster relation lets us move this projector through the preparation circuit. In the notation used throughout the note,

$$\Pi_x \, \Omega C_A(|\Psi\rangle_{BC}|0\rangle_A|+\rangle_a) \;=\; C_A \prod_v \frac{I + x_v X_v^a X_{d_0^a v}^A}{2} \Omega(|\Psi\rangle_{BC}|0\rangle_A|+\rangle_a). \tag{205}$$

This equation says: "instead of viewing the X_a measurements as acting after cluster preparation, we can pull their projectors through the cluster-preparation circuit." In that new viewpoint, each measured temporary qubit is accompanied by X's on all adjacent surviving A-qubits.

Write the initial state as

$$|\Phi_0\rangle = |\Psi\rangle_{BC}|0\rangle_A|+\rangle_a.$$

For a particular set of recorded measurement outcomes $x = \{x_v\}$, with $x_v = \pm 1$,

$$\Pi_x = \prod_v \Pi_v(x_v), \qquad \Pi_v(x_v) = \frac{I + x_v X_v^a}{2}.$$

This is just the projector onto the stated outcomes: if $x_v = +1$, it projects a_v onto $|+\rangle$; if $x_v = -1$, onto $|-\rangle$.

The cluster-preparation circuit C_A is a product of CNOTs from a temporary vertex qubit a_v to each neighbouring gauge-field qubit A_e. Its basic conjugation rule is

$$C_A^\dagger X_v^a C_A = X_v^a X_{d_0^a \mathbf{v}}^A.$$

Here $d_0^a \mathbf{v}$ is the binary list of edges incident on v. Thus, if v has neighbouring edges $e_1, \dots, e_r$,

$$X_{d_0^a \mathbf{v}}^A = X_{A_{e_1}} \cdots X_{A_{e_r}}.$$

For example, for a degree-three vertex,

$$C_A^\dagger X_v^a C_A = X_{a_v} X_{A_{e_1}} X_{A_{e_2}} X_{A_{e_3}}.$$

Now insert $I = C_A C_A^\dagger$ around each measurement projector:

$$\Pi_v(x_v) C_A = C_A (C_A^\dagger \Pi_v(x_v) C_A) = C_A \frac{I + x_v X_v^a X_{d_0^a \mathbf{v}}^A}{2}.$$

Doing this for every vertex gives the displayed equation:

$$\Pi_x \Omega C_A |\Phi_0\rangle = C_A \prod_v \frac{I + x_v X_v^a X_{d_0^a \mathbf{v}}^A}{2} \Omega |\Phi_0\rangle.$$

The enrichment circuit Ω commutes with C_A: both are controlled by the a-qubits, but they act on different targets (Ω on B, C, C_A on A). That is why their order can be exchanged here.

So the important content is not a new operation. It is a change of description:

measuring X_{a_v} after cluster preparation	$\longleftrightarrow$	measuring the cluster stabilizer $X_{a_v} X_{d_0^a \mathbf{v}}^A$ before it.

These are exact identities for unnormalized post-measurement states; the factors of $1/2$ have been retained. The crucial feature is the factor $X_v^a X_{d_0^a v}^A$: measuring an a qubit is tied to an X operation on the adjacent A-gauge qubits.

Expanding the product in Eq. (205) means choosing either the I term or the $X_v^a X_{d_0^a v}^A$ term at every vertex. A choice is a binary 0-cochain $\phi \in A^0$. Concretely, this means that ϕ is a binary label assigned to every temporary-ancilla vertex:

$$\phi : V_A \longrightarrow \mathbb{F}_2 = \{0, 1\}.$$

If there are N temporary a-qubits, ϕ is simply an N-bit string. In this equation it is not a new quantum state, a measurement outcome, or a physical excitation. It is an accounting device for one term in the algebraic expansion. At each vertex v, it records which of the two terms was selected:

$\phi(v)$	selected term
0	I
1	$x_v X_v^a X_{d_0^a v}^A$.

For example, for three vertices, $\phi = (1, 0, 1)$ means: select the non-identity term at vertices 1 and 3, and select I at vertex 2. The corresponding contribution is $x_1 x_3 K_1 K_3$, where $K_v = X_v^a X_{d_0^a v}^A$. Therefore

$$\prod_v \left(I + x_v X_v^a X_{d_0^a v}^A \right) = \sum_{\phi \in A^0} x_\phi \, X_\phi^a X_{d_0^a \phi}^A, \qquad x_\phi := \prod_v x_v^{\phi(v)}. \tag{206}$$

Explanation of the equation above

Start with exactly your expression:

$$\prod_{v \in V_A} \left(I + x_v X_v^a X_{d_0^a \mathbf{v}}^A \right).$$

Here x_v is a number, $+1$ or -1, while the uppercase X's are operators. We will expand the product first, then simplify the operators.

1. Write out two brackets to see what the product does.

For two vertices,

$$\left(I + x_1 X_1^a X_{d_0^a \mathbf{1}}^A \right) \left(I + x_2 X_2^a X_{d_0^a \mathbf{2}}^A \right).$$

Distributing gives four terms:

$$\begin{aligned} & I \\ & + x_1 X_1^a X^A_{d_0^a \mathbf{1}} \\ & + x_2 X_2^a X^A_{d_0^a \mathbf{2}} \\ & + x_1 x_2 X_1^a X^A_{d_0^a \mathbf{1}} X_2^a X^A_{d_0^a \mathbf{2}}. \end{aligned}$$

Each term comes from choosing **one contribution from each bracket**: either I, or the other contribution.

2. Introduce a bit to record each choice.

At every vertex v, define

$$\phi(v) = \begin{cases} 0 & \text{if we choose } I, \\ 1 & \text{if we choose } x_v X_v^a X^A_{d_0^a \mathbf{v}}. \end{cases}$$

For one specified set of choices ϕ, the resulting term is therefore

$$\prod_{v \in V_A} \left(x_v X_v^a X^A_{d_0^a \mathbf{v}} \right)^{\phi(v)}.$$

An exponent zero gives I; an exponent one keeps the contribution.

To include **every possible choice**, sum over all bit assignments:

$$\boxed{\prod_{v \in V_A} \left(I + x_v X_v^a X^A_{d_0^a \mathbf{v}} \right) = \sum_{\phi : V_A \to \{0,1\}} \prod_{v \in V_A} \left(x_v X_v^a X^A_{d_0^a \mathbf{v}} \right)^{\phi(v)}}.$$

This is where ϕ enters. "Zero cochain" simply means a bit assignment to vertices.

3. Separate each term into signs, temporary-qubit operators, and surviving-qubit operators.

All the operators here are products of Pauli X's, so they commute. We can regroup:

$$\boxed{\sum_{\phi} \left[\prod_{v \in V_A} x_v^{\phi(v)} \right] \left[\prod_{v \in V_A} (X_v^a)^{\phi(v)} \right] \left[\prod_{v \in V_A} \left(X^A_{d_0^a \mathbf{v}} \right)^{\phi(v)} \right]}.$$

At this stage, **all three products run over vertices**, independently.

The first two brackets have the abbreviations

$$x_\phi := \prod_{v \in V_A} x_v^{\phi(v)}, \qquad X_\phi^a := \prod_{v \in V_A} (X_v^a)^{\phi(v)}.$$

4. Unpack the third bracket.

For an ordinary graph,

$$X^A_{d_0^a \mathbf{v}} = \prod_{e \ni v} X_{A_e}.$$

It applies X to every edge qubit incident on vertex v.

Consequently, the third bracket is

$$\prod_{v\in V_A}\left(\prod_{e\ni v} X_{A_e}\right)^{\phi(v)}.$$

Consider one edge e with endpoints u, w. It receives one X_{A_e} from u if $\phi(u)=1$, and another from w if $\phi(w)=1$. Its total factor is

$$(X_{A_e})^{\phi(u)}(X_{A_e})^{\phi(w)} = (X_{A_e})^{\phi(u)+\phi(w)}.$$

Because $X_{A_e}^2 = I$, only the parity matters. Thus

$$\prod_{v\in V_A}\left(\prod_{e\ni v} X_{A_e}\right)^{\phi(v)} = \prod_{e\in E_A}(X_{A_e})^{[\phi(u_e)+\phi(w_e)] \bmod 2}.$$

The edge bit in this exponent is precisely

$$(d_0^a\phi)(e) = [\phi(u_e)+\phi(w_e)] \bmod 2.$$

So we abbreviate the edge product as

$$X^A_{d_0^a\phi} := \prod_{e\in E_A}(X_{A_e})^{(d_0^a\phi)(e)}.$$

5. Put the three abbreviations together.

We obtain equation (206):

$$\boxed{\prod_{v\in V_A}\left(I + x_v X_v^a X^A_{d_0^a \mathbf{v}}\right) = \sum_{\phi\in A^0} x_\phi\, X_\phi^a\, X^A_{d_0^a\phi}.}$$

Each summand represents one choice from every original bracket. Its three factors record the resulting sign, the selected vertex X's, and the surviving edge X's after cancellations.

The subscript on x_ϕ denotes a product of signs, whereas the entries of ϕ are exponents:

$$x_\phi := \prod_v x_v^{\phi(v)},$$

not $\prod_v x_{\phi(v)}$. The exponent is essential. Since $\phi(v)$ is either 0 or 1,

$$x_v^{\phi(v)} = \begin{cases} 1, & \phi(v)=0,\\ x_v, & \phi(v)=1.\end{cases}$$

Thus x_ϕ is simply the product of the recorded signs x_v over those vertices selected by ϕ.

This is the ordinary binomial expansion, with $K_v = X_v^a X^A_{d_0^a\mathbf{v}}$ and $X_\phi^a = \prod_v (X_v^a)^{\phi(v)}$. The two-vertex case displays every term explicitly.

For two vertices v_1, v_2, the left-hand side is

$$(I + x_1K_1)(I + x_2K_2) = I + x_1K_1 + x_2K_2 + x_1x_2K_1K_2.$$

The four terms correspond exactly to the four binary cochains:

$\phi = (\phi(v_1), \phi(v_2))$	x_ϕ	Pauli operator
$(0,0)$	1	I
$(1,0)$	x_1	K_1
$(0,1)$	x_2	K_2
$(1,1)$	x_1x_2	K_1K_2.

Finally, the compact form of the Pauli operator follows because d_0^a is linear over $\mathbb{F}_2$:

$$d_0^a\phi = \sum_v \phi(v)\, d_0^a\mathbf{v}.$$

Therefore, selecting K_v at every vertex with $\phi(v) = 1$ gives

$$\prod_{v:\phi(v)=1} K_v = \left[\prod_v (X_v^a)^{\phi(v)}\right] \left[\prod_v \left(X^A_{d_0^a\mathbf{v}}\right)^{\phi(v)}\right] = X_\phi^a X^A_{d_0^a\phi}.$$

The missing factors of $1/2$ are intentional: Eq. (206) expands the numerators of the measurement projectors. The full projector has an overall factor $2^{-|V_A|}$, which is irrelevant once the state is normalized.

This is not a new physical superposition of measurement records. It is an algebraic expansion of the fixed-record projector. It is useful because the co-boundary $d_0^a\phi$ will be tested by the later Z_A measurements.

2.20.5 Step 2: undo the cluster circuit, measure the edge qubits, and correct the offset

Step 1 measured the temporary vertex qubits a_v in the X basis, giving the recorded signs $x_v = \pm 1$. Step 2 performs three operations:

1. Apply $C_A^\dagger$, which undoes the cluster circuit.
2. Measure each surviving edge qubit A_e in the Z basis.
3. Use the edge measurement record to correct an offset on the temporary vertex qubits.

Let us first write the state between the first two operations, before discussing what the measurement does.

The state just before the Z measurements

Set

$$|\Phi_0\rangle = |\Psi\rangle_{BC}|0\rangle_A|+\rangle_a, \qquad N = |V_A| = \dim A^0.$$

Using Step 1's expansion, undoing the cluster circuit gives

$$\boxed{C_A^\dagger \Pi_x \Omega C_A |\Phi_0\rangle = 2^{-N} \sum_{\phi \in A^0} x_\phi X_\phi^a \Omega \Big(|\Psi\rangle_{BC} |d_0^a \phi\rangle_A |+\rangle_a \Big).}$$

The important new feature is the middle ket, $|d_0^a \phi\rangle_A$. It follows directly from

$$X^A_{d_0^a \phi} |0\rangle_A = |d_0^a \phi\rangle_A :$$

an X flips an edge qubit from $|0\rangle$ to $|1\rangle$ wherever the corresponding edge bit is one. The enrichment circuit Ω does not act on A, so it leaves this edge string unchanged.

Thus, in each term of the sum, the A register contains the edge-bit string $d_0^a \phi$. Different ϕ's can give the same edge string. The labels ϕ still index terms in the algebraic expansion; they are not additional recorded measurement outcomes.

What Z is, and what the record z means

Uppercase Z_{A_e} is the Pauli operator measured on edge qubit A_e:

$$Z|0\rangle = +|0\rangle, \qquad Z|1\rangle = -|1\rangle.$$

The measurement therefore returns a sign, $+1$ or -1, on each edge. We encode those signs using lowercase bits:

$$z(e) = \begin{cases} 0, & \text{if the measured sign is } +1, \\ 1, & \text{if the measured sign is } -1. \end{cases}$$

The full record z is simply the list of these bits over all edges. Calling it a **1-cochain** means that it assigns a bit to each edge. Both z and $d_0^a \phi$ belong to A^1, so we can compare their bits edge by edge. In homological gauging, the same statement applies to the surviving qubits indexed by A_1; the ordinary-graph interpretation is only needed for the example below.

For a specified record z, the measurement projector is

$$P_z^A = \prod_{e \in E_A} \frac{I + (-1)^{z(e)} Z_{A_e}}{2} = |z\rangle_A \langle z|_A,$$

with the identity on the other registers understood.

How the measurement selects terms in the sum

Apply this projector to the edge ket in one term:

$$P_z^A |d_0^a \phi\rangle_A = \begin{cases} |z\rangle_A, & d_0^a \phi = z, \\ 0, & d_0^a \phi \neq z. \end{cases}$$

So the measurement keeps exactly those terms whose edge string agrees with the observed record, on every edge. This gives the overall equation for the measurement part of Step 2:

$$\boxed{\begin{aligned} &P_z^A C_A^\dagger \Pi_x \Omega C_A |\Phi_0\rangle \\ &\qquad = 2^{-N} \sum_{\substack{\phi \in A^0 \\ d_0^a \phi = z}} x_\phi X_\phi^a \Omega \Big(|\Psi\rangle_{BC} |z\rangle_A |+\rangle_a \Big). \end{aligned}}$$

The change is in the sum: before measurement, we sum over every ϕ; after obtaining record z, we sum only over those satisfying $d_0^a \phi = z$. These are unnormalized states. For a normalized initial state, the squared norm of the displayed state gives the joint probability of the records x, z.

Every record that actually occurs in the ideal protocol is accepted. The condition in the sum describes the measurement's effect; it is not an extra postselection requirement. In particular, any occurring record belongs to $\operatorname{im} d_0^a$. On an ordinary graph, its bits sum to zero modulo two around any loop.

Why we introduce ϕ_0, and what the coset means

The edge measurement does not necessarily determine a unique vertex assignment. Choose one compatible assignment and call it ϕ_0:

$$d_0^a \phi_0 = z. \tag{207}$$

For any other compatible assignment ϕ, linearity over $\mathbb{F}_2$ gives

$$d_0^a \phi = d_0^a \phi_0 \quad \Longleftrightarrow \quad d_0^a (\phi + \phi_0) = 0. \tag{208}$$

Why zero appears here. The measurement does not require the surviving ϕ's to have zero coboundary: their coboundary is the recorded z. What must have zero coboundary is the difference $\phi + \phi_0$, because it cannot change any recorded edge bit. Indeed, $d_0^a(\phi + \phi_0) = z + z = 0$.

Writing $\mu = \phi + \phi_0$, every compatible assignment therefore has exactly the form

$$\phi = \phi_0 + \mu, \qquad \mu \in \ker d_0^a. \tag{209}$$

Here $\ker d_0^a$ is the set of vertex assignments with zero coboundary. The set $\phi_0 + \ker d_0^a$ is called a **coset**: it contains all vertex assignments compatible with the edge record that was measured. Unless $z = 0$, the compatible assignments do not themselves have zero coboundary; their differences do.

For example, on the three-vertex line, the record $z = (1, 1)$ requires

$$\phi_1 + \phi_2 = 1, \qquad \phi_2 + \phi_3 = 1 \quad \text{modulo two.}$$

There are two solutions, $(1, 0, 1)$ and $(0, 1, 0)$. Their difference, computed modulo two, is $(1, 1, 1)$, whose coboundary is zero. Thus, choosing $\phi_0 = (1, 0, 1)$, the two surviving labels are $\phi_0 + (0, 0, 0)$ and $\phi_0 + (1, 1, 1)$.

Correct the offset before Step 3

Substitute $\phi = \phi_0 + \mu$ in the surviving sum. The sign and operator products factor as

$$x_{\phi_0 + \mu} = x_{\phi_0} x_\mu, \qquad X^a_{\phi_0 + \mu} = X^a_{\phi_0} X^a_\mu.$$

Consequently,

$$\sum_{\phi:\, d_0^a \phi = z} x_\phi X_\phi^a = x_{\phi_0} X_{\phi_0}^a \sum_{\mu \in \ker d_0^a} x_\mu X_\mu^a.$$

We now apply $X_{\phi_0}^a$ to the temporary vertex qubits: apply an X at each vertex where $\phi_0(v) = 1$. This correction is chosen from the measured record, a procedure called **feedforward**. Since $(X_{\phi_0}^a)^2 = I$, it cancels the offset operator. The corrected state is

$$\boxed{2^{-N} x_{\phi_0} \sum_{\mu \in \ker d_0^a} x_\mu X_\mu^a \Omega \Big(|\Psi\rangle_{BC} |z\rangle_A |+\rangle_a \Big).} \tag{210}$$

The scalar x_{ϕ_0} is only a global sign. The A register remains in the measured state $|z\rangle_A$; it has not been reset to $|0\rangle_A$. The remaining sum is precisely what Step 3 needs before undoing Ω.

Why remove this offset? Every remaining operator is now labelled by $\mu \in \ker d_0^a$, the domain on which enrichment is guaranteed to produce the logical gates U_μ. Before correction, the labels were $\phi_0 + \mu$, which need not lie in that domain. This does not mean that conjugating those operators is impossible. In the CCZ construction, conjugation still attaches a physical matter circuit, but it can leave an extra, known circuit that is not guaranteed to preserve the original codespace. The correction cancels that byproduct before undoing enrichment. It does not create a logical projector from nothing; it removes the extra operation accompanying it. The worked conjugation below makes this physical-versus-logical distinction explicit.

The signs x_μ remain because they encode the logical measurement outcomes. We correct the offset ϕ_0, while retaining those signs for the logical projector revealed in Step 3.

2.20.6 Step 3: remove the ancilla and reveal the logical projector

Undo the enrichment circuit and use its defining symmetry-enrichment identity,

$$\Omega^\dagger X_\mu^a \Omega = X_\mu^a U_\mu. \tag{211}$$

What the kernel condition guarantees here. The μ's in this sum are zero-coboundary patterns because Step 2 corrected the offset. Equation (121) requires enrichment to associate a logical code symmetry U_μ with every such ancillary symmetry. The word *logical* is essential: a physical matter operator can exist without preserving the codespace. Equation (211) uses the guaranteed logical operators. It does not assert that every non-cocycle fails to produce a logical operator.

After conjugating through Ω, the remaining ancilla operators act on the original $|+\rangle_a$ factors. Hence each X_μ^a acts as $+1$, and the resulting state factors into an ancilla product state and a matter state. Choose a basis $\{\mu_i\}$ of the cocycle space. The output on the input codes is, up to normalization,

$$|\Psi\rangle_{BC} \longmapsto \prod_i \frac{I + x_{\mu_i} U_{\mu_i}}{2} |\Psi\rangle_{BC}. \tag{212}$$

2.20.6.1 What $\mu \in \ker d_0^a$ means here The kernel belongs to the map d_0^a: it is the vector space of inputs that this map sends to zero. Saying that μ belongs to it means

$$\mu \in \ker d_0^a \qquad \Longleftrightarrow \qquad d_0^a \mu = 0.$$

Why such a pattern is an ancillary symmetry. Recall the cluster stabilizer $K_e^A = Z_{A_e} Z^a_{\partial_1^a[e]}$. Conjugating it by X_μ^a changes its sign by

$$X_\mu^a K_e^A X_\mu^a = (-1)^{\mu(\partial_1^a[e])} K_e^A = (-1)^{(d_0^a \mu)(e)} K_e^A.$$

Thus zero coboundary means no edge stabilizer changes sign. The vertex stabilizers also commute with X_μ^a, because their temporary-qubit factors are X's. This is the concrete reason the kernel labels ancillary symmetries: they leave all the cluster constraints unchanged. Enrichment is designed to attach logical matter symmetries to these patterns.

For an ordinary ancilla graph, on an edge $e = \langle v, w \rangle$, this condition says

$$(d_0^a \mu)(e) = \mu(v) + \mu(w) = 0 \pmod 2.$$

Thus neighbouring vertices must have equal μ-bits. On a connected graph, μ must be constant everywhere, leaving just two possibilities:

```
mu = 0:       0 ---- 0 ---- 0 ---- ...
mu = mu_all:  1 ---- 1 ---- 1 ---- ...
```

For the three-vertex line used above,

$$d_0^a(m_1, m_2, m_3) = (m_1 + m_2,\ m_2 + m_3),$$

so $d_0^a \mu = 0$ requires $m_1 = m_2 = m_3$. Hence

$$\ker d_0^a = \{(0,0,0), (1,1,1)\}.$$

Why do precisely these differences remain after the offset correction in Step 2? Suppose one vertex cochain ϕ_0 explains the final edge record:

$$z = d_0^a \phi_0.$$

For any $\mu \in \ker d_0^a$,

$$d_0^a(\phi_0 + \mu) = d_0^a \phi_0 + d_0^a \mu = z.$$

Therefore ϕ_0 and $\phi_0 + \mu$ create exactly the same bit pattern on every surviving A-edge qubit. The final Z_A measurements cannot distinguish them. In a connected graph, adding μ_{all} simply flips every vertex bit; each edge still has the same endpoint *difference*. The kernel is therefore the residual global ambiguity left after the edge measurements have removed all local information.

Step 3 translates that ambiguity into a logical operation. The enrichment circuit was chosen so that every such ancilla symmetry obeys

$$\Omega^\dagger X^a_\mu \Omega = X^a_\mu U_\mu.$$

Moving X^a_μ backward through enrichment produces the same ancilla operator times the logical Clifford symmetry U_μ on the B, C code blocks. The a qubits are in $|+\rangle_a$, so X^a_μ acts as $+1$ and disappears. What remains is

$$\sum_{\mu \in \ker d^a_0} x_\mu U_\mu.$$

For a connected graph, the kernel has only 0 and μ_{all}. Dividing this operator sum by two gives the projector

$$\frac{I + x_{\text{all}} U_{\text{all}}}{2}, \qquad x_{\text{all}} = x_{\mu_{\text{all}}} = \prod_v x_v.$$

Although every temporary qubit was measured separately, only this global parity of their outcomes remains after ungauging. It is exactly the outcome of measuring the logical symmetry U_{all}.

This is Eq. (201), now derived. In the homological construction, each independent cocycle can measure a related, potentially addressable logical CZ. In the connected-graph construction, there is one independent cocycle and Eq. (212) is one logical-gate measurement.

2.20.6.2 Why a sum over cocycles becomes a product of projectors Let $k = \dim \ker d^a_0$. Choosing a basis means that every cocycle has one unique expression $\mu = \sum_{i=1}^k t_i \mu_i$ with bits t_i. In this CZ construction the symmetry operators are commuting involutions and obey $U_{\mu+\nu} = U_\mu U_\nu$; the outcome signs obey the same multiplication rule, since $x_v^2 = 1$. Therefore

$$\sum_{\mu \in \ker d^a_0} x_\mu U_\mu = \prod_{i=1}^k (I + x_{\mu_i} U_{\mu_i}).$$

For two basis cocycles this reads

$$I + x_{\mu_1} U_{\mu_1} + x_{\mu_2} U_{\mu_2} + x_{\mu_1} x_{\mu_2} U_{\mu_1} U_{\mu_2} = (I + x_{\mu_1} U_{\mu_1})(I + x_{\mu_2} U_{\mu_2}).$$

It is the same binomial expansion as Step 1, now restricted to the undistinguished symmetry labels. Each factor divided by two is a projector:

$$\left(\frac{I + xU}{2}\right)^2 = \frac{I + 2xU + U^2}{4} = \frac{I + xU}{2}, \qquad U \frac{I + xU}{2} = x \frac{I + xU}{2}.$$

The first identity says repeating the projection has no further effect; the second says its surviving states have eigenvalue x. Opposite-sign projectors are orthogonal, and their sum is I. The simultaneous projector is their product because these symmetries commute. Independent cocycles need not give independent logical gates: a symmetry acting as identity on the codespace has a fixed positive outcome.

For a normalized input, the probability of a logical record is $\|P_x|\Psi\rangle\|^2$, and the conditional output is $P_x|\Psi\rangle/\|P_x|\Psi\rangle\|$. The projector's factor of two and the state's normalization are different things. A general Clifford gate need not satisfy $U^2 = I$; the binary projector formula here applies to the involutions in this construction.

Proof of Step 3

There are **two separate steps** hidden in the compact expression:

1. Undoing Ω turns the ancilla operators X_μ^a into matter operators U_μ.
2. A separate algebraic identity rewrites the **sum over all** μ as a **product over a basis of** μ**'s**.

The sum-to-product conversion does not come from Ω. Let us derive both steps.

1. Start with the state at the end of Step 2

After correcting the offset, the state is

$$2^{-N} x_{\phi_0} \sum_{\mu \in \ker d_0^a} x_\mu X_\mu^a \Omega \Big(|\Psi\rangle_{BC} |z\rangle_A |+\rangle_a \Big).$$

For the moment, suppress the measured A register and the common scalar $2^{-N} x_{\phi_0}$. Neither affects the following algebra. What remains is

$$\boxed{\sum_{\mu \in \ker d_0^a} x_\mu X_\mu^a \Omega (|\Psi\rangle_{BC} |+\rangle_a).}$$

Here $|+\rangle_a$ means the product of $|+\rangle$ states on all temporary qubits.

2. Apply $\Omega^\dagger$ to undo enrichment

Applying $\Omega^\dagger$ to the whole state gives

$$\sum_{\mu \in \ker d_0^a} x_\mu \underbrace{\Omega^\dagger X_\mu^a \Omega}_{\text{conjugated operator}} (|\Psi\rangle_{BC} |+\rangle_a).$$

For each zero-coboundary pattern μ, the enrichment identity is

$$\Omega^\dagger X_\mu^a \Omega = X_\mu^a U_\mu.$$

Why the zero-coboundary qualification matters. It identifies the patterns for which the attached matter operator is guaranteed to be a logical gate. In the CCZ construction, let V_v be the product of matter CZ gates controlled by a_v. For an arbitrary pattern ϕ, including one with nonzero coboundary, the one-gate CCZ identity gives

$$\Omega^\dagger X_\phi^a \Omega = X_\phi^a V_\phi, \qquad V_\phi = \prod_{v:\, \phi(v)=1} V_v.$$

This identity holds for all patterns. What is special about $\mu \in \ker d_0^a$ is the requirement that $V_\mu = U_\mu$ implements a logical symmetry of the input codes. For a general ϕ, V_ϕ is only a physical CZ circuit

and need not preserve those codes. Zero coboundary alone does not prove that a gate is logical; the construction of Ω must satisfy the enrichment condition.

For example, the uncorrected three-vertex sum, after undoing enrichment, would give the matter state

$$[x_1x_3V_{101} + x_2V_{010}]|\Psi\rangle = x_1x_3V_{101}[I + (x_1x_2x_3)U_{111}]|\Psi\rangle.$$

Here $V_{010} = V_{101}V_{111} = V_{101}U_{111}$, since these CZ products commute and square to identity. The desired logical projector is already inside the brackets, but an extra physical byproduct V_{101} remains. Correcting X^a_{101} before undoing enrichment removes that byproduct. Equivalently, in this CCZ construction, one could cancel V_{101} on the matter after undoing enrichment. Neither approach requires the measurement record itself to be zero.

Thus the corrected state becomes

$$\sum_{\mu\in\ker d^a_0} x_\mu X^a_\mu U_\mu(|\Psi\rangle_{BC}|+\rangle_a).$$

The two operators act on different registers:

- U_μ acts on the matter BC.
- X^a_μ acts on the temporary qubits a.

Since $X|+\rangle = |+\rangle$, any product of those ancilla X's leaves $|+\rangle_a$ unchanged:

$$X^a_\mu|+\rangle_a = |+\rangle_a.$$

This last equality holds for every X pattern, not just a cocycle. The kernel condition was needed to identify the attached matter operation as a guaranteed logical symmetry, not to make the ancilla X's act trivially on $|+\rangle_a$.

Consequently,

$$\boxed{\left[\sum_{\mu\in\ker d^a_0} x_\mu U_\mu\right]|\Psi\rangle_{BC}\otimes|+\rangle_a.}$$

This is everything that undoing Ω has done. We still have a sum.

3. Write every μ using a basis

Let the kernel have dimension k, and choose a basis

$$\mu_1, \dots, \mu_k.$$

Every element of the kernel can be written uniquely as

$$\mu = b_1\mu_1 + \cdots + b_k\mu_k, \qquad b_i \in \{0, 1\}.$$

So summing over μ means summing over all 2^k choices of the bits $b_1, \dots, b_k$.

For example, with two basis elements, the kernel consists of

$$0, \qquad \mu_1, \qquad \mu_2, \qquad \mu_1 + \mu_2.$$

We now need to understand the sign and operator associated with each combination.

4. Both the signs and the operators multiply when the labels add

Recall

$$x_\mu = \prod_v x_v^{\mu(v)}, \qquad x_v = \pm 1.$$

Because $x_v^2 = 1$, adding labels modulo two gives

$$x_{\mu+\nu} = x_\mu x_\nu.$$

Likewise,

$$U_{\mu+\nu} = U_\mu U_\nu.$$

This operator identity follows from the enrichment identity. Indeed, the ancilla operators satisfy

$$X^a_{\mu+\nu} = X^a_\mu X^a_\nu.$$

Conjugating both sides by Ω gives

$$\begin{aligned} X^a_{\mu+\nu} U_{\mu+\nu} &= \Omega^\dagger X^a_\mu X^a_\nu \Omega \\ &= (\Omega^\dagger X^a_\mu \Omega)(\Omega^\dagger X^a_\nu \Omega) \\ &= (X^a_\mu U_\mu)(X^a_\nu U_\nu) \\ &= X^a_{\mu+\nu} U_\mu U_\nu. \end{aligned}$$

Cancel the common ancilla operator to obtain the claimed identity. The kernel is closed under addition: $d_0^a(\mu + \nu) = d_0^a \mu + d_0^a \nu = 0$. Thus all three labels in this argument remain in the domain where the logical enrichment identity is guaranteed. This is why we can use that identity consistently to turn the sum over the kernel into a product of logical projectors.

Therefore, if $\mu = \sum_i b_i \mu_i$,

$$x_\mu U_\mu = \prod_{i=1}^{k} (x_{\mu_i} U_{\mu_i})^{b_i}.$$

5. Expand a product and recognize the sum

First take $k = 2$. Our sum is

$$\sum_{\mu\in\ker d_0^a} x_\mu U_\mu = I + x_{\mu_1}U_{\mu_1} + x_{\mu_2}U_{\mu_2} + x_{\mu_1}x_{\mu_2}U_{\mu_1}U_{\mu_2}.$$

But these are exactly the four terms obtained by expanding

$$\boxed{(I + x_{\mu_1}U_{\mu_1})(I + x_{\mu_2}U_{\mu_2}).}$$

For arbitrary k, the same reasoning gives

$$\boxed{\sum_{\mu\in\ker d_0^a} x_\mu U_\mu = \prod_{i=1}^{k}(I + x_{\mu_i}U_{\mu_i}).}$$

Each bracket contributes either I or $x_{\mu_i}U_{\mu_i}$. A choice from all brackets corresponds to one basis-bit assignment $(b_1, \ldots, b_k)$, hence to exactly one μ.

This is **the same binomial expansion used in Step 1, now read in reverse**.

6. Where do the divisions by two come from?

The exact identity above contains no divisions by two. To write each factor as a measurement projector, use

$$I + x_{\mu_i}U_{\mu_i} = 2\left(\frac{I + x_{\mu_i}U_{\mu_i}}{2}\right).$$

There are k factors, so

$$\boxed{\sum_{\mu\in\ker d_0^a} x_\mu U_\mu = 2^k \prod_{i=1}^{k} \frac{I + x_{\mu_i}U_{\mu_i}}{2}.}$$

Restoring the scalar and registers suppressed at the start, the exact unnormalized output is

$$\boxed{2^{k-N} x_{\phi_0} \left[\prod_{i=1}^{k} \frac{I + x_{\mu_i}U_{\mu_i}}{2}\right] |\Psi\rangle_{BC} \otimes |z\rangle_A \otimes |+\rangle_a.}$$

Equation (212) omits the common scalar because it describes the conditional matter state up to normalization.

7. Why these factors are projectors

The U_{μ_i}'s commute and satisfy $U_{\mu_i}^2 = I$. A factor

$$\frac{I + x_{\mu_i}U_{\mu_i}}{2}$$

keeps the component with U_{μ_i} eigenvalue x_{μ_i}, and removes the component with the opposite eigenvalue. Their product imposes all those conditions together.

For the connected three-vertex example, the kernel has only two elements, 000 and 111, so $k = 1$. The entire calculation reduces to

$$\sum_{\mu\in\{000,111\}} x_\mu U_\mu = I + (x_1x_2x_3)U_{111}$$
$$= 2\,\frac{I + (x_1x_2x_3)U_{111}}{2}.$$

There, the "sum becoming a product" is simply a two-term sum becoming a product with **one factor**.

2.20.7 The complete protocol with the ancillary registers retained

The following calculation keeps the normalization factors and the measured A register, so that every equality has the same input and output spaces. Set $N = \dim A^0$, $k = \dim\ker d_0^a$, and define

$$|\Phi_0\rangle = |\Psi\rangle_{BC}|0\rangle_A|+\rangle_a, \qquad P_z^A = \prod_e \frac{I + (-1)^{z(e)}Z_{A_e}}{2}.$$

For the fixed records x, z, the unnormalized state after undoing the cluster entangler and measuring A is

$$P_z^A C_A^\dagger \Pi_x \Omega C_A|\Phi_0\rangle = 2^{-N} \sum_{\phi:\, d_0^a\phi = z} x_\phi X_\phi^a \Omega(|\Psi\rangle_{BC}|z\rangle_A|+\rangle_a)$$
$$= 2^{-N} x_{\phi_0} X_{\phi_0}^a \sum_{\mu\in\ker d_0^a} x_\mu X_\mu^a \Omega(|\Psi\rangle_{BC}|z\rangle_A|+\rangle_a).$$

The first line is Step 1's expansion followed by Step 2's selection rule. The second substitutes $\phi = \phi_0 + \mu$. Applying the correction $X_{\phi_0}^a$ and then $\Omega^\dagger$ gives exactly

$$2^{k-N} x_{\phi_0} \left[\prod_{i=1}^k \frac{I + x_{\mu_i}U_{\mu_i}}{2}\right] |\Psi\rangle_{BC} \otimes |z\rangle_A \otimes |+\rangle_a.$$

The factor x_{ϕ_0} is a global sign. The scalar 2^{k-N} contributes to the probability of the *full* ancillary record; it does not change the normalized conditional matter state. The A qubits are left in the measured string $|z\rangle$, not automatically $|0\rangle$; they can be discarded or reset by X_z^A. The a qubits have returned to $|+\rangle$ and can also be discarded.

The physical roles of the three steps are now explicit. The temporary measurements introduce the sum over cochains. The gauge-qubit measurements identify their coboundaries, leaving a coset of the kernel. Correcting its offset and undoing enrichment converts the remaining symmetry sum into logical projectors. In a connected graph, the only logical measurement bit left is $x_{\text{all}} = \prod_v x_v$.

2.20.8 Worked example: all three steps on a three-vertex line with encoded matter

We now carry out the whole calculation on the line

```
v1 ---- e12 ---- v2 ---- e23 ---- v3
```

There are three temporary vertex qubits a_1, a_2, a_3 and two surviving edge qubits A_{12}, A_{23}. To make the matter operation explicit, we attach two simple encoded blocks B, C. This is a finite toy example of the measurement algebra, not a three-vertex realization of the ALP topological code or its anyons.

1. Specify the codes and the operation we want to measure

Each block is a three-qubit repetition code:

$$|\overline{0}\rangle_B = |000\rangle_B, \qquad |\overline{1}\rangle_B = |111\rangle_B,$$

and likewise for C. The checks on B are $Z_{B_1}Z_{B_2}$ and $Z_{B_2}Z_{B_3}$; their common $+1$ eigenspace is spanned by these two states. The same checks define C. This code protects against a single bit flip, but is not a full single-error-correcting quantum code.

Attach the matter pair (B_i, C_i) to vertex v_i. Define

$$V_i = \mathsf{CZ}_{B_i,C_i}, \qquad U = V_1V_2V_3.$$

On encoded basis states, for $b, c \in \{0, 1\}$, each pair contains the bits (b, c). Therefore

$$U|\bar{b}\rangle_B|\bar{c}\rangle_C = (-1)^{3bc}|\bar{b}\rangle_B|\bar{c}\rangle_C = (-1)^{bc}|\bar{b}\rangle_B|\bar{c}\rangle_C.$$

Thus U implements a logical $\overline{\mathsf{CZ}}$. The protocol will **measure** this logical operator, producing its $+1$ or -1 eigenspace projection; it does not simply apply the unitary U.

Keep $|\Psi\rangle_{BC}$ arbitrary in the two-block codespace until we evaluate a particular input at the end.

2. Specify the cluster and enrichment circuits

Initialize the edge qubits in $|00\rangle_A$ and the temporary qubits in $|++\rangle_a$:

$$|\Phi_0\rangle = |\Psi\rangle_{BC}|00\rangle_A|++\rangle_a.$$

The cluster circuit has four CNOTs:

$$C_A = \mathsf{CNOT}_{a_1\to A_{12}}\mathsf{CNOT}_{a_2\to A_{12}}\mathsf{CNOT}_{a_2\to A_{23}}\mathsf{CNOT}_{a_3\to A_{23}}.$$

These gates commute. On a temporary computational-basis string $r = (r_1, r_2, r_3)$, they write the endpoint differences onto the edges:

$$C_A|00\rangle_A|r_1r_2r_3\rangle_a = |r_1 + r_2,\ r_2 + r_3\rangle_A|r_1r_2r_3\rangle_a,$$

where the additions are modulo two.

The enrichment circuit is

$$\Omega = \prod_{i=1}^{3} \mathsf{CCZ}_{a_i,B_i,C_i}.$$

For a temporary bit $r_i = 0$, its CCZ does nothing to the matter pair; for $r_i = 1$, it applies V_i. Explicitly,

$$\Omega(|\Psi\rangle_{BC}|r\rangle_a) = (V_1^{r_1}V_2^{r_2}V_3^{r_3}|\Psi\rangle_{BC}) \otimes |r\rangle_a.$$

Both circuits use the temporary qubits as controls and act on different targets, so they commute. The prepared state is $\Omega C_A|\Phi_0\rangle$.

Translation to chains and cochains. A vertex chain is a binary selection of vertices, and an edge chain is a binary selection of edges. Their spaces are $A_0 = \mathbb{F}_2^3$ and $A_1 = \mathbb{F}_2^2$. The boundary sends an edge to its endpoints:

$$\partial_1^a e_{12} = v_1 + v_2, \qquad \partial_1^a e_{23} = v_2 + v_3, \qquad \partial_1^a = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

A vertex cochain ϕ assigns bits to vertices. Its coboundary assigns bits to edges by evaluating ϕ on each edge's boundary:

$$(d_0^a\phi)(e) = \phi(\partial_1^a e), \qquad d_0^a = (\partial_1^a)^{\mathsf{T}} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Thus $d_0^a\phi = (\phi_1 + \phi_2, \phi_2 + \phi_3)$. This is exactly the endpoint-difference calculation implemented by the cluster CNOTs. A computational-basis string r can be identified with a vertex cochain in these coordinates; below, ϕ instead labels terms in a projector expansion, not an additional measured string.

3. Step 1: measure the temporary qubits and expand the projectors

Suppose the three X measurements give

$$(x_1, x_2, x_3) = (+1, -1, +1).$$

The projector describing this record is

$$\Pi_x = \frac{I + X_{a_1}}{2}\frac{I - X_{a_2}}{2}\frac{I + X_{a_3}}{2}.$$

Pulling it through C_A replaces each X_{a_i} by

$$K_1 = X_{a_1}X_{A_{12}}, \qquad K_2 = X_{a_2}X_{A_{12}}X_{A_{23}}, \qquad K_3 = X_{a_3}X_{A_{23}}.$$

The unnormalized post-measurement state is therefore

$$\Pi_x\Omega C_A|\Phi_0\rangle = \frac{1}{8}C_A(I + K_1)(I - K_2)(I + K_3)\Omega|\Phi_0\rangle.$$

Distributing the brackets gives all eight terms:

$$\frac{1}{8}C_A\left[I + K_1 - K_2 + K_3 - K_1K_2 + K_1K_3 - K_2K_3 - K_1K_2K_3\right]\Omega|\Phi_0\rangle.$$

The following table spells out their cochain labels and edge strings. Here X_i in the third column means X_{a_i}.

Vertex cochain ϕ	Sign $x_\phi = (-1)^{\phi_2}$	Temporary operator X^a_ϕ	Edge string $d^a_0\phi$
000	$+1$	I	00
100	$+1$	X_1	10
010	-1	X_2	11
001	$+1$	X_3	01
110	-1	X_1X_2	01
101	$+1$	X_1X_3	11
011	-1	X_2X_3	10
111	-1	$X_1X_2X_3$	00

For instance, K_1K_2 contains two $X_{A_{12}}$'s, which cancel, leaving one $X_{A_{23}}$. That is the 110 row with edge string 01.

Translation to the proof. The three bits of $\phi \in A^0$ say which brackets contributed their nonidentity term. The product of the chosen edge operators is $X^A_{d^a_0\phi}$. Cancellation of duplicate X's is addition modulo two in the coboundary calculation. Nothing has measured ϕ; the eight rows are terms of one algebraic sum.

4. Step 2: undo the cluster circuit and measure the two edges

Apply $C_A^\dagger$, canceling the leftmost C_A. The state becomes

$$\frac{1}{8}\sum_{\phi\in\{0,1\}^3} x_\phi X^a_\phi \Omega(|\Psi\rangle_{BC}|d^a_0\phi\rangle_A|++\rangle_a).$$

Reading the edge ket $|d^a_0\phi\rangle_A$. For each term in this sum, $\phi = (\phi_1, \phi_2, \phi_3)$ is a choice of vertex bits, and $d^a_0\phi$ is the resulting list of edge bits. The subscript A on the ket identifies the edge-qubit register; it does not add another operation. Explicitly,

$$|d^a_0\phi\rangle_A = |\phi_1 + \phi_2\rangle_{A_{12}} \otimes |\phi_2 + \phi_3\rangle_{A_{23}},$$

with both sums taken modulo two. This ket was obtained by applying the edge operators from Step 1 to $|00\rangle_A$:

$$\begin{aligned} X^A_{d^a_0\phi}|00\rangle_A &= (X_{A_{12}})^{\phi_1}(X_{A_{12}}X_{A_{23}})^{\phi_2}(X_{A_{23}})^{\phi_3}|00\rangle_A \\ &= (X_{A_{12}})^{\phi_1+\phi_2}(X_{A_{23}})^{\phi_2+\phi_3}|00\rangle_A \\ &= |d^a_0\phi\rangle_A. \end{aligned}$$

Each vertex with $\phi_i = 1$ contributes an X to every incident edge. If both endpoints contribute to the same edge, the two X's cancel because $X^2 = I$. Thus an edge bit is one precisely when **exactly one of its endpoints is selected**, rather than whenever at least one endpoint is selected. For example, $\phi = 110$ flips A_{12} twice and A_{23} once, producing $|01\rangle_A$. These operations describe the terms already present in the state; they are not additional gates applied during the following measurement.

Now measure $Z_{A_{12}}$ and $Z_{A_{23}}$. Suppose both signs are -1, so the binary record is $z = (1, 1)$. Its projector is

$$P_{11}^A = \frac{I - Z_{A_{12}}}{2} \frac{I - Z_{A_{23}}}{2} = |11\rangle_A \langle 11|_A.$$

For each term, compare its two edge bits with the observed record. The projector acts as

$$P_{11}^A |d_0^a \phi\rangle_A = \langle 11 | d_0^a \phi\rangle_A \, |11\rangle_A.$$

The overlap is one if both bits agree, and zero if either bit disagrees. This is how the ordinary Z-basis measurement projects out the incompatible terms; no separate search or measurement of ϕ is performed physically.

Look at the last column of the table: only $\phi = 101$ and $\phi = 010$ match 11. All other rows are annihilated. Thus

$$\boxed{|\psi_{x,11}\rangle = \frac{1}{8} \left(X_{a_1} X_{a_3} - X_{a_2} \right) \Omega (|\Psi\rangle_{BC} |11\rangle_A |++\rangle_a).}$$

These two contributions remain coherent: the edge measurement cannot distinguish them because they have the same edge string.

Translation to the proof. The edge record is a cochain $z \in A^1$. The selection rule is $d_0^a \phi = z$, or explicitly $\phi_1 + \phi_2 = 1$, $\phi_2 + \phi_3 = 1$. Its two solutions form a coset. Choose $\phi_0 = 101$; then

$$\ker d_0^a = \{000, 111\}, \qquad \{101, 010\} = 101 + \{000, 111\}.$$

The kernel is the remaining ambiguity: flipping all three vertex bits leaves both endpoint differences unchanged.

5. Finish Step 2: remove the offset with two X gates

Our choice $\phi_0 = 101$ tells us to apply $X_{a_1} X_{a_3}$. This operation acts on the whole state, including both surviving terms:

$$\begin{aligned} (X_{a_1} X_{a_3})(X_{a_1} X_{a_3}) &= I, \\ (X_{a_1} X_{a_3}) X_{a_2} &= X_{a_1} X_{a_2} X_{a_3}. \end{aligned}$$

The corrected state is

$$\boxed{|\psi_{\text{corrected}}\rangle = \frac{1}{8} \left(I - X_{a_1} X_{a_2} X_{a_3} \right) \Omega (|\Psi\rangle_{BC} |11\rangle_A |++\rangle_a).}$$

The edge register stays in $|11\rangle_A$. We have changed the operator patterns on the temporary register from $101, 010$ to $000, 111$, not changed the recorded edge outcomes.

Translation to the proof. Multiplication by $X^a_{\phi_0}$ adds ϕ_0 modulo two to each operator label: $101 + 101 = 000$, $010 + 101 = 111$. The labels now lie in the kernel. Here $x_{\phi_0} = x_1 x_3 = +1$, and $x_{111} = x_1 x_2 x_3 = -1$, explaining the retained minus sign.

6. Step 3: undo enrichment, one CCZ at a time

For the CCZ at vertex i, the exact identity is

$$\mathsf{CCZ}_{a_i,B_i,C_i} X_{a_i} \mathsf{CCZ}_{a_i,B_i,C_i} = X_{a_i} V_i.$$

One way to see it is to apply the left side to bits (r, b, c). The two CCZ phases multiply to $(-1)^{rbc+(r+1)bc} = (-1)^{bc}$, while X_{a_i} flips r. These are exactly the actions of $X_{a_i} \mathsf{CZ}_{B_i,C_i}$.

Combining the three identities gives

$$\Omega^\dagger X_{a_1} X_{a_2} X_{a_3} \Omega = X_{a_1} X_{a_2} X_{a_3} V_1 V_2 V_3 = X_{a_1} X_{a_2} X_{a_3} U.$$

Apply $\Omega^\dagger$ to the corrected state:

$$\begin{aligned} |\psi_{\text{out}}\rangle &= \frac{1}{8} \left(I - X_{a_1} X_{a_2} X_{a_3} U \right) \left(|\Psi\rangle_{BC} |11\rangle_A |++\rangle_a \right) \\ &= \frac{1}{8} (I - U) |\Psi\rangle_{BC} \otimes |11\rangle_A \otimes |++\rangle_a \\ &= \frac{1}{4} \left(\frac{I - U}{2} \right) |\Psi\rangle_{BC} \otimes |11\rangle_A \otimes |++\rangle_a. \end{aligned}$$

The second line uses $X_{a_1} X_{a_2} X_{a_3} |++\rangle_a = |++\rangle_a$. The matter has undergone the -1 eigenspace projection of logical $\overline{\mathsf{CZ}}$.

Translation to the proof. The kernel has dimension $k = 1$, with basis $\mu_1 = 111$. Its two-element sum is $I + x_{111} U_{111} = I - U$; the product of projectors has just one factor, $(I - U)/2$. With $N = 3$, the prefactor is exactly $2^{k-N} x_{\phi_0} = 1/4$, as in the complete protocol above.

7. Evaluate an actual encoded input and its probability

Take

$$|\Psi\rangle_{BC} = |\overline{+}\rangle_B |\overline{+}\rangle_C = \frac{1}{2} \left(|\overline{00}\rangle + |\overline{01}\rangle + |\overline{10}\rangle + |\overline{11}\rangle \right).$$

Logical CZ has eigenvalue $+1$ on the first three basis states and -1 on $|\overline{11}\rangle$. Hence

$$\frac{I - U}{2} |\Psi\rangle_{BC} = \frac{1}{2} |\overline{11}\rangle, \qquad |\psi_{\text{out}}\rangle = \frac{1}{8} |\overline{11}\rangle_{BC} |11\rangle_A |++\rangle_a.$$

The normalized conditional matter output is therefore $|\overline{11}\rangle = |111\rangle_B |111\rangle_C$. The probability of this particular full record, $x = (+1, -1, +1)$ and $z = (1, 1)$, is its squared norm, $1/64$. The probability of the logical -1 outcome alone is $1/4$: many full ancillary records have the same logical parity.

Every ideal ancillary record is handled by the same procedure: solve for one compatible ϕ_0, correct it, and retain $x_{111} = x_1x_2x_3$. A positive parity would instead produce $(I+U)/2$, whose normalized output for this input is $(|\overline{00}\rangle + |\overline{01}\rangle + |\overline{10}\rangle)/\sqrt{3}$.

What this example says about the offset. The edge measurement keeps all vertex labels ϕ satisfying $d_0^a\phi = z$. Choose one compatible label ϕ_0, so that $d_0^a\phi_0 = z$. Every other compatible label can be written uniquely, for this choice of ϕ_0, as

$$\phi = \phi_0 + \mu, \qquad \mu \in \ker d_0^a.$$

Adding a kernel element does not change the edge record, because

$$d_0^a(\phi_0 + \mu) = d_0^a\phi_0 + d_0^a\mu = z + 0 = z.$$

The correction is $X^a_{\phi_0}$, with lowercase a: it acts on the temporary **vertex qubits**, not the surviving edge qubits. Applying it to the whole state multiplies the operator in each surviving term:

$$X^a_{\phi_0}X^a_{\phi} = X^a_{\phi_0+\phi}.$$

The labels add modulo two because two X's on the same qubit cancel. Substituting $\phi = \phi_0 + \mu$ gives

$$\phi_0 + \phi = \phi_0 + (\phi_0 + \mu) = \mu,$$

since $\phi_0 + \phi_0 = 0$. Therefore the correction removes the same offset from every term and leaves only kernel labels. In our example,

$$101 + 101 = 000, \qquad 101 + 010 = 111.$$

The whole sum still contains **all the kernel elements**; the correction does not select just one. It transforms the coset $\phi_0 + \ker d_0^a$ into $\ker d_0^a$, while leaving the recorded edge state $|z\rangle_A$ unchanged. On this connected graph, the remaining operator patterns are the identity and the flip of all temporary qubits. These are precisely the ancilla symmetry patterns used in Step 3 to reveal the logical projector.

The graph bookkeeping and the code action have different jobs: the boundary and coboundary determine which vertex patterns share an edge record; the enrichment and the chosen matter code determine which logical operator the remaining kernel symmetry produces.

2.20.9 The magic-state example

Suppose the measured symmetry acts as one logical $\overline{\mathsf{CZ}}$ on a chosen pair of encoded qubits. This is a simple logical example of the preceding result, not an assertion that the whole-block ALP gate acts on only one pair: each ALP block encodes many qubits. If a logical gate acts on several pairs, one must analyze that full action, use an addressable construction, or fix suitable spectator logical qubits so the action reduces to the stated pair.

For that pair, choose the input logical state $|\overline{+}\rangle_B|\overline{+}\rangle_C$. A $+1$ logical CZ outcome gives

$$\frac{I+\overline{\mathsf{CZ}}}{2}|\overline{+}\rangle_B|\overline{+}\rangle_C \propto |\overline{00}\rangle + |\overline{01}\rangle + |\overline{10}\rangle,$$
$$|\overline{M}_{\mathsf{CZ}}\rangle := \frac{|\overline{00}\rangle + |\overline{01}\rangle + |\overline{10}\rangle}{\sqrt{3}}. \tag{213}$$

The $|\overline{11}\rangle$ component is absent because it is the unique computational-basis component with CZ eigenvalue -1. The state in Eq. (213) is non-stabilizer: a stabilizer state has a computational-basis support of size a power of two when its nonzero amplitudes have equal magnitude, whereas this state has three equal nonzero amplitudes. It is therefore a magic-state resource.

For this particular input, the $+1$ outcome occurs with probability $3/4$; the -1 outcome gives $|\overline{11}\rangle$. Thus this simple demonstration is heralded rather than a deterministic magic-state factory: retain a $+1$ trial or repeat the preparation. The paper's central achievement is that the required logical CZ measurement has been compiled into local enrichment gates, gauging and ungauging measurements, classical decoding, and feedforward on qLDPC code blocks.

2.20.10 Where does the magic actually come from?

At first sight, the last calculation can sound paradoxical. The measured operator $\overline{\mathsf{CZ}}$ is a Clifford gate, yet its $+1$ outcome produces a non-stabilizer state. Does the protocol somehow create magic using only Clifford operations?

It does not. The important distinction is between **applying a Clifford unitary** and **measuring a Clifford operator**. Applying CZ to a stabilizer state only changes signs and correlations; the result is still a stabilizer state. Measuring CZ, however, uses the two projectors

$$P_+ = \frac{I+\mathsf{CZ}}{2}, \qquad P_- = \frac{I-\mathsf{CZ}}{2}.$$

These projectors are not Clifford unitaries: they are not unitary at all. They remove the component of a state lying in the opposite eigenspace. For example,

$$|++\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2},$$

and $|11\rangle$ is the only basis vector on which CZ has eigenvalue -1. Consequently,

$$P_+|++\rangle = \frac{|00\rangle + |01\rangle + |10\rangle}{2},$$

before normalization. This projection deletes one amplitude; merely applying CZ would instead change the sign of that amplitude. A measurement of a Clifford operator is therefore not generally a Pauli measurement or a stabilizer operation.

There must nevertheless be a non-Clifford physical ingredient somewhere. In this construction it is the enrichment circuit Ω, which contains local physical CCZ gates. This is consistent with the general principle that stabilizer-state preparation, Clifford unitaries, Pauli measurements and classical feedforward cannot produce a magic state by themselves. Schematically, the resource conversion is

$$\boxed{\text{local physical } \mathsf{CCZ} \text{ resources}} \longrightarrow \boxed{\text{encoded logical } \mathsf{CZ} \text{ measurement}} \longrightarrow \boxed{\text{encoded non-stabilizer state}}.$$

The first box is where the magic enters. The last box is the useful encoded form in which the protocol delivers it.

2.20.11 What the construction gains—and what it has not yet proved

The physical CCZ gates in Ω are local, bounded-weight gates: each one couples a temporary ancilla to one nearby pair of data qubits. The protocol does not assume that the input qLDPC code already has a transversal **logical** CCZ. Instead, gauging, measuring, decoding and ungauging combine many local physical enrichment gates into a measurement of an encoded logical CZ, without decoding the logical information into unprotected physical qubits. This conversion is the conceptual contribution.

Whether it is also an economical fault-tolerant procedure is a separate question. Suppose a hardware platform can perform local physical CCZ gates directly and with tolerable noise. The construction might then provide a useful route from those elementary operations to an encoded magic state in a qLDPC code for which direct logical non-Clifford operations are difficult. Locality, parallel execution and remaining inside the encoded architecture could all be advantages.

On the other hand, suppose every physical CCZ must itself be implemented by consuming a carefully prepared magic state. The present argument does not show that the total magic-state cost is reduced; it may only have moved that cost into the enrichment circuit. Using a non-Clifford resource to prepare another non-Clifford resource is useful only if the input resource is easier to supply—for example, if it may be noisier, more local, or cheaper than the encoded state produced at the end.

The paper establishes the ideal algebraic transformation, but it does not yet give a complete fault-tolerance theorem for this protocol. A practical scheme would additionally have to show that

1. errors in the physical CCZ gates do not spread into uncorrectable errors;
2. noisy ancilla preparation and noisy measurements can be decoded;
3. the logical failure probability decreases as the code grows;
4. the required connectivity and parallel gates are physically realistic;
5. the complete resource cost compares favourably with alternatives such as ordinary magic-state injection and distillation.

The paper explicitly describes fault tolerance as its largest unaddressed question. Its conclusion should therefore be read carefully: it provides a general method for **converting local non-Clifford physical resources into an encoded magic state through logical Clifford measurement**. It does not yet prove that this conversion is a lower-overhead magic-state factory, nor that the resulting computation is fault tolerant. Those are the next questions that must be answered before assessing its practical advantage.

2.21 Sources and scope

- Target text: arXiv-2602.12228v1/main.tex, Sections II.A–III.C, "Two Approaches to Gauging qLDPC Codes," and the introduction, "Graph requirements," and "Example: Gauging transversal SWAP gate" of "Non-Abelian Codes from Gauging with Graph States," plus the opening graph-gauged ALP fracton example and "Measurement Preparation of Non-Abelian qLDPC Codes," and "Logical Clifford Measurement from Gauging and Ungauging" (Section VI.A).
- ALP illustrations: the original TikZ diagrams from source labels `eq:qubit-ALP-stabilizers`, `eq:cubic-CZ-sym`, and `eq:cubic-symm-enrich`, rendered with their original geometry and colours. Reproducible extracts are in figures/alp-paper.
- Context for gauging and $\mathcal{D}(D_4)$: arXiv-2209.06202v2/main.tex, especially its discussion of Abelian Kramers-Wannier gauging and D_4 quantum doubles.

- Experimental and twisted-$\mathbb{Z}_2^3$ presentation: arXiv-2305.03766v2/main.tex, especially "The model," "Ground State Preparation," and its table equating $\mathcal{D}(D_4)$ with $\mathcal{D}^\alpha(\mathbb{Z}_2^3)$.